EXP 801 DS Wheat Heat Drought

— Ed Chaplin (USyd), *Apr–June 2025* —

Porometer DATA - ANALYSIS OF LIGHT OFF PORTION

- Analysis of gs decrease, when light switched off for the second time

Load the libraries

```
library(knitr)
#knitr::spin("Exp801.porometer.anal_d.v1.r")
library(ggplot2)
opts_chunk$set("tidy" = FALSE, "dev" = "png", "dpi" = 300,
               "fig.width" = 7.5, "fig.height" = 10, "comment"=NA, fig.path = "figures/increase/")
library(dplyr)
library(asreml)
library(brms) ##package used to fit bayesian regression model using Stan
```

Set up constants and files

```
options(width=90)
load("../data/porometer.v1.Rdata")
saveFile <- "../data/porometer.anal_d.v1.Rdata"
```

Fit a model to the entire dataset, for the light off portion, following light on

```
##find the minimum time when the light was on
min(li600.dat$time.elapsed[li600.dat$Light == "On"]) ##38.46667
```

```
[1] 38.46667
```

```
max(li600.dat$time.elapsed[li600.dat$Light == "On"]) ##101.5667
```

```
[1] 101.5667
```

```
hist(li600.dat$time.elapsed[li600.dat$Light == "Off" &
                            li600.dat$time.elapsed > 38.46667 &
                            li600.dat$time.elapsed < 101.5667])
```

```
li600.off.dat <- subset(li600.dat, Light == "Off" &
                        time.elapsed > 60)

##using full dataset, figure out the earliest time at which the light was turned off for each day
li600.off.dat <- li600.off.dat %>% group_by(Date) %>% mutate(min.time = min(time.elapsed))
##calculate adjusted time by subtracting the earliest time the light was on each day
li600.off.dat$time.adj <- li600.off.dat$time.elapsed - li600.off.dat$min.time

#head(li600.dat)
# unique(li600.off.dat$CEBRoom)
# unique(li600.off.dat$CEBLane)
# unique(li600.off.dat$CEBPosition)
# unique(li600.off.dat$CEBSide)
# unique(li600.off.dat$CEBLocation)
# unique(li600.off.dat$CEBBlock)
```

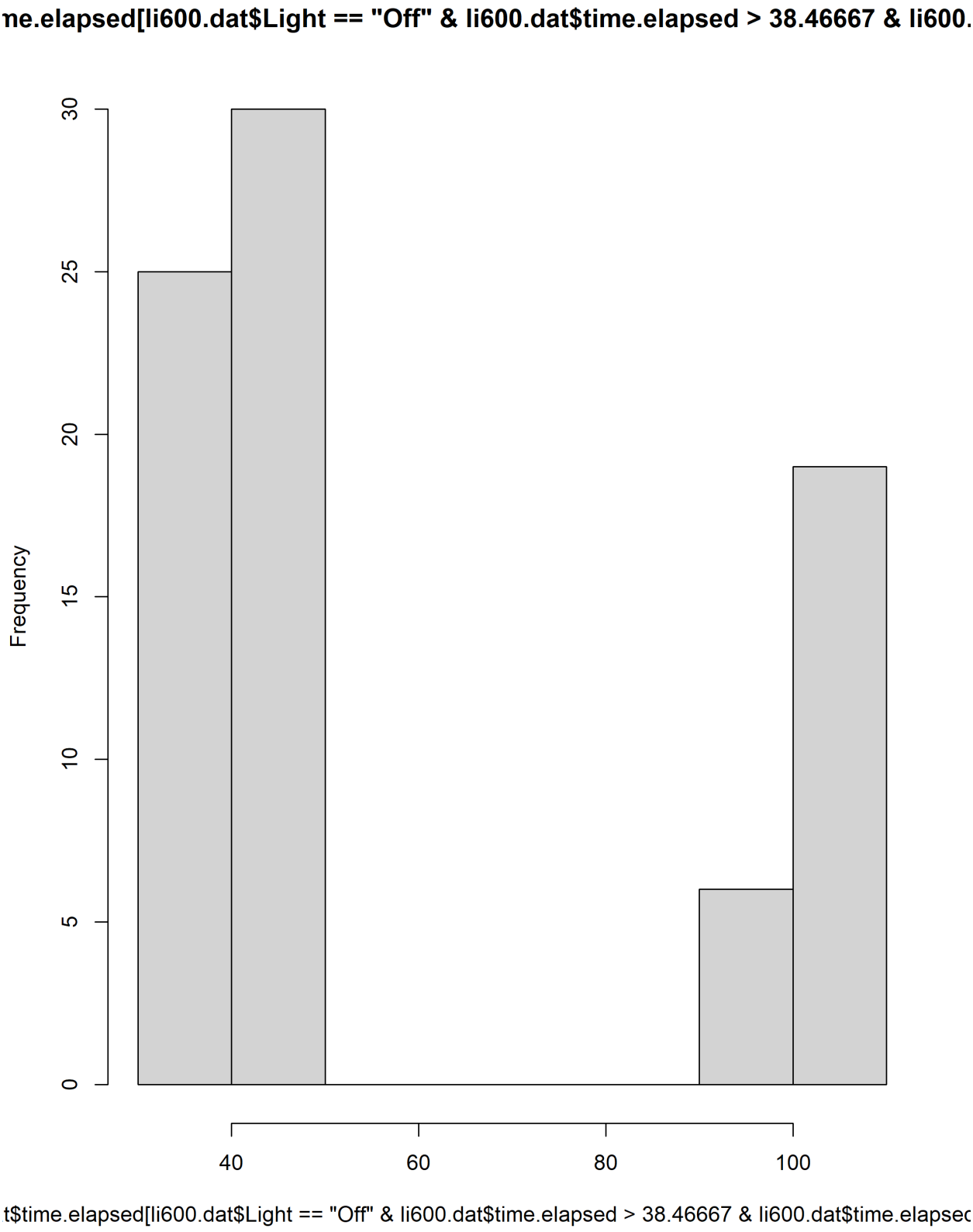


Figure 1: plot of chunk unnamed-chunk-3

```
# unique(li600.off.dat$CEBSubUnit)
# unique(li600.off.dat$CEBCell)
# ggplot(li600.off.dat, aes(x=CEBLane, y = CEBPosition)) +
#   geom_tile(fill = "white") +
#   geom_text(aes(label = CEBSide)) +
#   facet_wrap(~CEBRoom)

##Fit a variance component model to the time course data
# li600.off.varcomp <- asreml(gsw ~ (Room_Temperature*Device_Watering*Line) +
#                             time.elapsed +
#                             Room_Temperature:time.elapsed +
#                             Device_Watering:time.elapsed +
#                             Line:time.elapsed +
#                             Room_Temperature:Device_Watering:time.elapsed +
#                             Room_Temperature:Line:time.elapsed +
#                             Device_Watering:Line:time.elapsed,# +
#                             #Room_Temperature:Device_Watering:Line:time.elapsed,
#                           random = ~ spl(time.elapsed) +
#                             Room_Temperature:spl(time.elapsed) +
#                             Device_Watering:spl(time.elapsed) +
#                             Line:spl(time.elapsed) +
#                             Room_Temperature:Device_Watering:spl(time.elapsed) +
#                             Room_Temperature:Line:spl(time.elapsed) +
#                             Device_Watering:Line:spl(time.elapsed) +
#                             #Room_Temperature:Device_Watering:Line:spl(time.elapsed) +
#                             CEBRoom:CEBSide +
#                             CEBRoom:CEBSide:CEBLocation +
#                             CEBRoom:CEBSide:CEBLocation:CEBSubUnit +
#                             CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell,
#                           residual = ~ units,
#                           data = li600.off.dat,
#                           workspace = "5gb")
#
# li600.off.varcomp <- update(li600.off.varcomp)
# summary(li600.off.varcomp)$varcomp
# wald(li600.off.varcomp, denDF = "algebraic", ssType = "conditional")

# plot(li600.off.varcomp)

# temp.asr <- asreml(gsw ~ (Room_Temperature*Device_Watering*Line) +
#                                time.elapsed +
#                                Room_Temperature:time.elapsed +
#                                Device_Watering:time.elapsed +
#                                Line:time.elapsed +
#                                Room_Temperature:Device_Watering:time.elapsed +
#                                Room_Temperature:Line:time.elapsed +
#                                Device_Watering:Line:time.elapsed,# +
#                                #Room_Temperature:Device_Watering:Line:time.elapsed,
#                              random = ~ spl(time.elapsed) +
#                                Room_Temperature:spl(time.elapsed) +
#                                Device_Watering:spl(time.elapsed) +
#                                Line:spl(time.elapsed) +
#                                Room_Temperature:Device_Watering:spl(time.elapsed) +
```

```
#                                 Room_Temperature:Line:spl(time.elapsed) +
#                                 Device_Watering:Line:spl(time.elapsed) +
#                                 #Room_Temperature:Device_Watering:Line:spl(time.elapsed) +
#                                 CEBRoom:CEBSide +
#                                 CEBRoom:CEBSide:CEBLocation +
#                                 CEBRoom:CEBSide:CEBLocation:CEBSubUnit +
#                                 CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell,
#                               residual = ~ units,
#                               data = li600.off.dat,
#                               workspace = "5gb")
# temp.asr <- update(temp.asr)
# lrt(li600.off.varcomp, temp.asr)

# li600.off.varcomp.pred_d1c8 <- predict(li600.off.varcomp,
#                                        "Room_Temperature:Device_Watering:Line:time.elapsed",
#                                        levels = list(Device_Watering = "D1_Unlimited",
#                                                      Room_Temperature = "CER8_Ambient",
#                                                      time.elapsed = seq(100, 165, 1)),
#                                    workspace = "5gb",
#                                    pworkspace = "5gb")
# head(li600.off.varcomp.pred_d1c8$pvals, 30)
# li600.off.varcomp.pred_d1c7 <- predict(li600.off.varcomp,
#                                    "Room_Temperature:Device_Watering:Line:time.elapsed",
#                                    levels = list(Device_Watering = "D1_Unlimited",
#                                                  Room_Temperature = "CER7_Heat",
#                                                  time.elapsed = seq(100, 165, 1)),
#                                    workspace = "5gb",
#                                    pworkspace = "5gb")
# head(li600.off.varcomp.pred_d1c7$pvals, 30)
# li600.off.varcomp.pred_d2c8 <- predict(li600.off.varcomp,
#                                    "Room_Temperature:Device_Watering:Line:time.elapsed",
#                                    levels = list(Device_Watering = "D2_Limited",
#                                                  Room_Temperature = "CER8_Ambient",
#                                                  time.elapsed = seq(100, 165, 1)),
#                                    workspace = "5gb",
#                                    pworkspace = "5gb")
# head(li600.off.varcomp.pred_d2c8$pvals, 30)
# li600.off.varcomp.pred_d2c7 <- predict(li600.off.varcomp,
#                                    "Room_Temperature:Device_Watering:Line:time.elapsed",
#                                    levels = list(Device_Watering = "D2_Limited",
#                                                  Room_Temperature = "CER7_Heat",
#                                                  time.elapsed = seq(100, 165, 1)),
#                                    workspace = "5gb",
#                                    pworkspace = "5gb")
# head(li600.off.varcomp.pred_d1c8$pvals, 30)
# li600.off.varcomp.pred <- bind_rows(li600.off.varcomp.pred_d1c8$pvals,
#                                    li600.off.varcomp.pred_d1c7$pvals,
#                                    li600.off.varcomp.pred_d2c8$pvals,
#                                    li600.off.varcomp.pred_d2c7$pvals)
#
# ggplot(li600.off.varcomp.pred, aes(x=time.elapsed, y = predicted.value)) +
#   facet_grid(Room_Temperature + Device_Watering ~ Line) +
#   geom_point(data = li600.off.dat, aes(x=time.elapsed, y = gsw,
```

```
#                                                colour = Replicate)) +
#   geom_point(colour = "red")
```

Stomatal conductance model - McAusland from McAusland et al 2016 Now let's try the Baesian approach that was fitted to the li-6800 data

```
nlform <- bf(gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - time.adj)/k)+1)),
             g0 ~ 1 + Line*Device_Watering*Room_Temperature +
               (1 | CEBRoom:CEBSide + CEBRoom:CEBSide:CEBLocation +
                  CEBRoom:CEBSide:CEBLocation:CEBSubUnit +
                  CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell),
             g1 ~ 1 + Line*Device_Watering*Room_Temperature +
               (1 | CEBRoom:CEBSide + CEBRoom:CEBSide:CEBLocation +
                  CEBRoom:CEBSide:CEBLocation:CEBSubUnit +
                  CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell),
             k ~ 1 + Line*Device_Watering*Room_Temperature,
             lambda ~ 1 + Line*Device_Watering*Room_Temperature,
             nl = TRUE)

priors <- c(prior(normal(0.2, 0.5), nlpar = "g0"),
            prior(normal(0, 0.02), nlpar = "g1"),
            prior(normal(3, 0.1), nlpar = "k", lb = 0),
            prior(normal(0.5, 0.25), nlpar = "lambda", lb = 0),
            prior(normal(0, 0.15), class = "sd", nlpar = "g0"),
            prior(normal(0, 0.01), class = "sd", nlpar = "g1"),
            prior(student_t(3, 0, 0.005), class = "sigma"))

plot(0.2 + (0 - 0.2) * exp(-exp(((0.5 - seq(0, 60, 0.5))/3) + 1)) ~
       seq(0, 60, 0.5))
```

```
# Fit the model
fit_off_max <- brm(
  formula = nlform,
  data = li600.off.dat,
  prior = priors,
  chains = 6,
  cores = 6,
  iter = 3000,
  warmup = 2000,
  control = list(
    adapt_delta = 0.99,
    max_treedepth = 12,
    metric = "diag_e"
  )
)
```

```
Compiling Stan program...
```

```
Start sampling
```

```
print(summary(fit_off_max))
```

```
 Family: gaussian
  Links: mu = identity
Formula: gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - time.adj)/k) + 1))
```

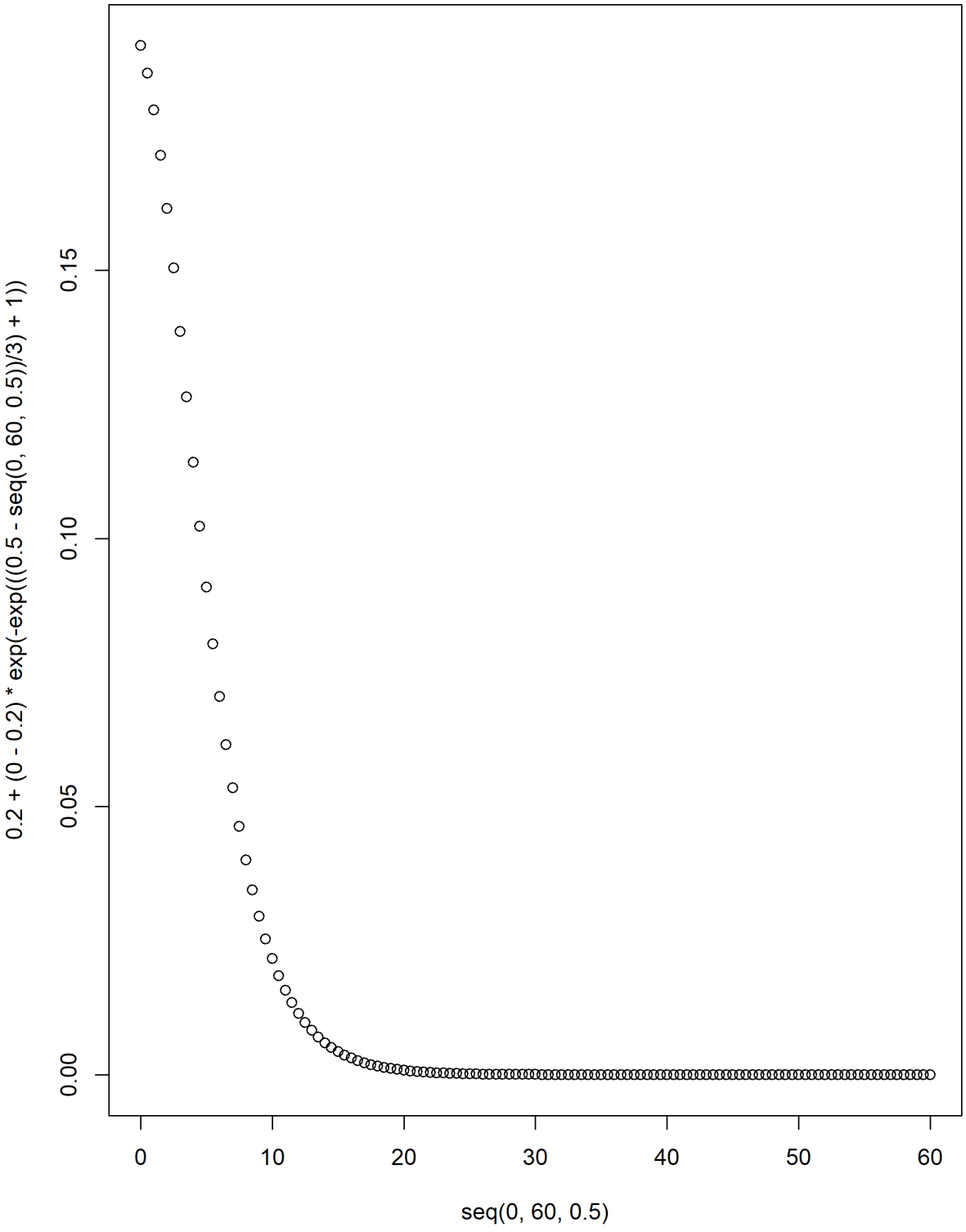


Figure 2: plot of chunk unnamed-chunk-4

```
        g0 ~ 1 + Line * Device_Watering * Room_Temperature + (1 | CEBRoom:CEBSide + CEBRoom:CEBSide:CE
        g1 ~ 1 + Line * Device_Watering * Room_Temperature + (1 | CEBRoom:CEBSide + CEBRoom:CEBSide:CE
        k ~ 1 + Line * Device_Watering * Room_Temperature
        lambda ~ 1 + Line * Device_Watering * Room_Temperature
   Data: li600.off.dat (Number of observations: 1959)
  Draws: 6 chains, each with iter = 3000; warmup = 2000; thin = 1;
         total post-warmup draws = 6000

Multilevel Hyperparameters:
~CEBRoom:CEBSide (Number of levels: 4)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.04      0.04     0.00     0.16 1.00     2735     3203
sd(g1_Intercept)     0.01      0.00     0.00     0.02 1.00     3711     3126

~CEBRoom:CEBSide:CEBLocation (Number of levels: 8)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.02      0.02     0.00     0.06 1.00     2759     3367
sd(g1_Intercept)     0.01      0.00     0.00     0.02 1.00     3054     3359

~CEBRoom:CEBSide:CEBLocation:CEBSubUnit (Number of levels: 16)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.03      0.01     0.00     0.06 1.00     1307     2023
sd(g1_Intercept)     0.02      0.00     0.01     0.02 1.00     4044     4538

~CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell (Number of levels: 159)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.07      0.01     0.06     0.09 1.00     2033     3147
sd(g1_Intercept)     0.00      0.00     0.00     0.01 1.00     3030     3458

Regression Coefficients:
                                                                                                  Estimate
g0_Intercept                                                                                          0.38
g0_LinePBI17N002M0CM0NM0NM010NM16N                                                                   -0.29
g0_LinePBI17N006M0NM0NM89N                                                                           -0.08
g0_LinePBI17N007M0CM0NM0NM010NM8N                                                                    -0.12
g0_LinePBI17N013M0CM0NM0NM010NM8N                                                                    -0.11
g0_LinePBI17N016M0CM0NM0NM010NM27N                                                                   -0.29
g0_LinePBI19N007M0NM3N                                                                               -0.02
g0_LinePBIC15034M0CM24NM010NM1NM0N                                                                   -0.17
g0_LineROCKSTAR                                                                                      -0.09
g0_LineSUNCHASER                                                                                     -0.07
g0_Device_WateringD2_Limited                                                                         -0.36
g0_Room_TemperatureCER8_Ambient                                                                       0.18
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited                                          0.28
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited                                                  0.09
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited                                           0.10
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited                                           0.17
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited                                          0.28
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited                                                      0.10
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited                                          0.20
g0_LineROCKSTAR:Device_WateringD2_Limited                                                             0.19
g0_LineSUNCHASER:Device_WateringD2_Limited                                                            0.10
g0_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient                                      -0.18
g0_LinePBI17N006M0NM0NM89N:Room_TemperatureCER8_Ambient                                              -0.26
```

g0_LinePBI17N007M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	-0.30
g0_LinePBI17N013M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	-0.22
g0_LinePBI17N016M0CM0NM0NM010NM27N:Room_TemperatureCER8_Ambient	-0.18
g0_LinePBI19N007M0NM3N:Room_TemperatureCER8_Ambient	-0.22
g0_LinePBIC15034M0CM24NM010NM1NM0N:Room_TemperatureCER8_Ambient	-0.16
g0_LineROCKSTAR:Room_TemperatureCER8_Ambient	-0.25
g0_LineSUNCHASER:Room_TemperatureCER8_Ambient	-0.22
g0_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.09
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.10
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.18
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.22
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.12
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.15
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.10
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.06
g0_LineROCKSTAR:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.17
g0_LineSUNCHASER:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.15
g1_Intercept	0.00
g1_LinePBI17N002M0CM0NM0NM010NM16N	-0.01
g1_LinePBI17N006M0NM0NM89N	-0.02
g1_LinePBI17N007M0CM0NM0NM010NM8N	-0.01
g1_LinePBI17N013M0CM0NM0NM010NM8N	-0.02
g1_LinePBI17N016M0CM0NM0NM010NM27N	-0.01
g1_LinePBI19N007M0NM3N	-0.01
g1_LinePBIC15034M0CM24NM010NM1NM0N	-0.02
g1_LineROCKSTAR	-0.01
g1_LineSUNCHASER	-0.01
g1_Device_WateringD2_Limited	-0.00
g1_Room_TemperatureCER8_Ambient	-0.01
g1_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	-0.00
g1_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	0.00
g1_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	0.00
g1_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	-0.00
g1_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	-0.00
g1_LinePBI19N007M0NM3N:Device_WateringD2_Limited	-0.01
g1_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	0.00
g1_LineROCKSTAR:Device_WateringD2_Limited	-0.02
g1_LineSUNCHASER:Device_WateringD2_Limited	-0.00
g1_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient	0.01
g1_LinePBI17N006M0NM0NM89N:Room_TemperatureCER8_Ambient	-0.02
g1_LinePBI17N007M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	-0.00
g1_LinePBI17N013M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	-0.01
g1_LinePBI17N016M0CM0NM0NM010NM27N:Room_TemperatureCER8_Ambient	-0.00
g1_LinePBI19N007M0NM3N:Room_TemperatureCER8_Ambient	-0.02
g1_LinePBIC15034M0CM24NM010NM1NM0N:Room_TemperatureCER8_Ambient	-0.01
g1_LineROCKSTAR:Room_TemperatureCER8_Ambient	-0.02
g1_LineSUNCHASER:Room_TemperatureCER8_Ambient	-0.02
g1_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.00
g1_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.00
g1_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.01
g1_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.01
g1_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.01
g1_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.01
g1_LinePBI19N007M0NM3N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.01

g1_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.00
g1_LineROCKSTAR:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.02
g1_LineSUNCHASER:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.01
k_Intercept	1.98
k_LinePBI17N002M0CM0NM0NM010NM16N	2.97
k_LinePBI17N006M0NM0NM89N	2.80
k_LinePBI17N007M0CM0NM0NM010NM8N	2.93
k_LinePBI17N013M0CM0NM0NM010NM8N	2.88
k_LinePBI17N016M0CM0NM0NM010NM27N	2.97
k_LinePBI19N007M0NM3N	2.74
k_LinePBIC15034M0CM24NM010NM1NM0N	2.86
k_LineROCKSTAR	2.83
k_LineSUNCHASER	2.84
k_Device_WateringD2_Limited	2.89
k_Room_TemperatureCER8_Ambient	2.54
k_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	3.00
k_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	3.00
k_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	3.00
k_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	2.99
k_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	2.99
k_LinePBI19N007M0NM3N:Device_WateringD2_Limited	2.98
k_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	2.99
k_LineROCKSTAR:Device_WateringD2_Limited	2.95
k_LineSUNCHASER:Device_WateringD2_Limited	2.99
k_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient	2.98
k_LinePBI17N006M0NM0NM89N:Room_TemperatureCER8_Ambient	2.93
k_LinePBI17N007M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	2.98
k_LinePBI17N013M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	2.93
k_LinePBI17N016M0CM0NM0NM010NM27N:Room_TemperatureCER8_Ambient	2.99
k_LinePBI19N007M0NM3N:Room_TemperatureCER8_Ambient	2.90
k_LinePBIC15034M0CM24NM010NM1NM0N:Room_TemperatureCER8_Ambient	2.92
k_LineROCKSTAR:Room_TemperatureCER8_Ambient	2.94
k_LineSUNCHASER:Room_TemperatureCER8_Ambient	2.92
k_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	2.96
k_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	3.00
k_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	3.00
k_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	3.00
k_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	3.00
k_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	2.99
k_LinePBI19N007M0NM3N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	3.00
k_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	3.00
k_LineROCKSTAR:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	2.98
k_LineSUNCHASER:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	3.00
lambda_Intercept	0.02
lambda_LinePBI17N002M0CM0NM0NM010NM16N	0.42
lambda_LinePBI17N006M0NM0NM89N	0.16
lambda_LinePBI17N007M0CM0NM0NM010NM8N	0.35
lambda_LinePBI17N013M0CM0NM0NM010NM8N	0.28
lambda_LinePBI17N016M0CM0NM0NM010NM27N	0.41
lambda_LinePBI19N007M0NM3N	0.12
lambda_LinePBIC15034M0CM24NM010NM1NM0N	0.25
lambda_LineROCKSTAR	0.18
lambda_LineSUNCHASER	0.22
lambda_Device_WateringD2_Limited	0.18

lambda_Room_TemperatureCER8_Ambient	0.06
lambda_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	0.50
lambda_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	0.49
lambda_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	0.50
lambda_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	0.48
lambda_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	0.48
lambda_LinePBI19N007M0NM3N:Device_WateringD2_Limited	0.45
lambda_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	0.48
lambda_LineROCKSTAR:Device_WateringD2_Limited	0.37
lambda_LineSUNCHASER:Device_WateringD2_Limited	0.45
lambda_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient	0.45
	Tail_ESS
g0_Intercept	2356
g0_LinePBI17N002M0CM0NM0NM010NM16N	3040
g0_LinePBI17N006M0NM0NM89N	2855
g0_LinePBI17N007M0CM0NM0NM010NM8N	3149
g0_LinePBI17N013M0CM0NM0NM010NM8N	3030
g0_LinePBI17N016M0CM0NM0NM010NM27N	2468
g0_LinePBI19N007M0NM3N	2847
g0_LinePBIC15034M0CM24NM010NM1NM0N	2365
g0_LineROCKSTAR	2662
g0_LineSUNCHASER	2548
g0_Device_WateringD2_Limited	2555
g0_Room_TemperatureCER8_Ambient	2514
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	3093
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	2531
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	2911
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	2873
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	2922
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited	2515
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	2888
g0_LineROCKSTAR:Device_WateringD2_Limited	2935
g0_LineSUNCHASER:Device_WateringD2_Limited	3013
g0_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient	2674
g0_LinePBI17N006M0NM0NM89N:Room_TemperatureCER8_Ambient	2600
g0_LinePBI17N007M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	2715
g0_LinePBI17N013M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	3208
g0_LinePBI17N016M0CM0NM0NM010NM27N:Room_TemperatureCER8_Ambient	2526
g0_LinePBI19N007M0NM3N:Room_TemperatureCER8_Ambient	2909
g0_LinePBIC15034M0CM24NM010NM1NM0N:Room_TemperatureCER8_Ambient	2560
g0_LineROCKSTAR:Room_TemperatureCER8_Ambient	2710
g0_LineSUNCHASER:Room_TemperatureCER8_Ambient	3088
g0_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	2561
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	2251
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	2349
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	3122
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	3099
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	2957
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	2563
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	2719
g0_LineROCKSTAR:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	2845
g0_LineSUNCHASER:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	2581
g1_Intercept	3948
g1_LinePBI17N002M0CM0NM0NM010NM16N	4738

g1_LinePBI17N006M0NM0NM89N	4406
g1_LinePBI17N007M0CM0NM0NM010NM8N	4757
g1_LinePBI17N013M0CM0NM0NM010NM8N	4973
g1_LinePBI17N016M0CM0NM0NM010NM27N	4935
g1_LinePBI19N007M0NM3N	4887
g1_LinePBIC15034M0CM24NM010NM1NM0N	4910
g1_LineROCKSTAR	4696
g1_LineSUNCHASER	5103
g1_Device_WateringD2_Limited	3891
g1_Room_TemperatureCER8_Ambient	4250
g1_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	5274
g1_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	4884
g1_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	4597
g1_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	4977
g1_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	4493
g1_LinePBI19N007M0NM3N:Device_WateringD2_Limited	5347
g1_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	5240
g1_LineROCKSTAR:Device_WateringD2_Limited	5123
g1_LineSUNCHASER:Device_WateringD2_Limited	4835
g1_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient	4932
g1_LinePBI17N006M0NM0NM89N:Room_TemperatureCER8_Ambient	4946
g1_LinePBI17N007M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	5082
g1_LinePBI17N013M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	4712
g1_LinePBI17N016M0CM0NM0NM010NM27N:Room_TemperatureCER8_Ambient	4497
g1_LinePBI19N007M0NM3N:Room_TemperatureCER8_Ambient	4796
g1_LinePBIC15034M0CM24NM010NM1NM0N:Room_TemperatureCER8_Ambient	4884
g1_LineROCKSTAR:Room_TemperatureCER8_Ambient	4630
g1_LineSUNCHASER:Room_TemperatureCER8_Ambient	5302
g1_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	4436
g1_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	4051
g1_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	4719
g1_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	4621
g1_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	4922
g1_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	4530
g1_LinePBI19N007M0NM3N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	4770
g1_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	4894
g1_LineROCKSTAR:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	4600
g1_LineSUNCHASER:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	4417
k_Intercept	4896
k_LinePBI17N002M0CM0NM0NM010NM16N	4166
k_LinePBI17N006M0NM0NM89N	4797
k_LinePBI17N007M0CM0NM0NM010NM8N	4186
k_LinePBI17N013M0CM0NM0NM010NM8N	4390
k_LinePBI17N016M0CM0NM0NM010NM27N	4294
k_LinePBI19N007M0NM3N	4105
k_LinePBIC15034M0CM24NM010NM1NM0N	4327
k_LineROCKSTAR	4484
k_LineSUNCHASER	4144
k_Device_WateringD2_Limited	4438
k_Room_TemperatureCER8_Ambient	4381
k_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	4506
k_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	4314
k_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	4305
k_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	4195

```
k_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited                                  3937
k_LinePBI19N007M0NM3N:Device_WateringD2_Limited                                              3884
k_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited                                  4034
k_LineROCKSTAR:Device_WateringD2_Limited                                                     4497
k_LineSUNCHASER:Device_WateringD2_Limited                                                    4267
k_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient                               3724
k_LinePBI17N006M0NM0NM89N:Room_TemperatureCER8_Ambient                                       4290
k_LinePBI17N007M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient                                4181
k_LinePBI17N013M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient                                4048
k_LinePBI17N016M0CM0NM0NM010NM27N:Room_TemperatureCER8_Ambient                               3865
k_LinePBI19N007M0NM3N:Room_TemperatureCER8_Ambient                                           4388
k_LinePBIC15034M0CM24NM010NM1NM0N:Room_TemperatureCER8_Ambient                               4555
k_LineROCKSTAR:Room_TemperatureCER8_Ambient                                                  4271
k_LineSUNCHASER:Room_TemperatureCER8_Ambient                                                 4075
k_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                                     3920
k_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient     4229
k_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient             4368
k_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient      4229
k_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient      4147
k_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient     4429
k_LinePBI19N007M0NM3N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                 4285
k_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient     4360
k_LineROCKSTAR:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                        4032
k_LineSUNCHASER:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                       4375
lambda_Intercept                                                                             3349
lambda_LinePBI17N002M0CM0NM0NM010NM16N                                                       1703
lambda_LinePBI17N006M0NM0NM89N                                                               3287
lambda_LinePBI17N007M0CM0NM0NM010NM8N                                                        2588
lambda_LinePBI17N013M0CM0NM0NM010NM8N                                                        3302
lambda_LinePBI17N016M0CM0NM0NM010NM27N                                                       2363
lambda_LinePBI19N007M0NM3N                                                                   3287
lambda_LinePBIC15034M0CM24NM010NM1NM0N                                                       2699
lambda_LineROCKSTAR                                                                          3238
lambda_LineSUNCHASER                                                                         2972
lambda_Device_WateringD2_Limited                                                             2828
lambda_Room_TemperatureCER8_Ambient                                                          3473
lambda_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited                             1798
lambda_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited                                     2277
lambda_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited                              2334
lambda_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited                              1827
lambda_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited                             1859
lambda_LinePBI19N007M0NM3N:Device_WateringD2_Limited                                         1801
lambda_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited                             2360
lambda_LineROCKSTAR:Device_WateringD2_Limited                                                2716
lambda_LineSUNCHASER:Device_WateringD2_Limited                                               2406
lambda_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient                          2105
 [ reached 'max' / getOption("max.print") -- omitted 18 rows ]

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.04      0.00     0.04     0.04 1.00     6566     4943

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
```

```
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
plot(fit_off_max, ask = FALSE)
```

```
#pairs(fit_off_max)
#mcmc_plot(fit_off_max, type = "trace")
rstan::check_hmc_diagnostics(fit_off_max$fit)
```

```
Divergences:
0 of 6000 iterations ended with a divergence.

Tree depth:
0 of 6000 iterations saturated the maximum tree depth of 12.

Energy:
E-BFMI indicated no pathological behavior.
```

```
loo(fit_off_max)
```

```
Warning: Found 61 observations with a pareto_k > 0.7 in model 'fit_off_max'. We recommend to set 'momen
matching for problematic observations.

Computed from 6000 by 1959 log-likelihood matrix.

         Estimate    SE
elpd_loo   3199.2  85.0
p_loo       395.2  35.9
looic     -6398.3 169.9
------
MCSE of elpd_loo is NA.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.5, 1.6]).

Pareto k diagnostic values:
                         Count Pct.    Min. ESS
(-Inf, 0.7]   (good)     1898  96.9%   133
   (0.7, 1]   (bad)        41   2.1%   <NA>
   (1, Inf)   (very bad)   20   1.0%   <NA>
See help('pareto-k-diagnostic') for details.
```

```
pp_check(fit_off_max)  # compares density of the observed data with data generated from posterior predi
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

```
pp_check(fit_off_max, type = "error_hist", ndraws = 12)
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
pp_check(fit_off_max, type = "scatter_avg", ndraws = 1000)
```

```
pp_check(fit_off_max, type = "stat_2d")
```

```
Using all posterior draws for ppc type 'stat_2d' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest.
```

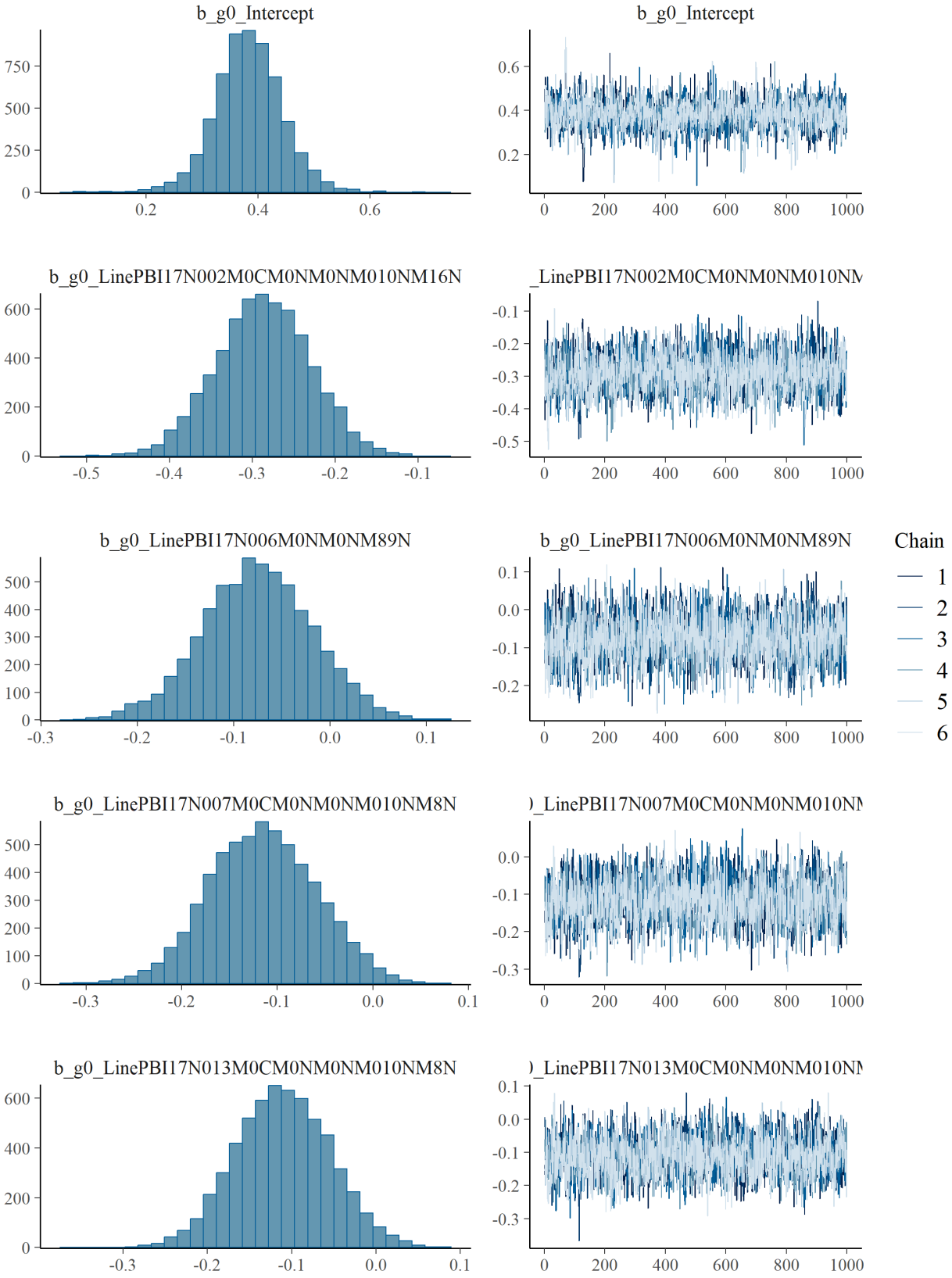


Figure 3: plot of chunk unnamed-chunk-4

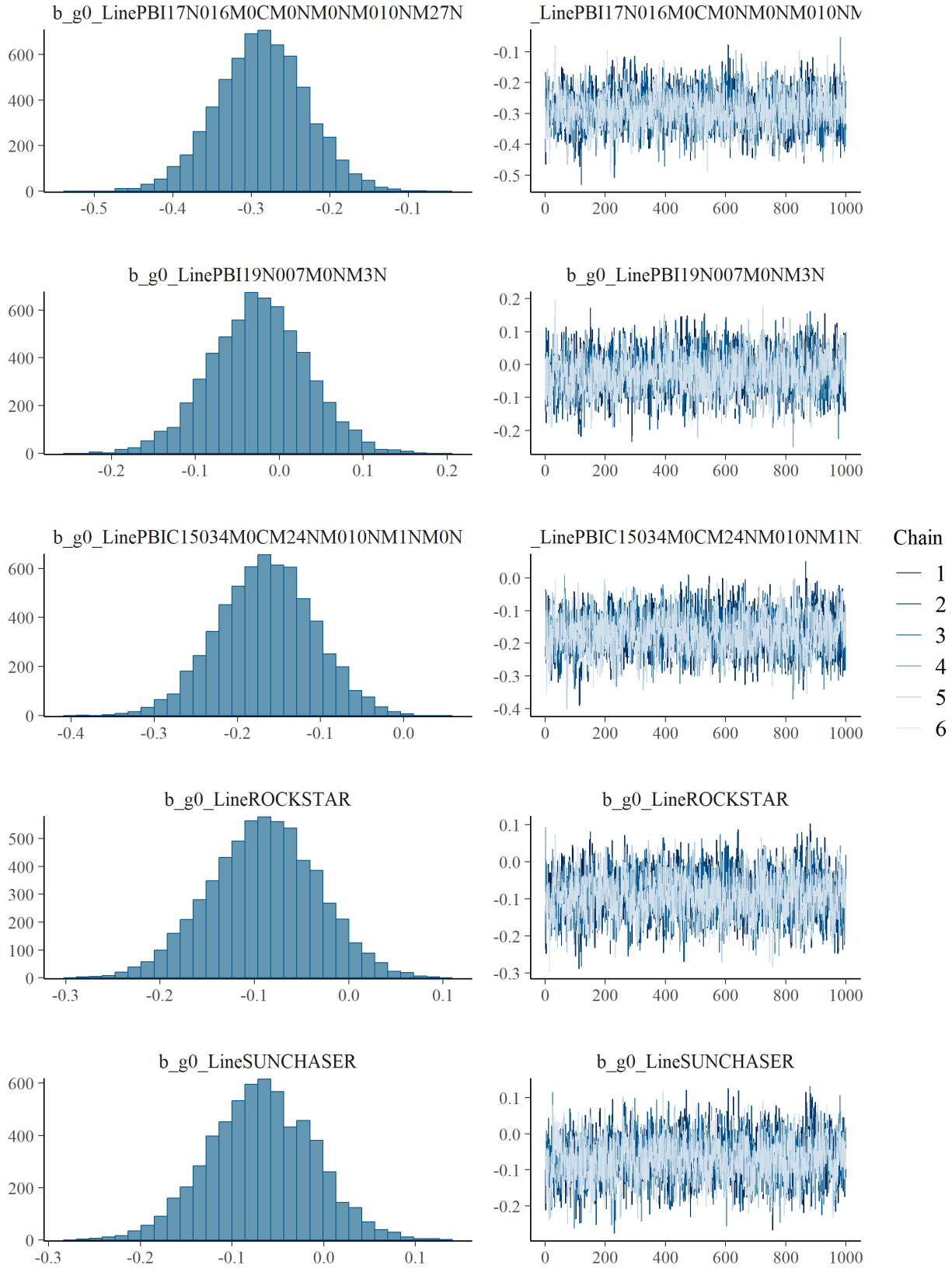


Figure 4: plot of chunk unnamed-chunk-4

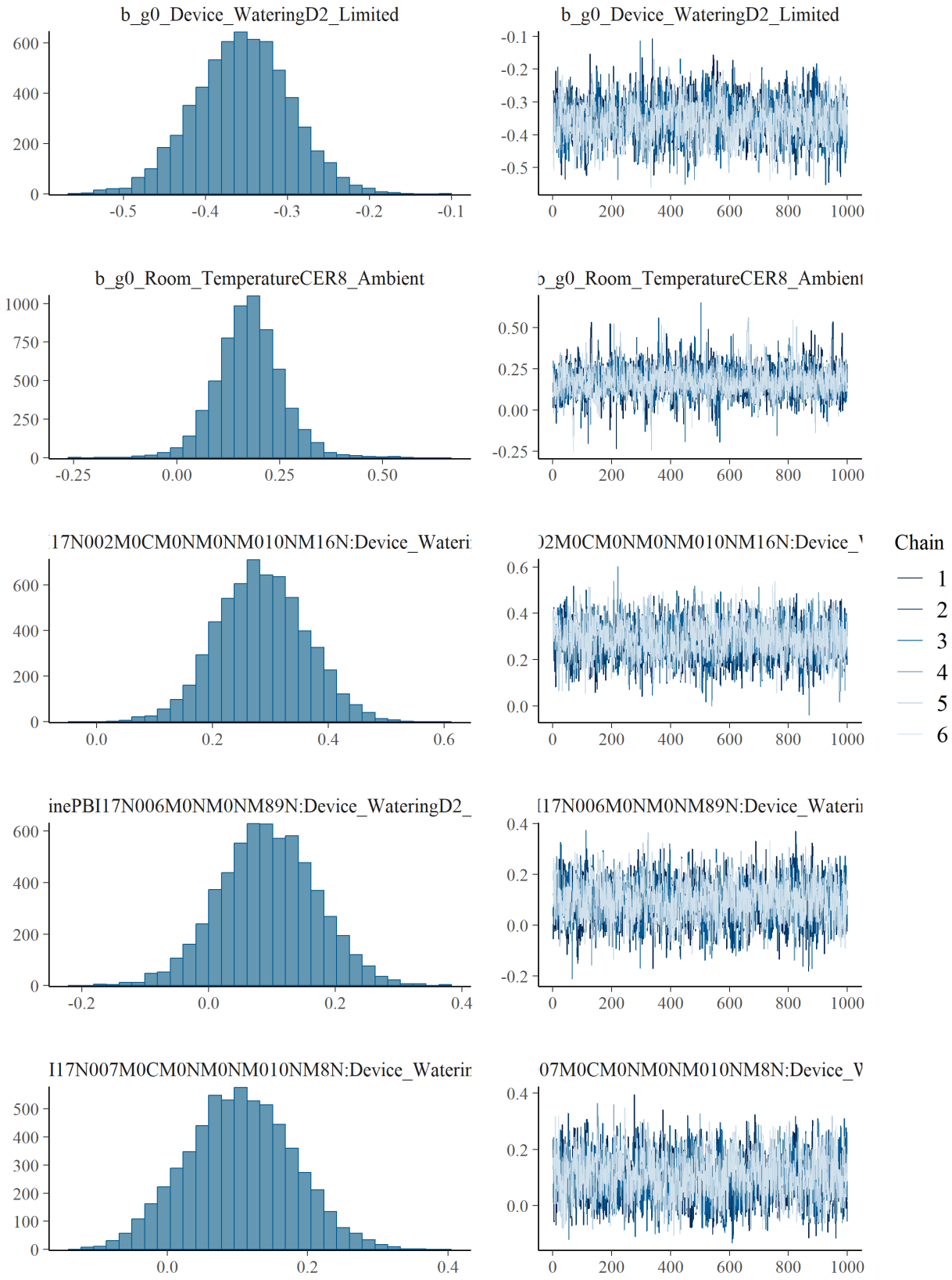


Figure 5: plot of chunk unnamed-chunk-4

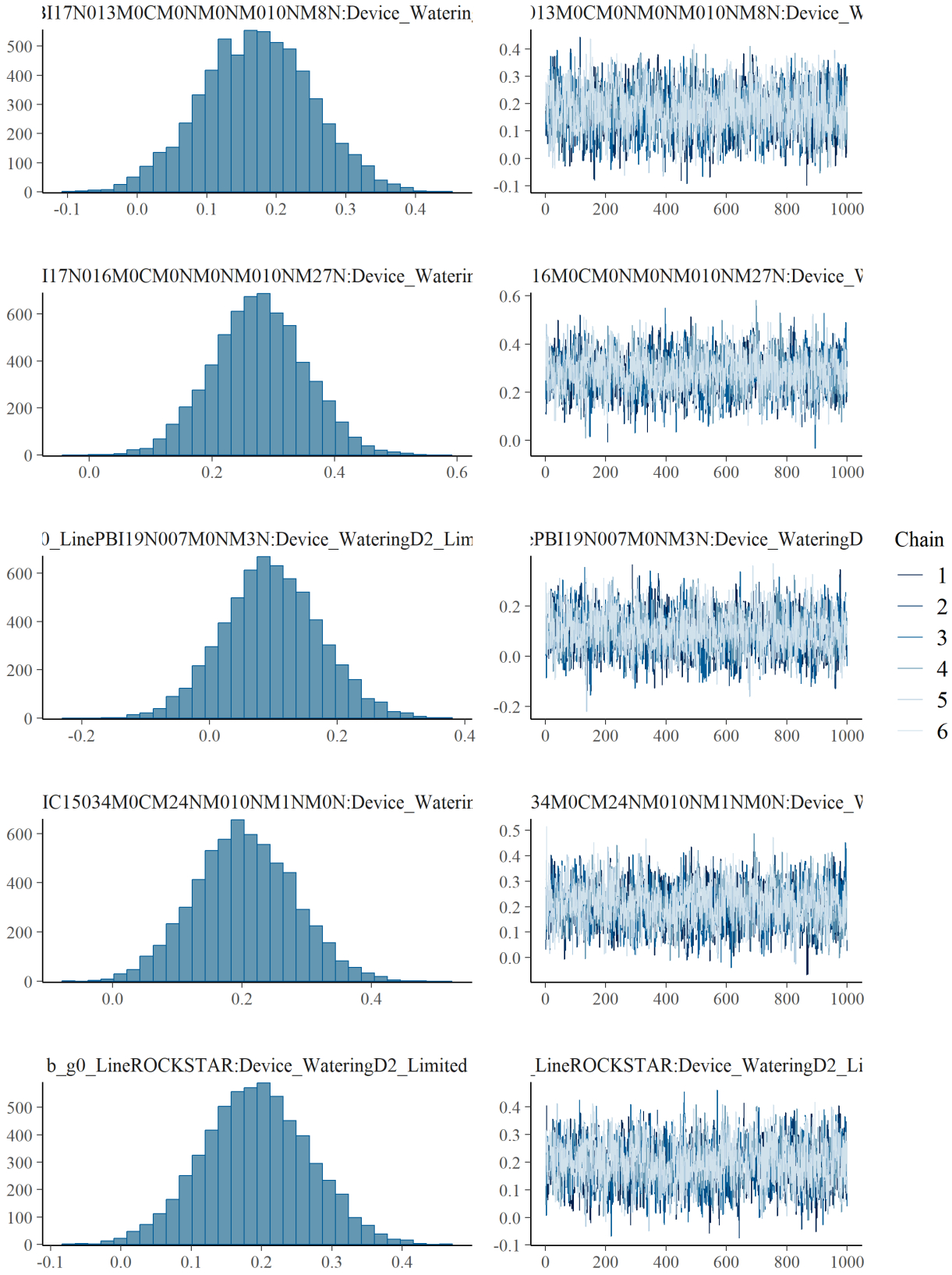


Figure 6: plot of chunk unnamed-chunk-4

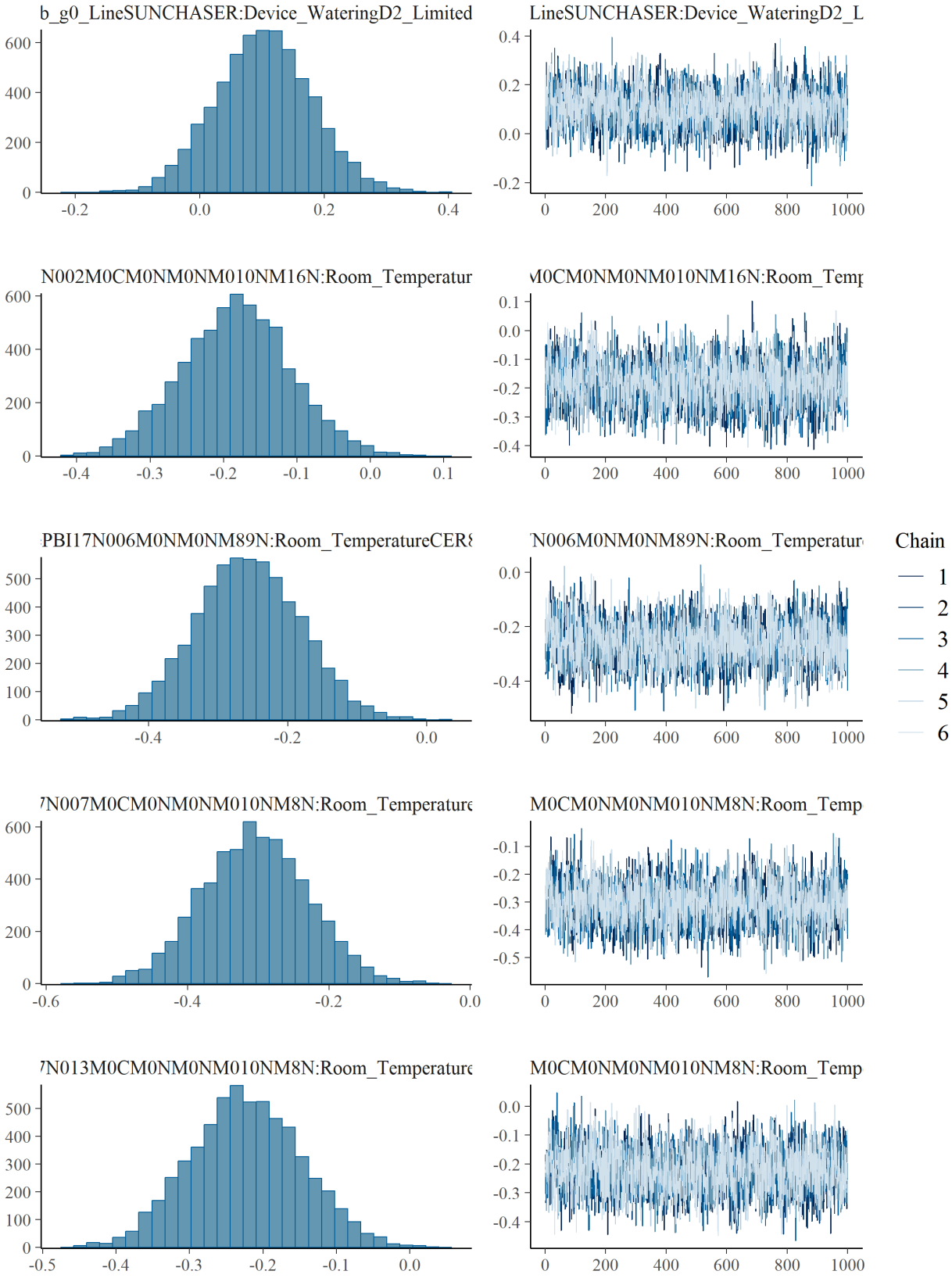


Figure 7: plot of chunk unnamed-chunk-4

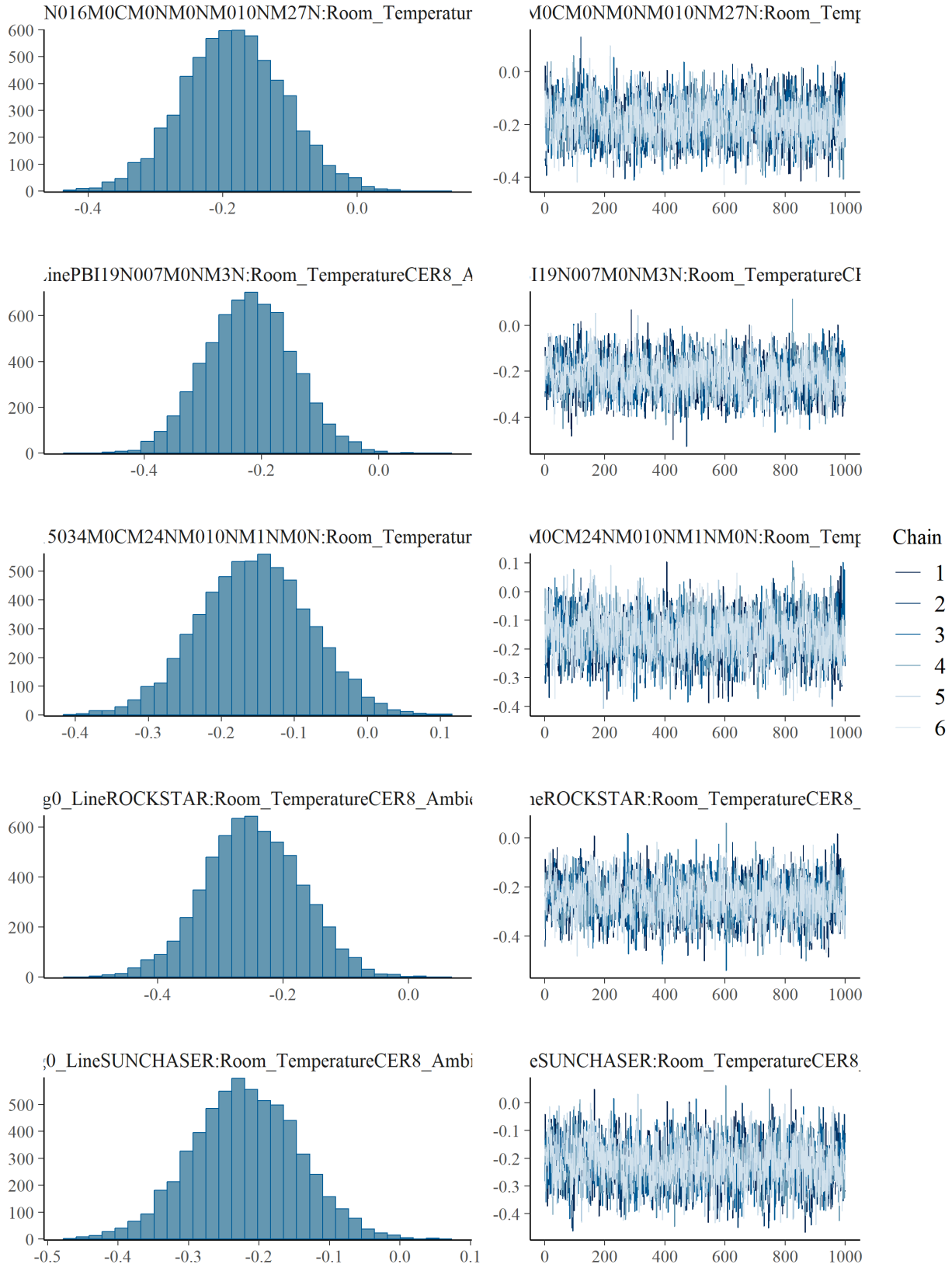


Figure 8: plot of chunk unnamed-chunk-4

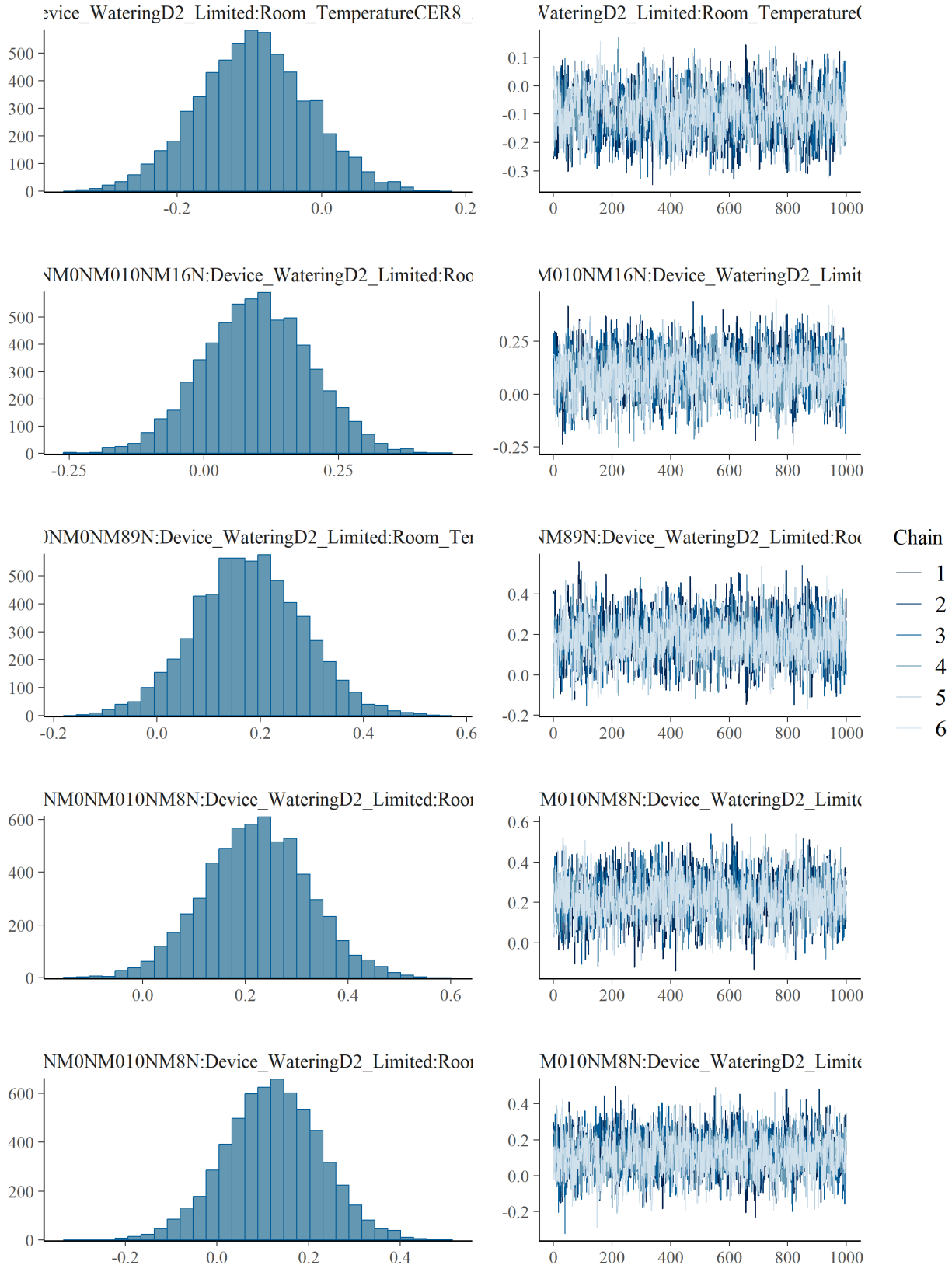


Figure 9: plot of chunk unnamed-chunk-4

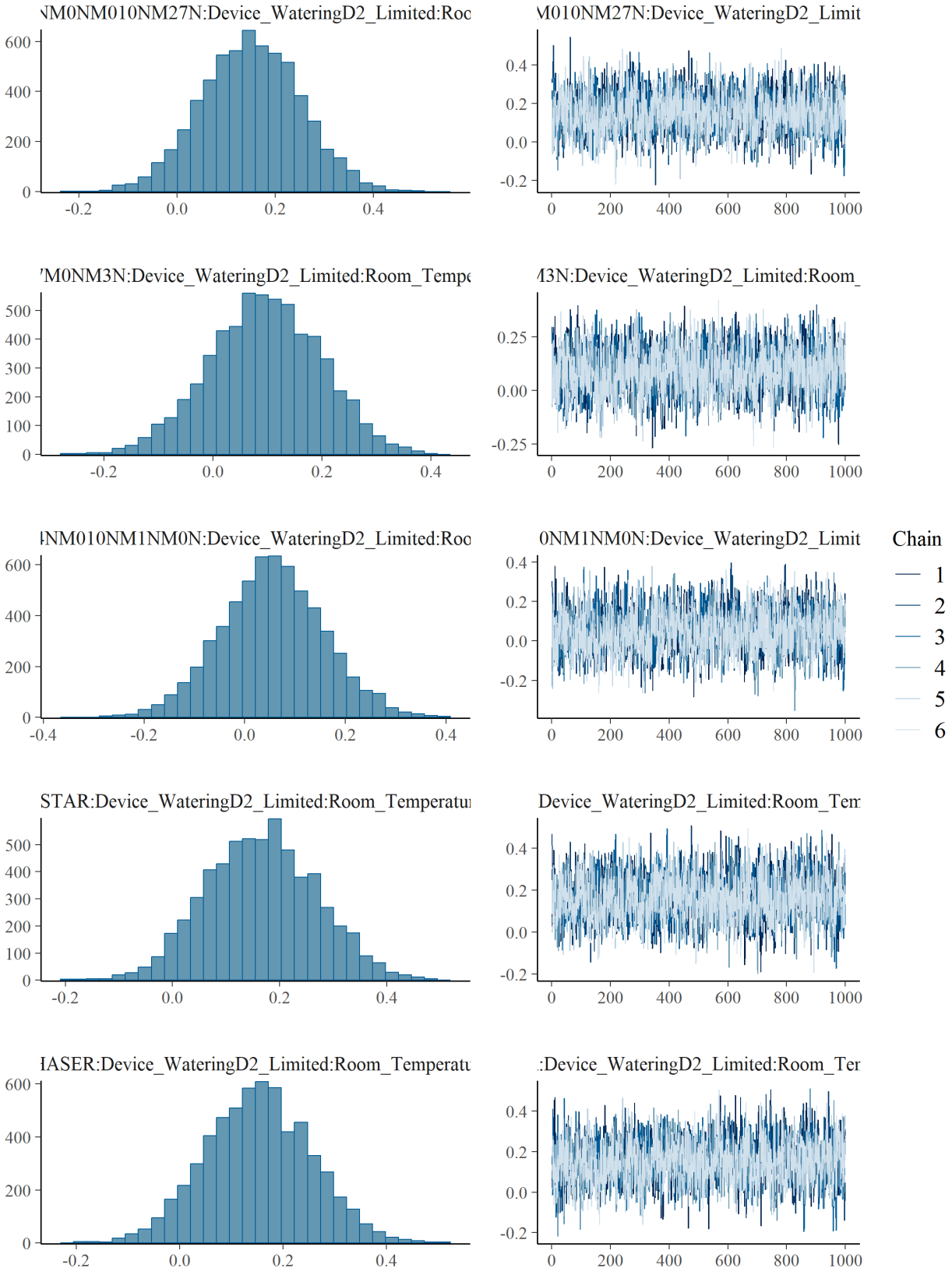


Figure 10: plot of chunk unnamed-chunk-4

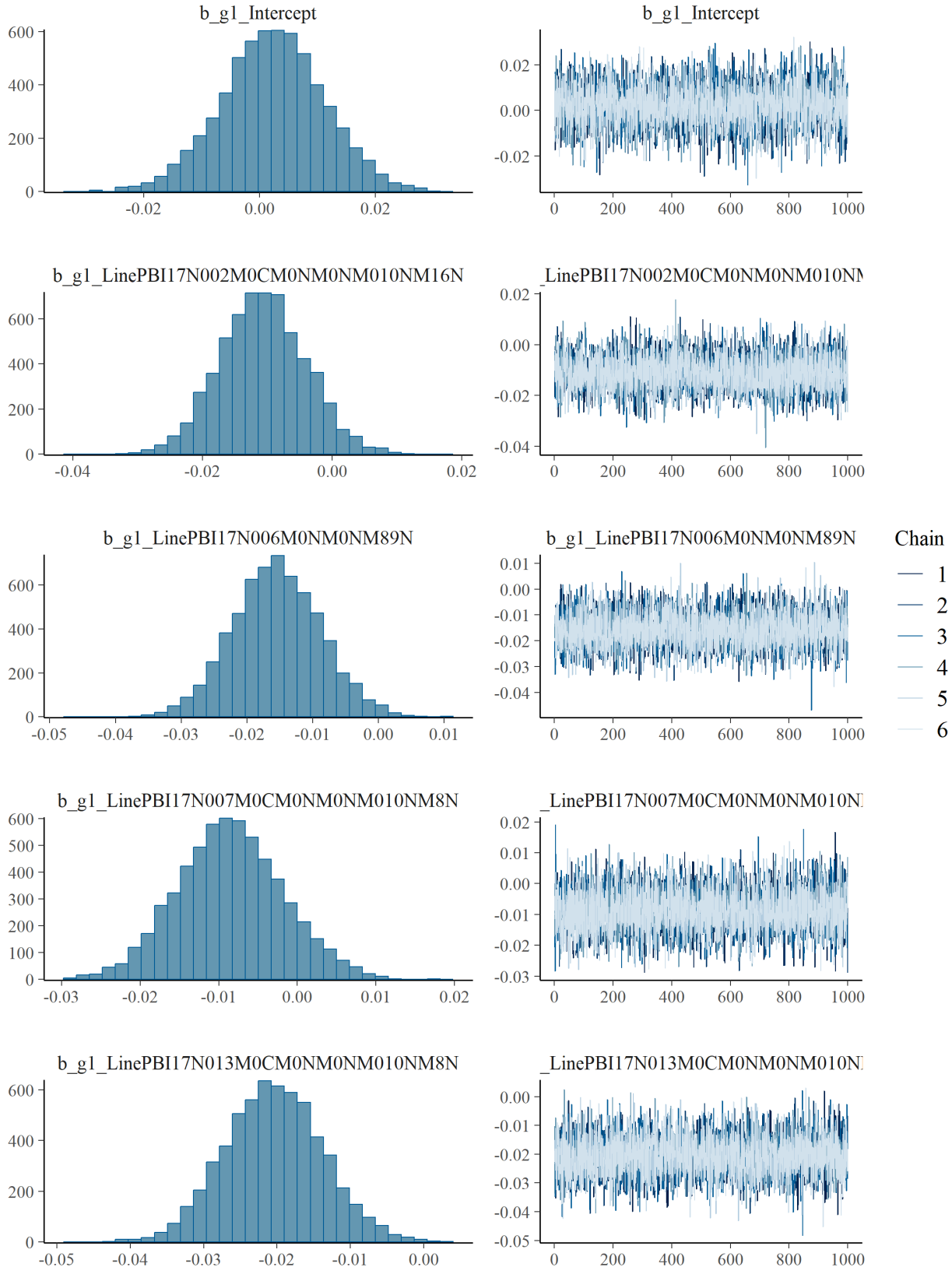


Figure 11: plot of chunk unnamed-chunk-4

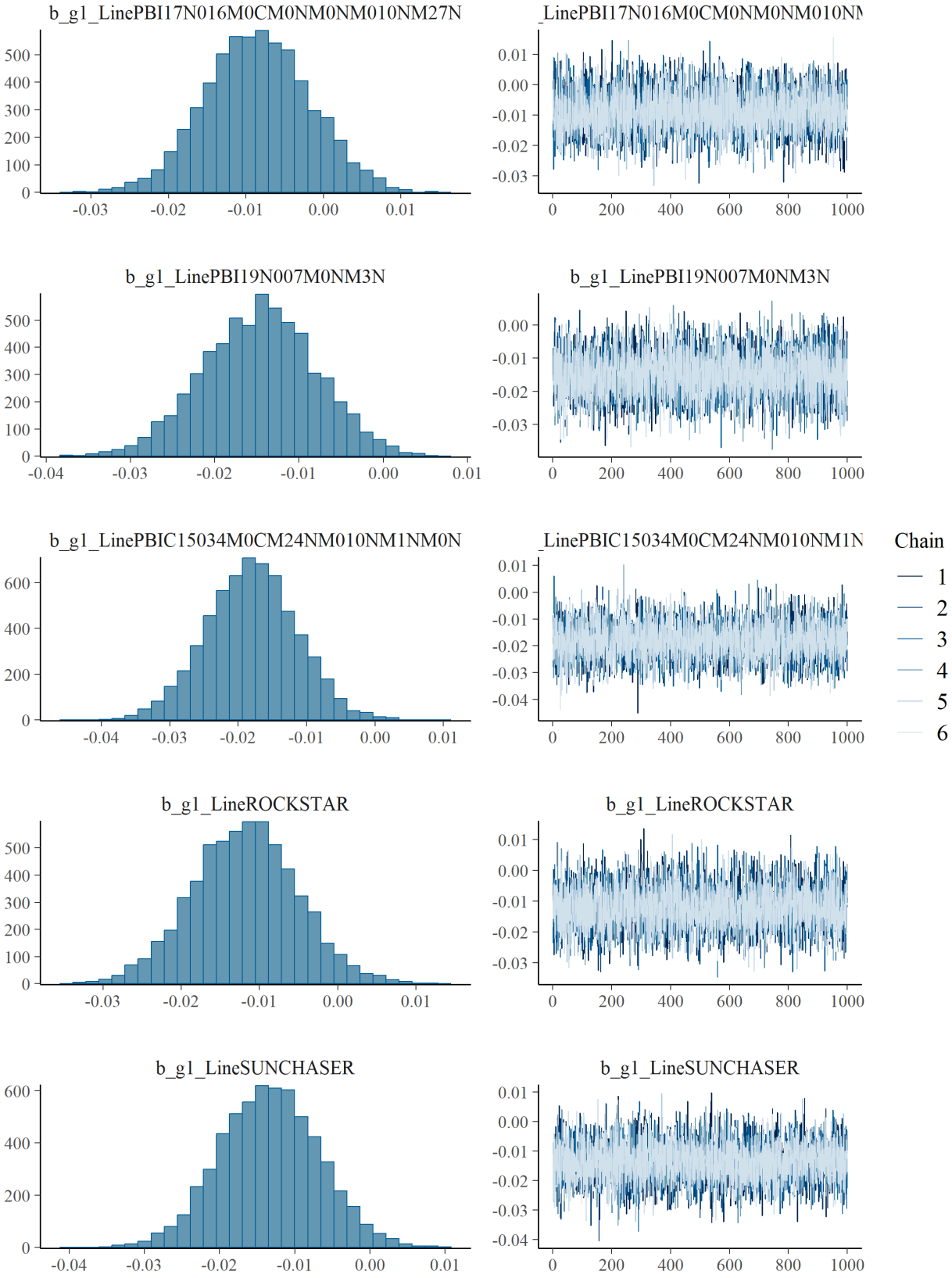


Figure 12: plot of chunk unnamed-chunk-4

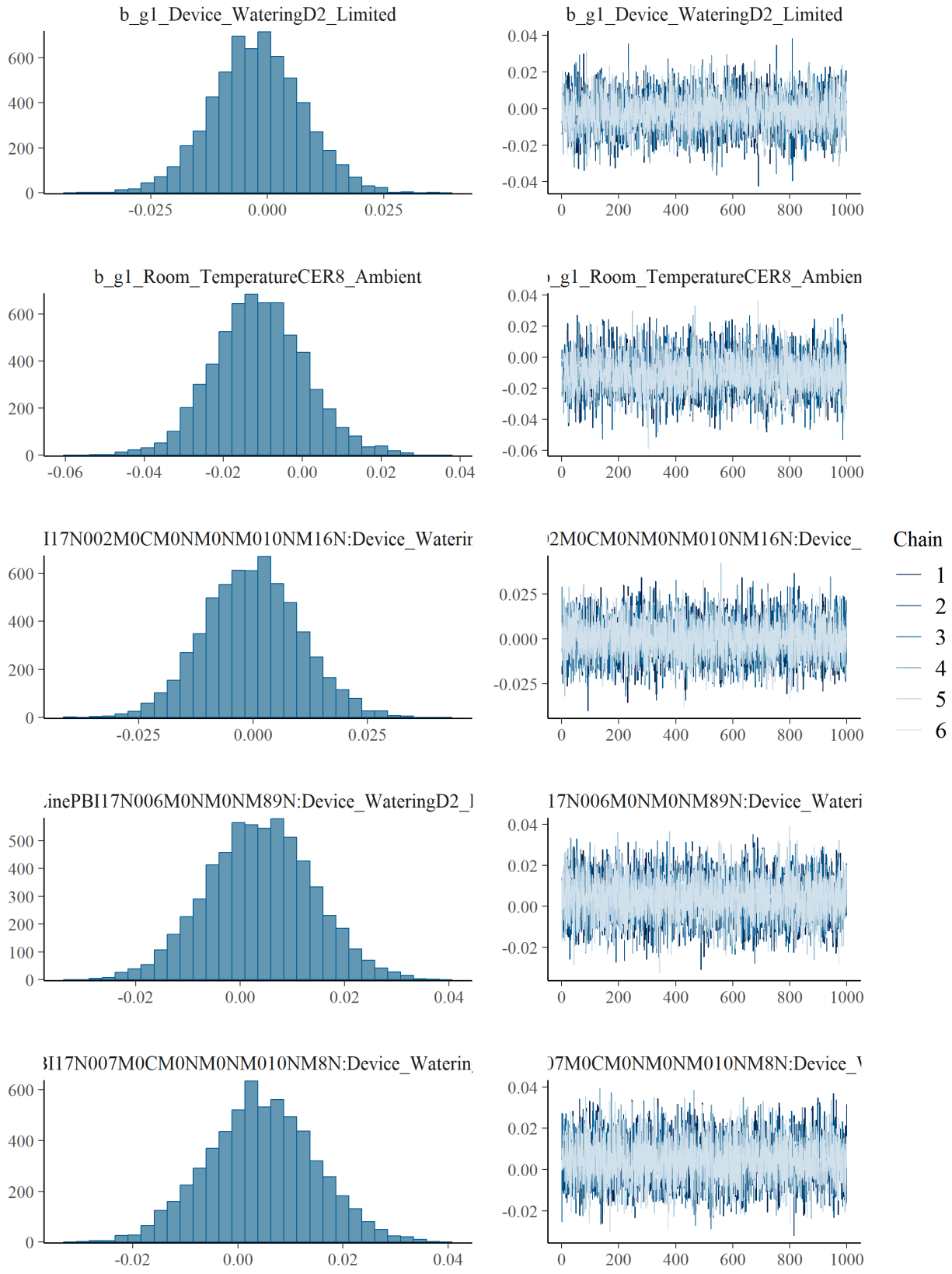


Figure 13: plot of chunk unnamed-chunk-4

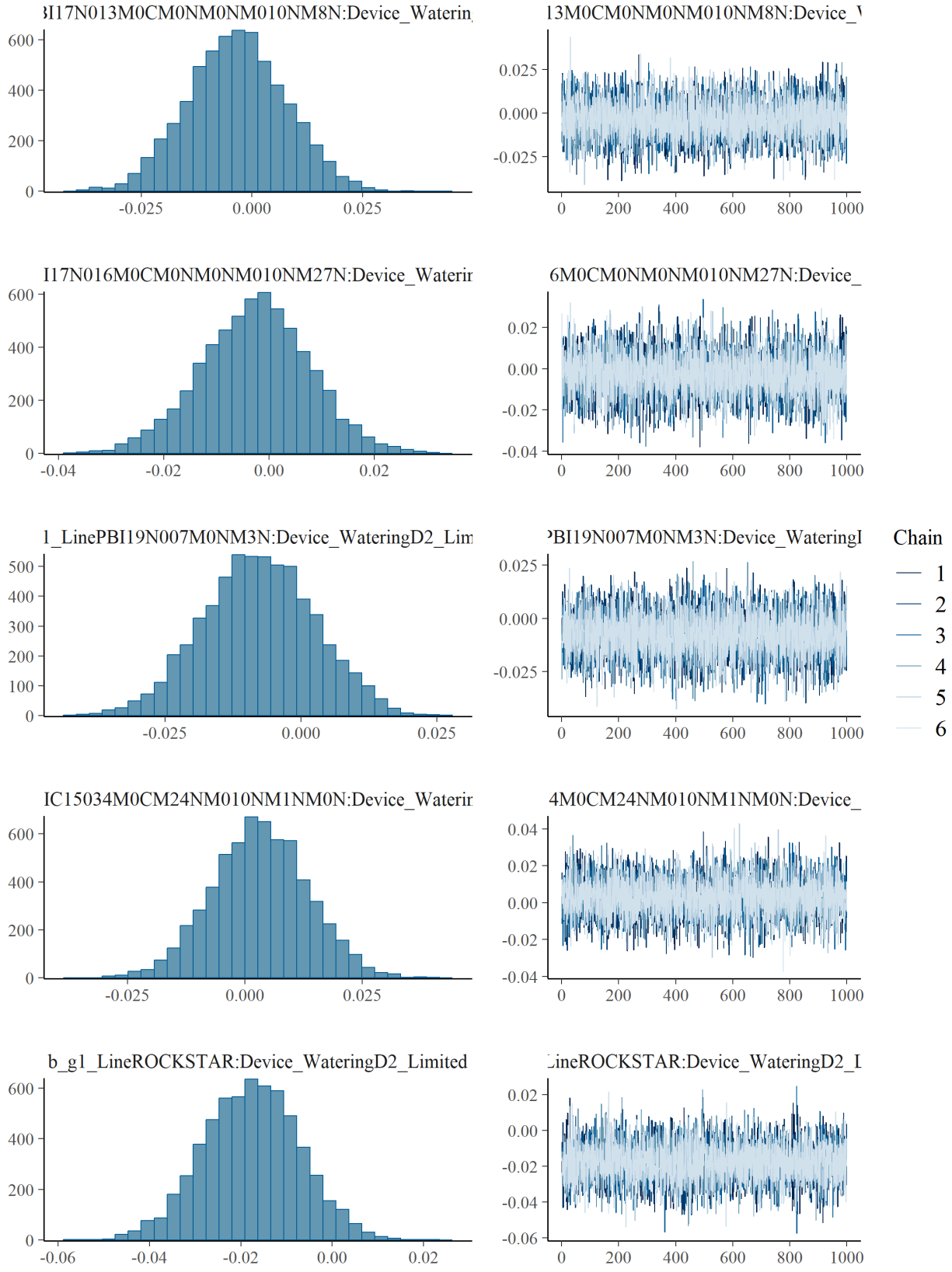


Figure 14: plot of chunk unnamed-chunk-4

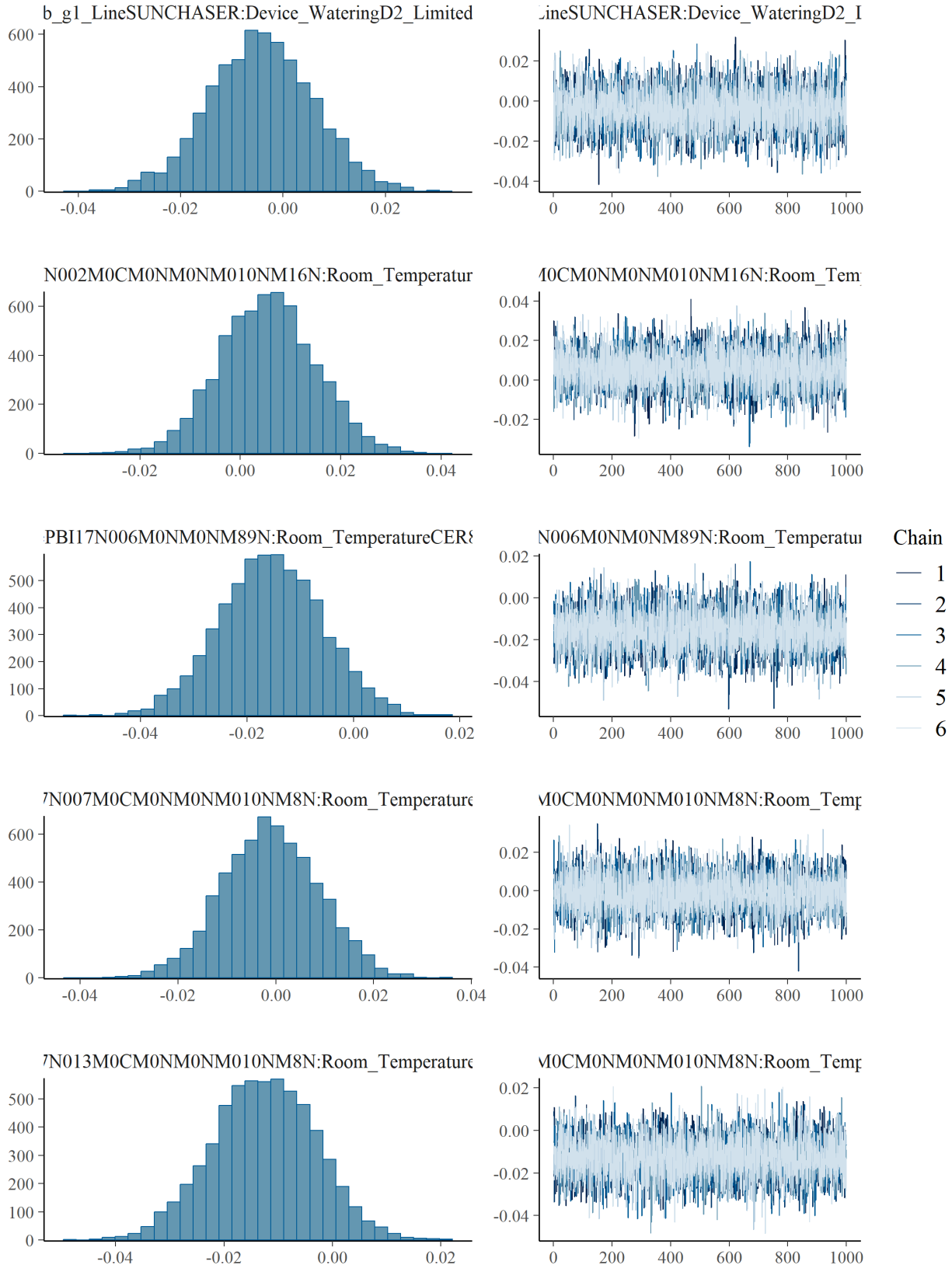


Figure 15: plot of chunk unnamed-chunk-4

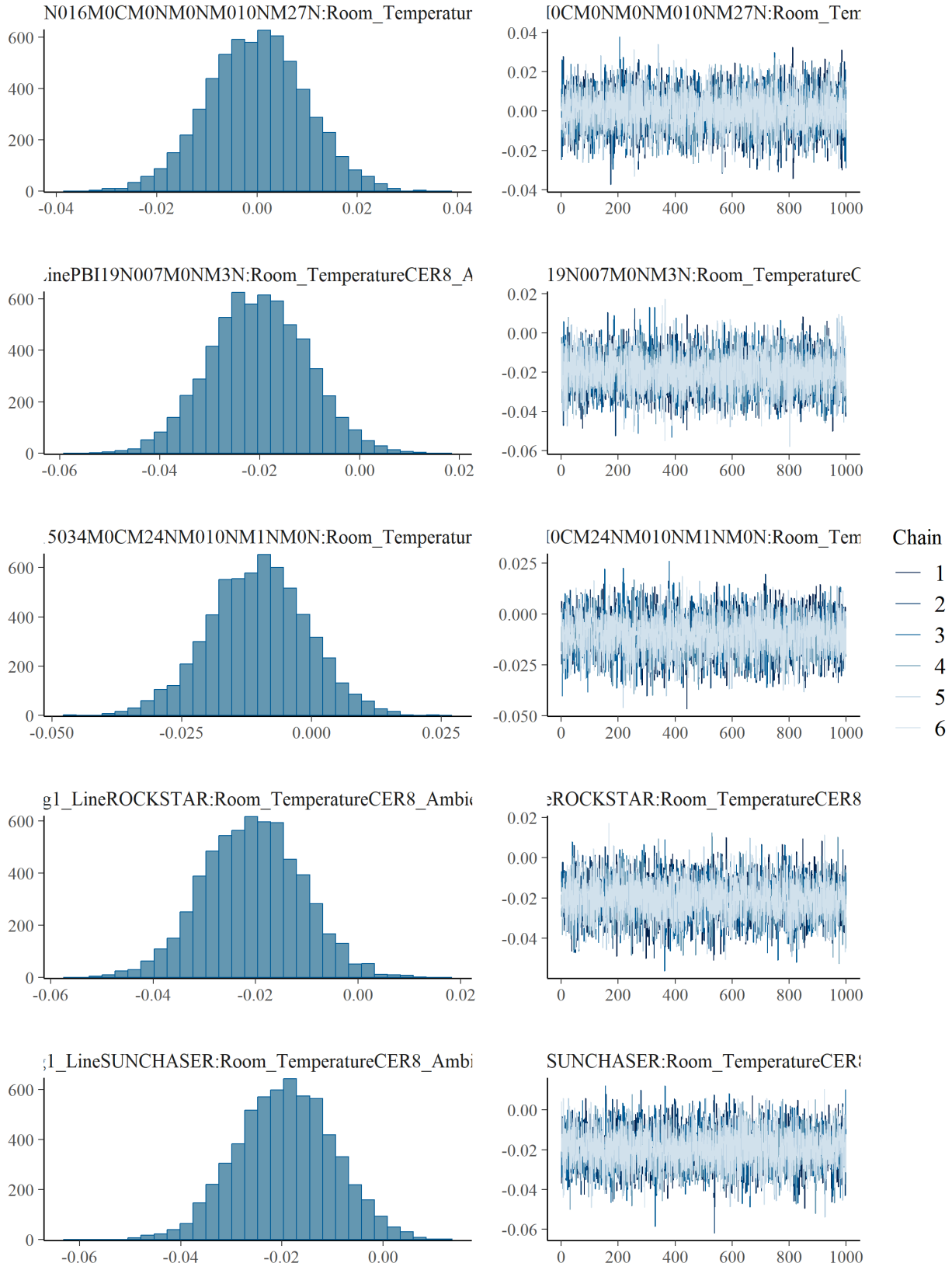


Figure 16: plot of chunk unnamed-chunk-4

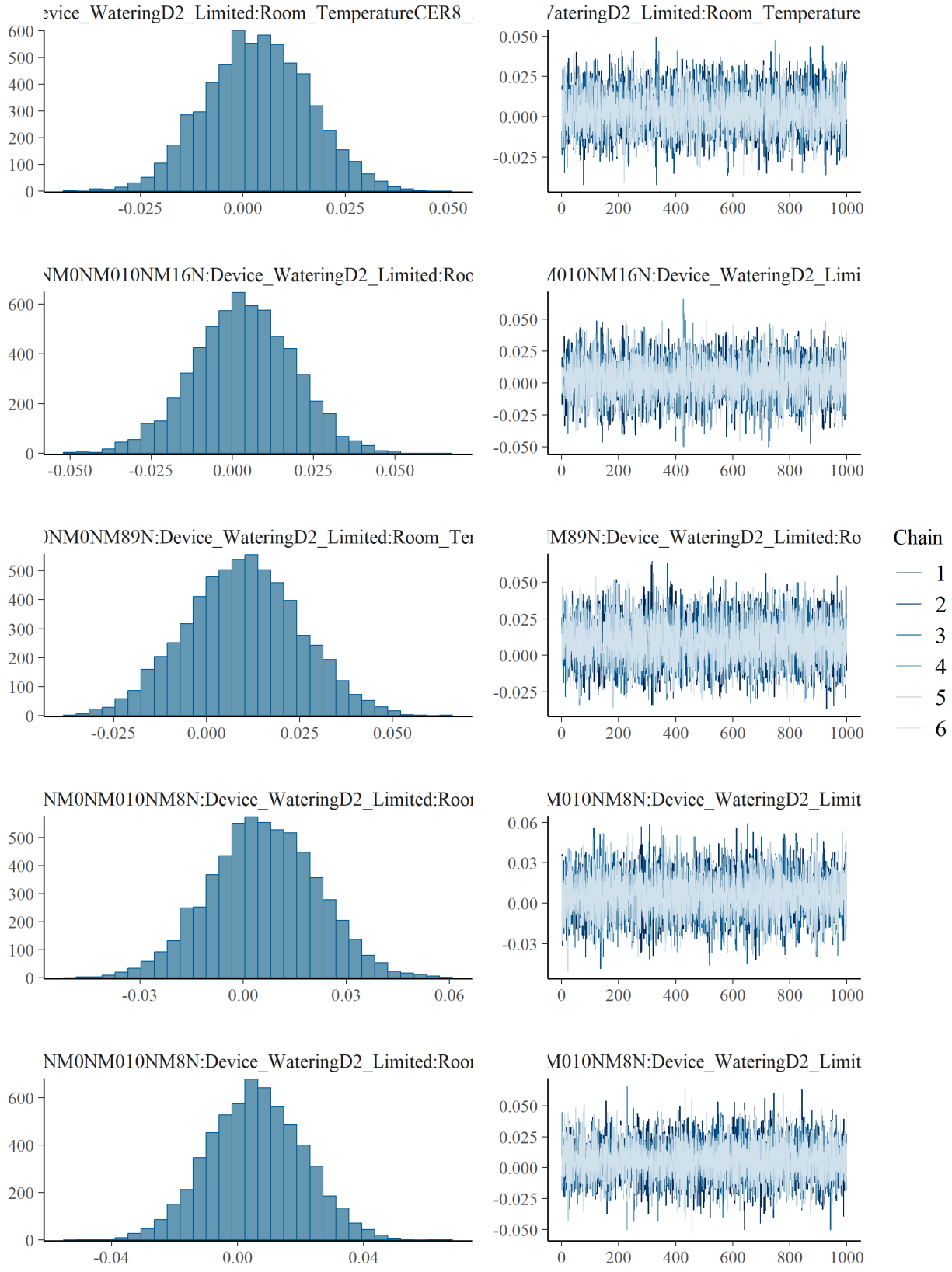


Figure 17: plot of chunk unnamed-chunk-4

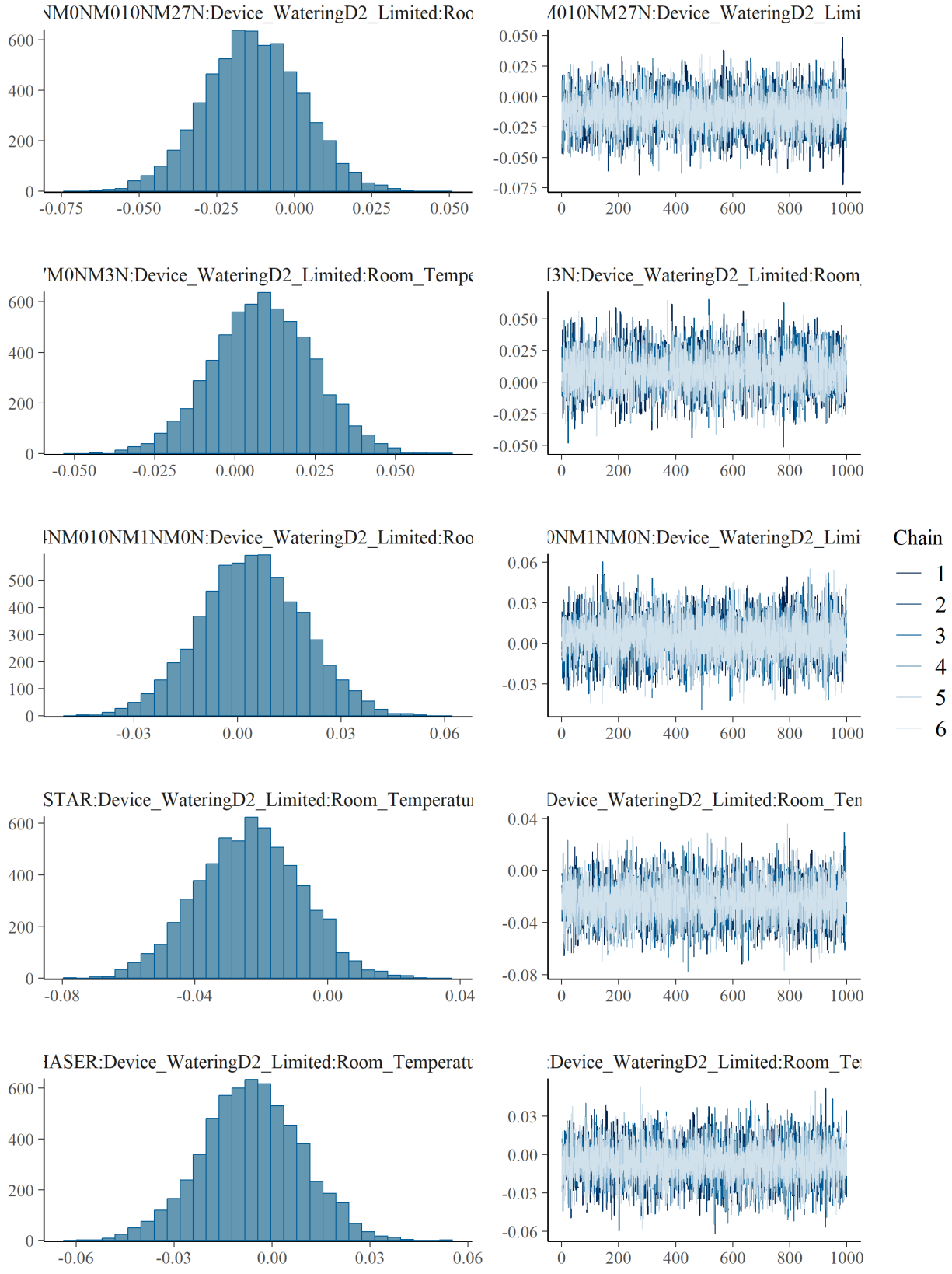


Figure 18: plot of chunk unnamed-chunk-4

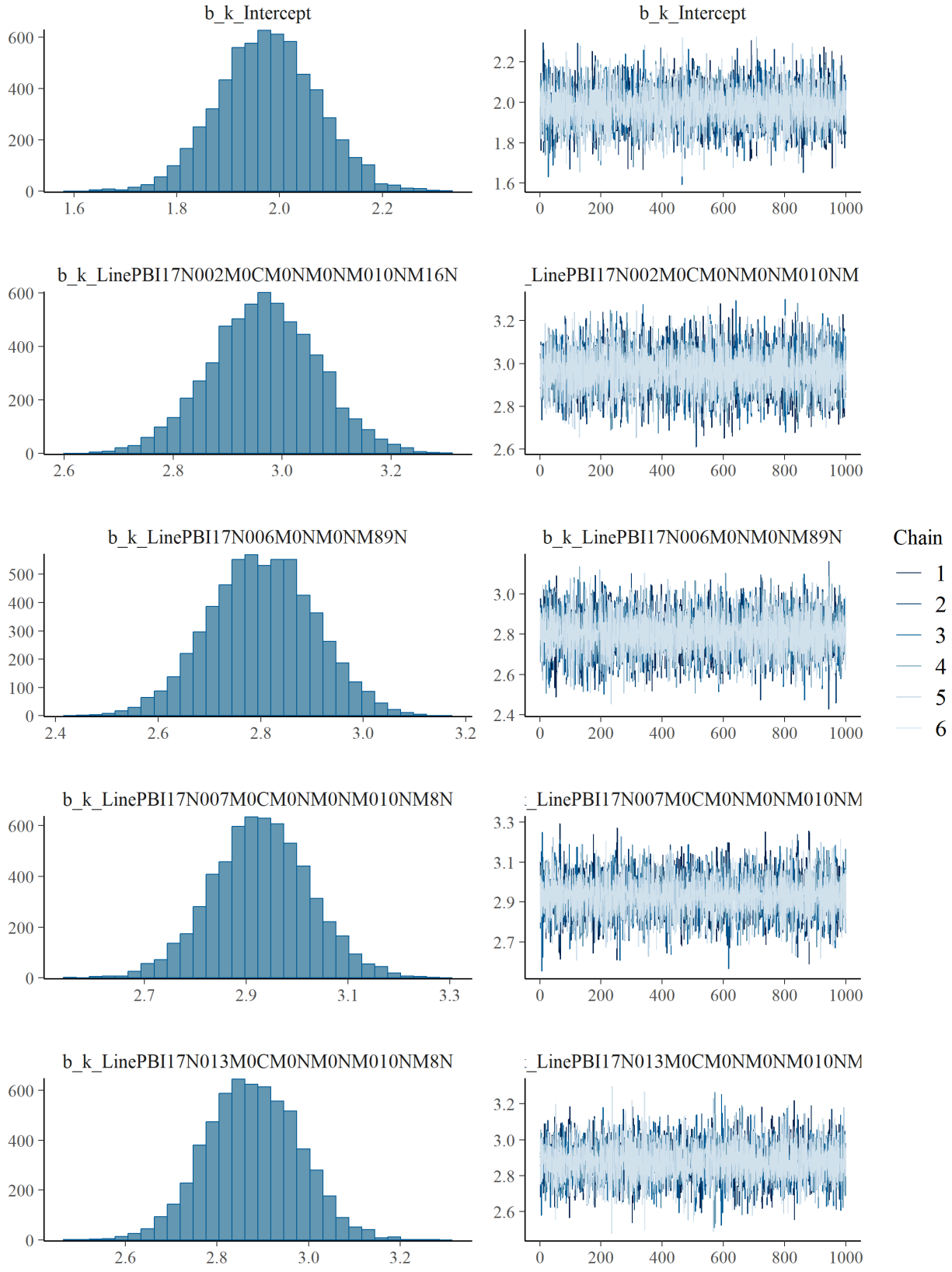


Figure 19: plot of chunk unnamed-chunk-4

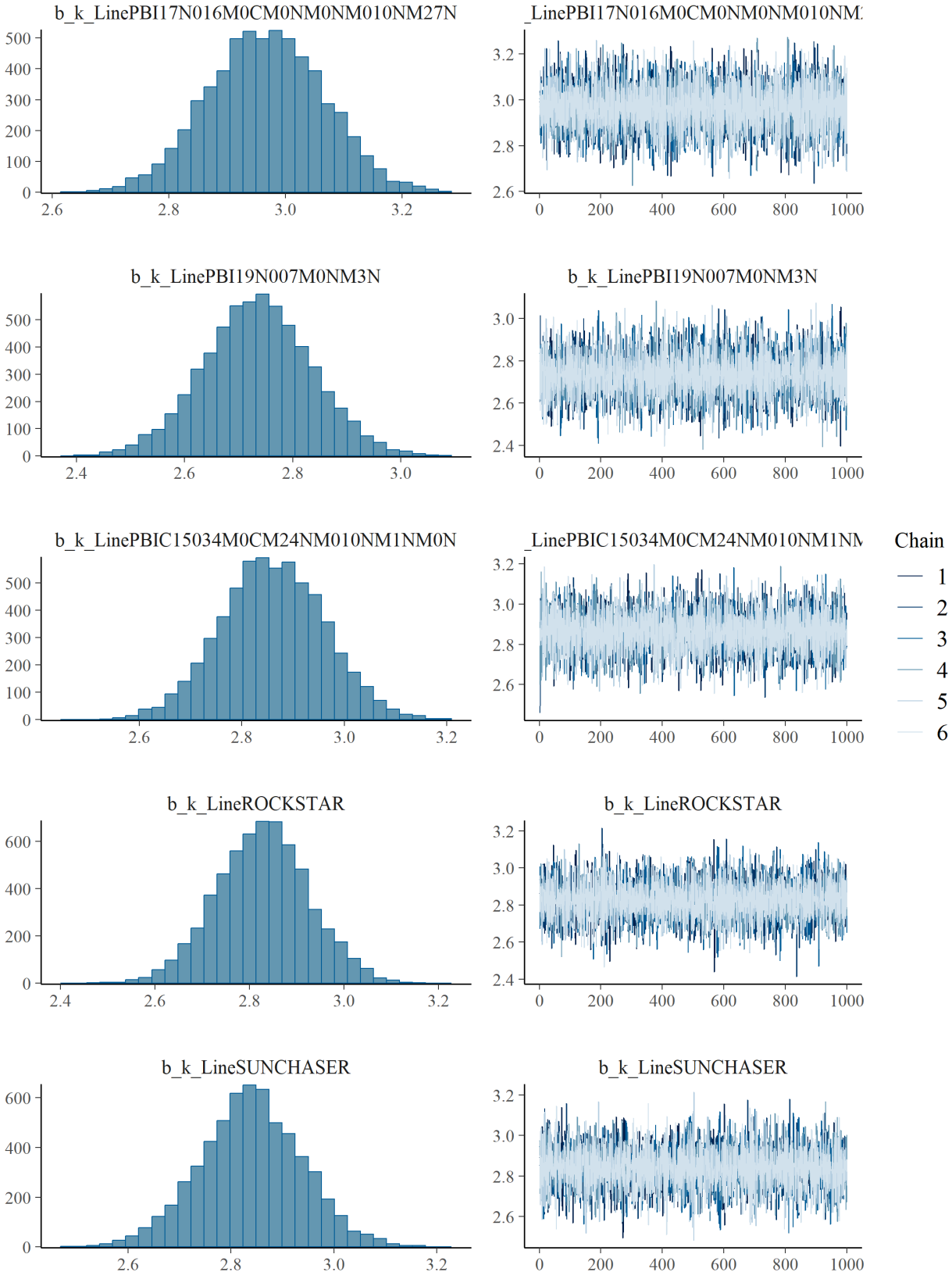


Figure 20: plot of chunk unnamed-chunk-4

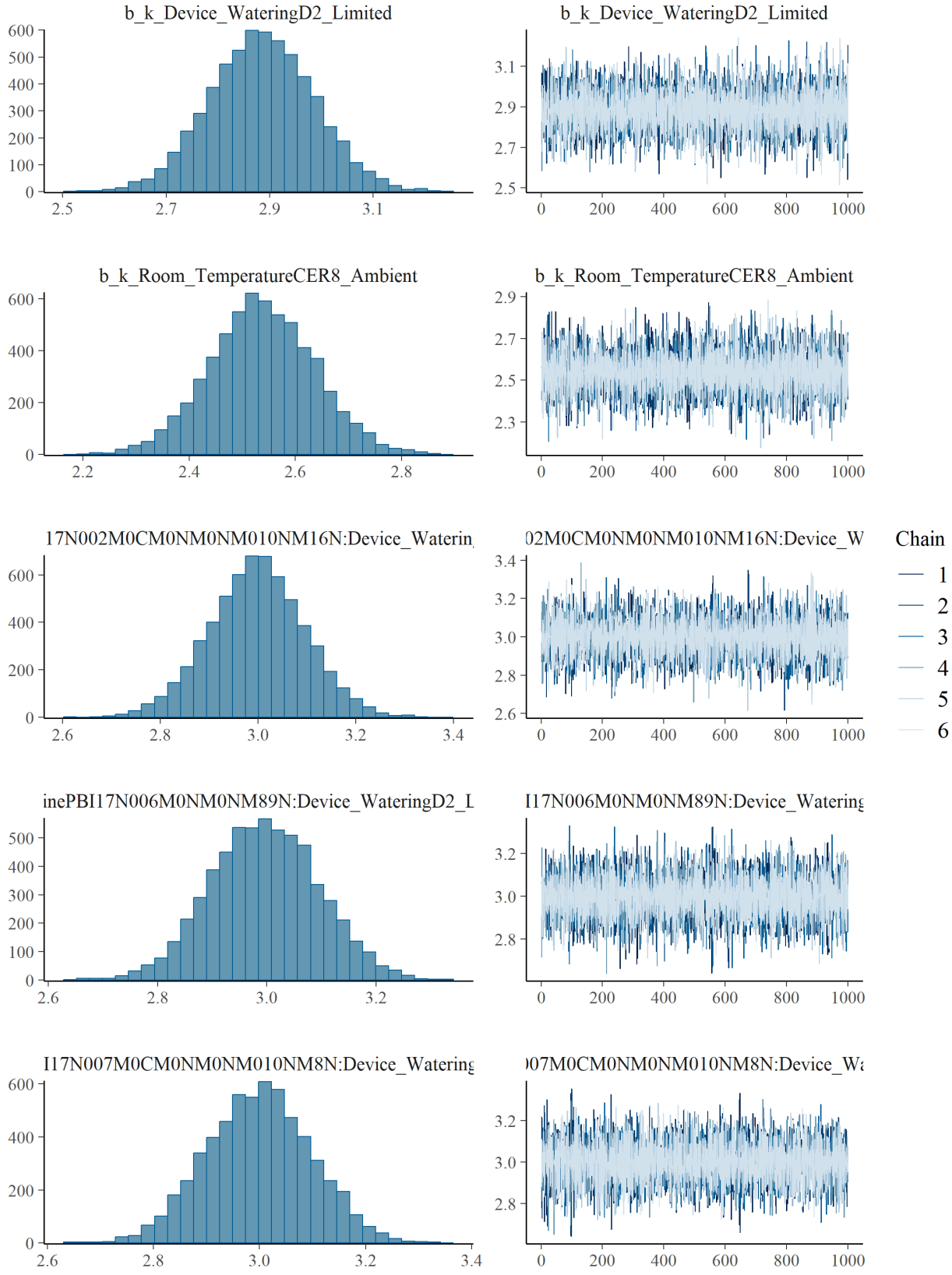


Figure 21: plot of chunk unnamed-chunk-4

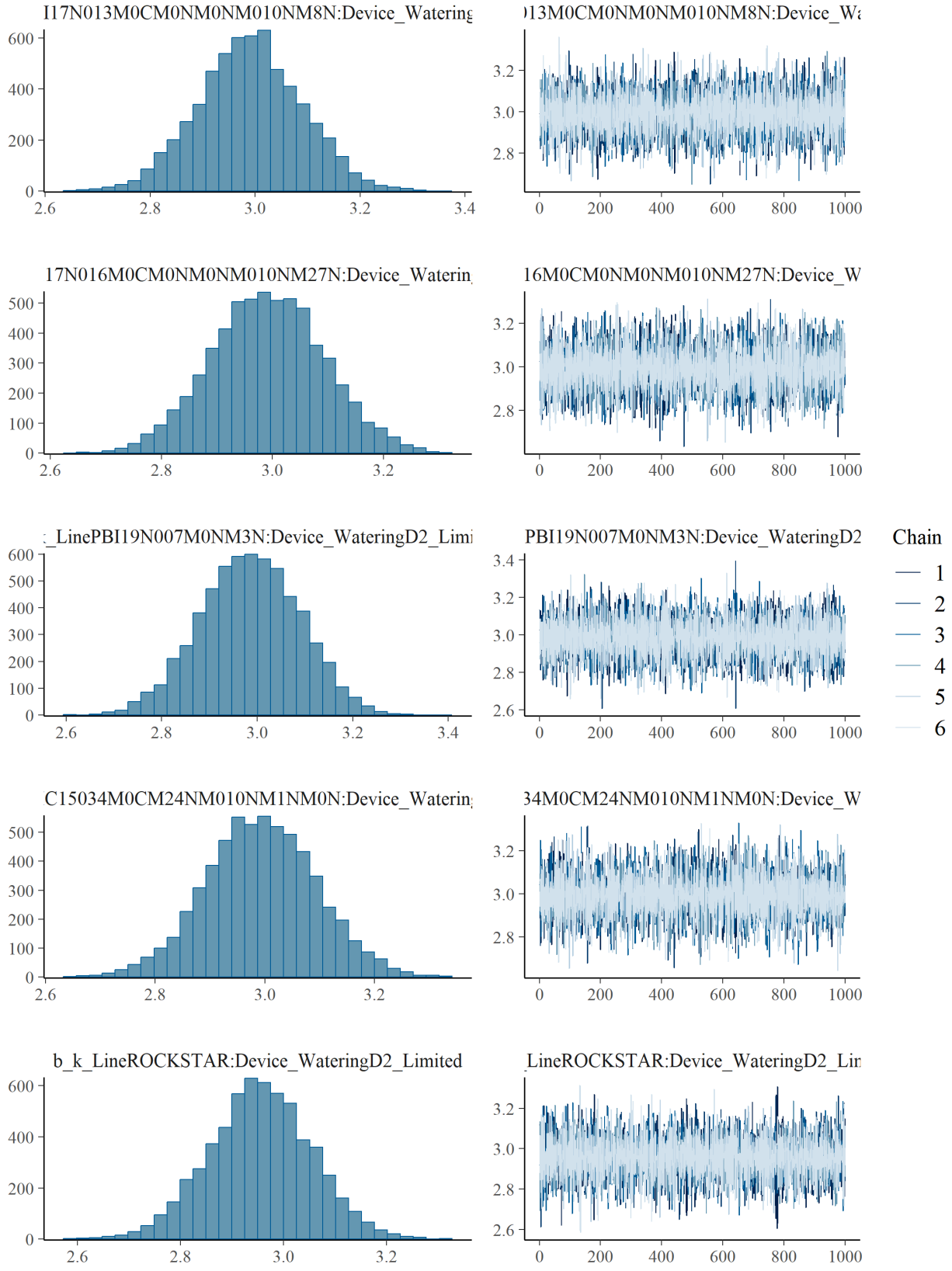


Figure 22: plot of chunk unnamed-chunk-4

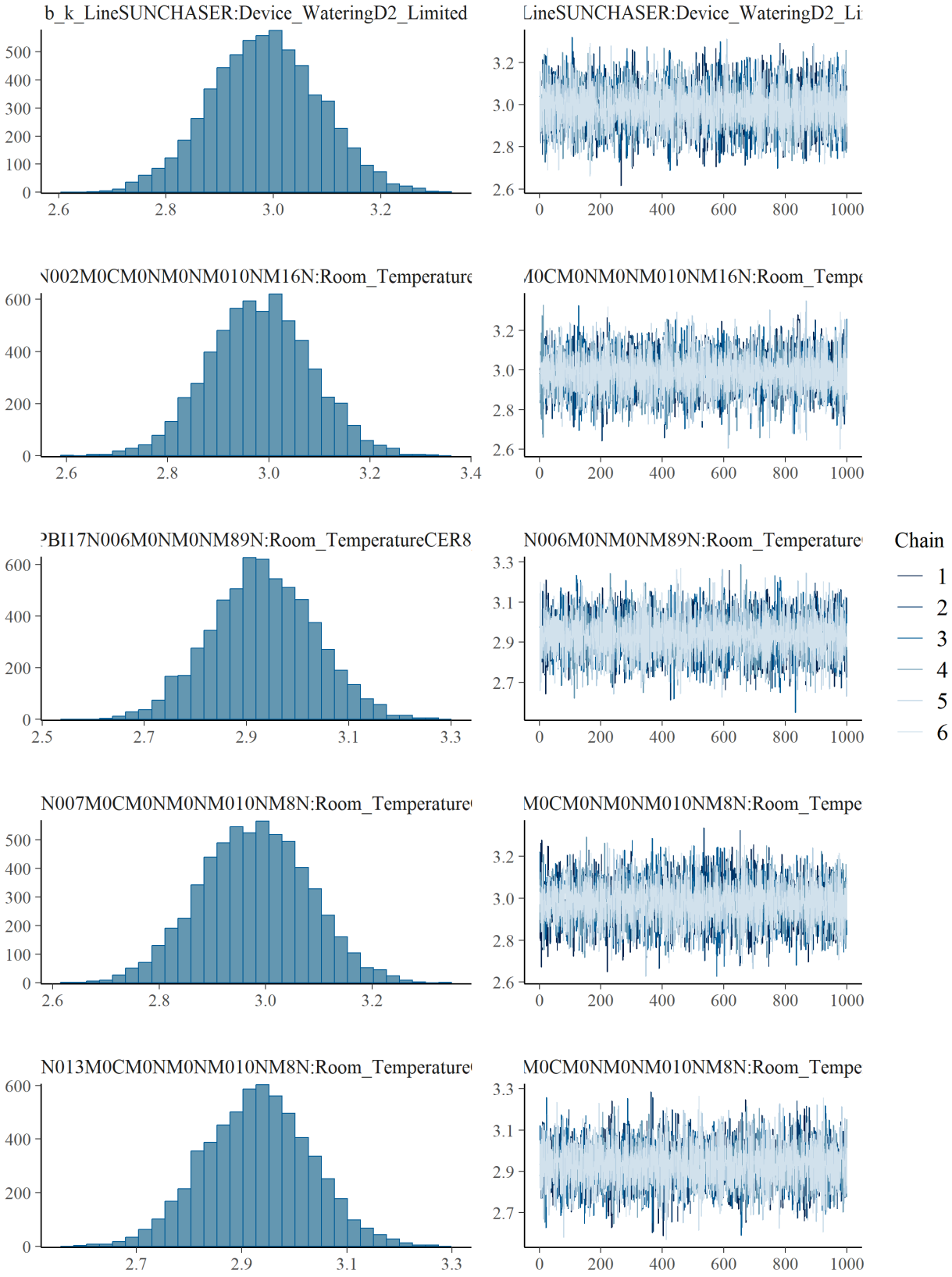


Figure 23: plot of chunk unnamed-chunk-4

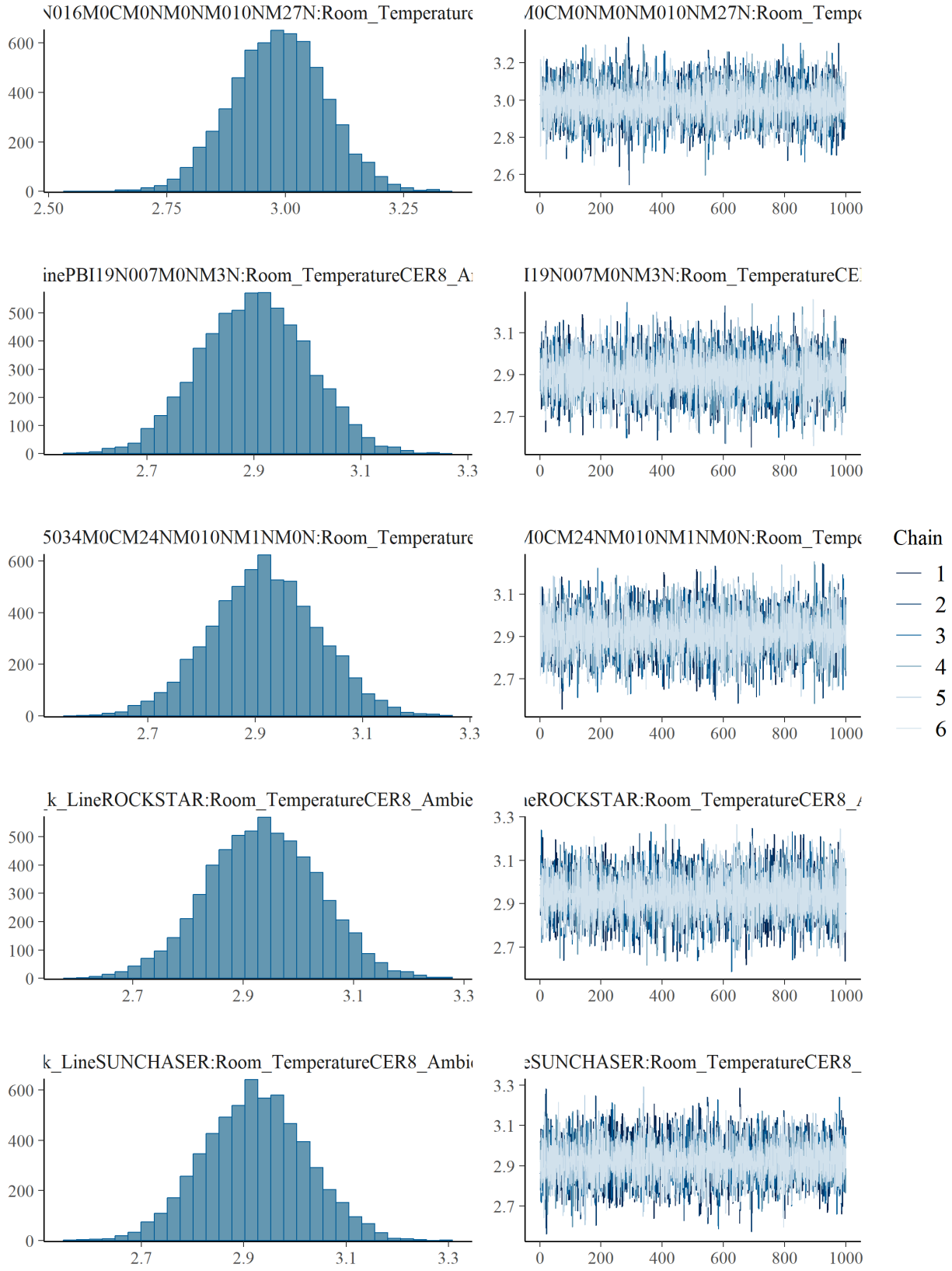


Figure 24: plot of chunk unnamed-chunk-4

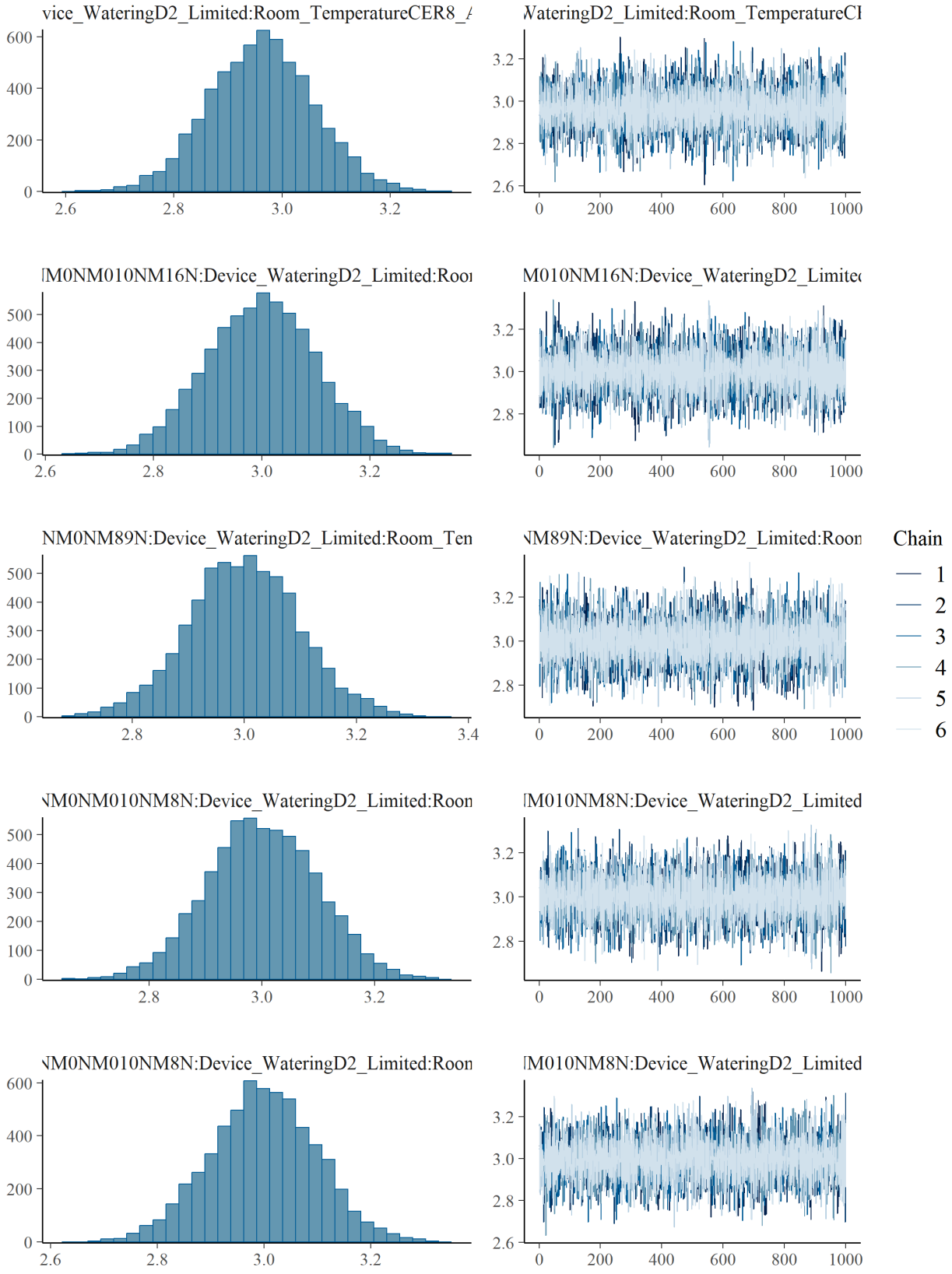


Figure 25: plot of chunk unnamed-chunk-4

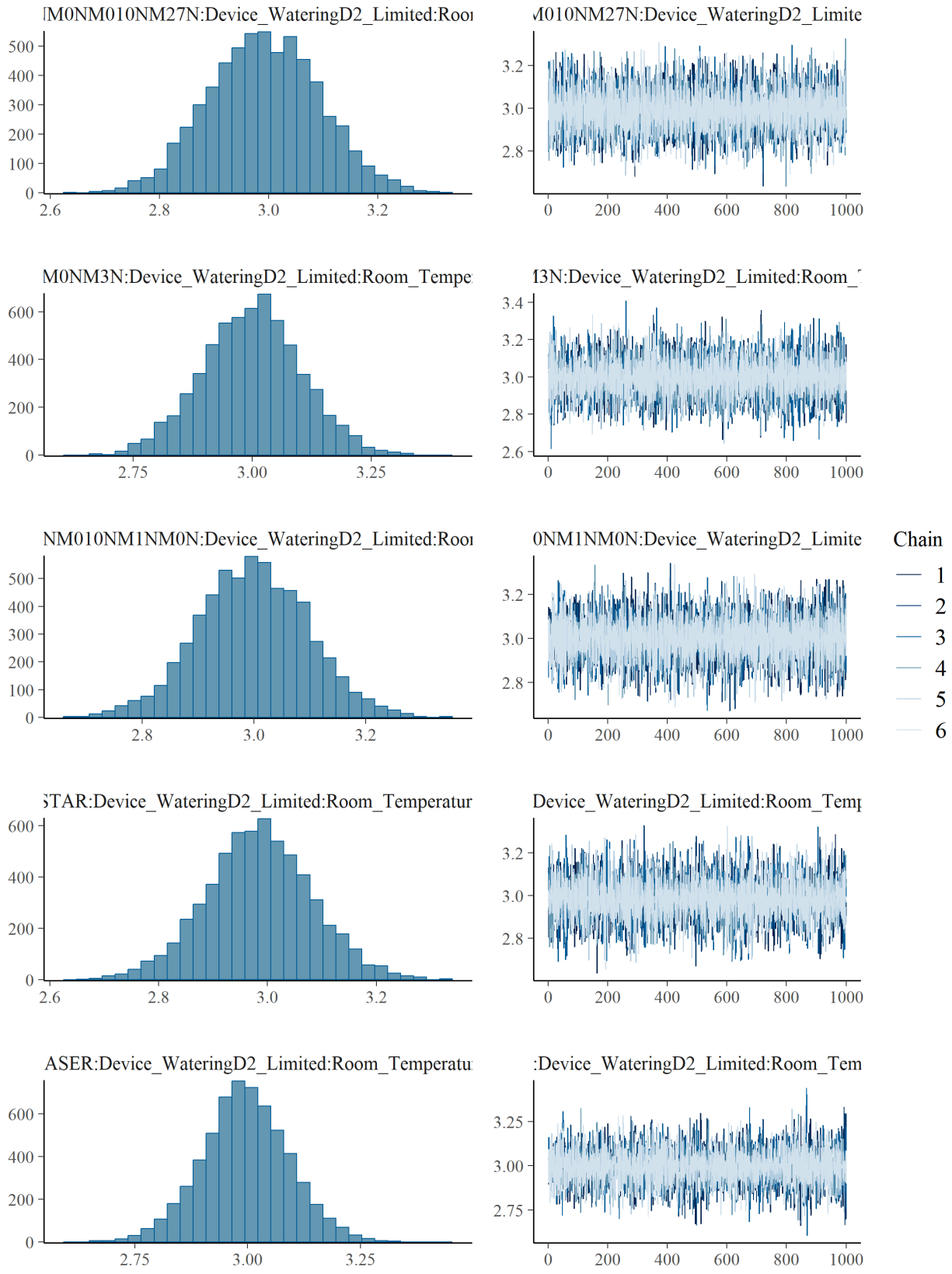


Figure 26: plot of chunk unnamed-chunk-4

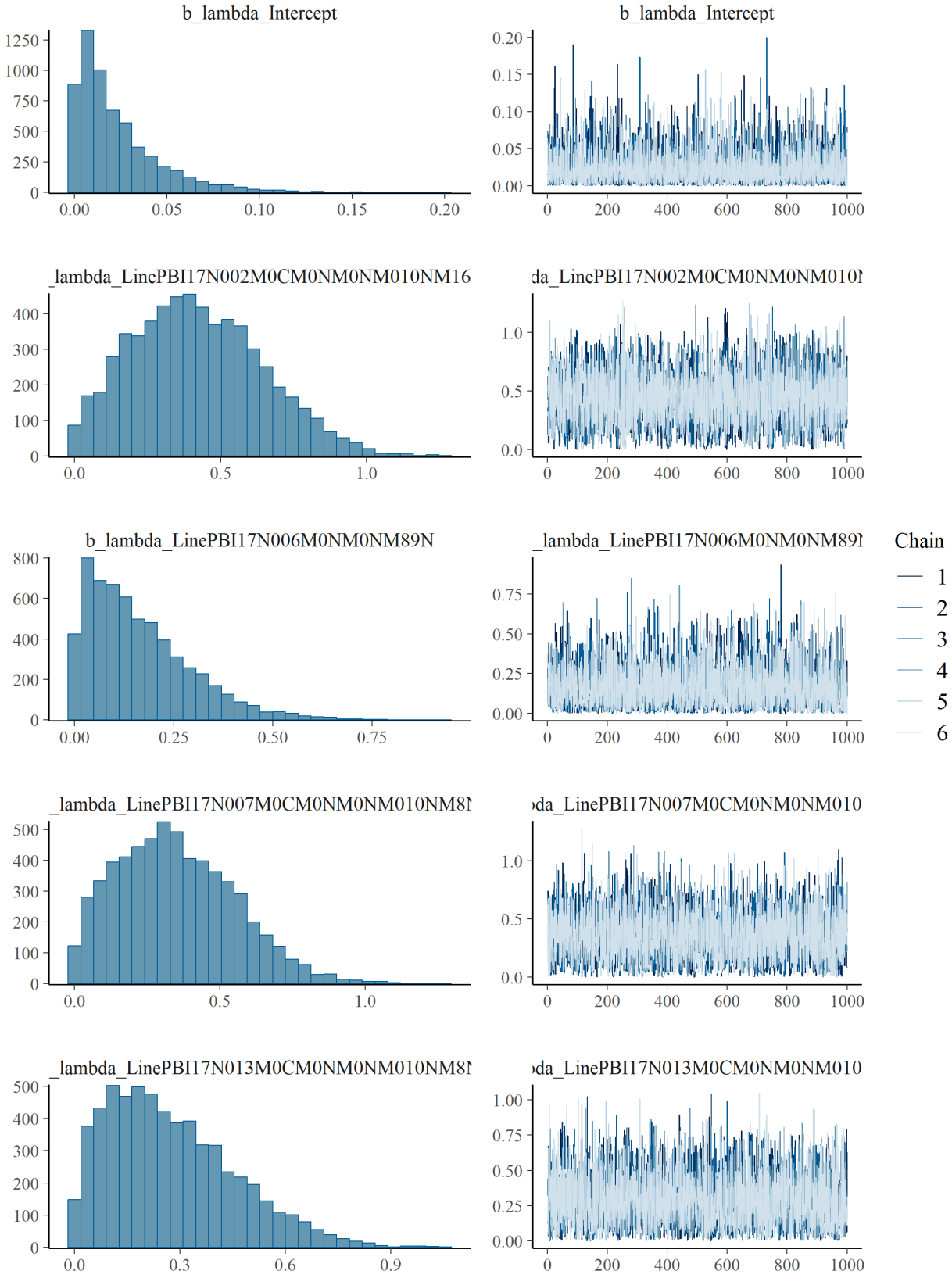


Figure 27: plot of chunk unnamed-chunk-4

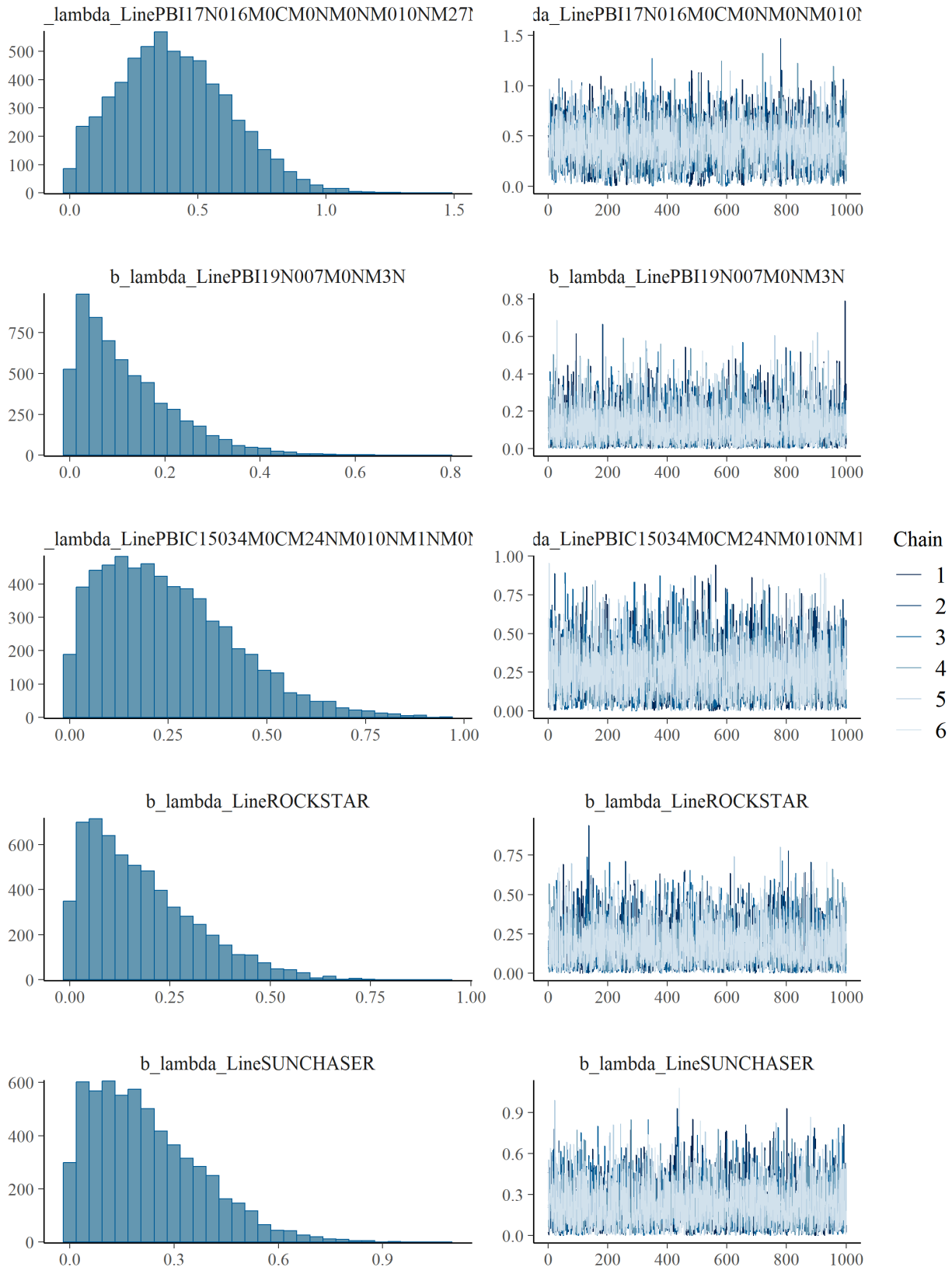


Figure 28: plot of chunk unnamed-chunk-4

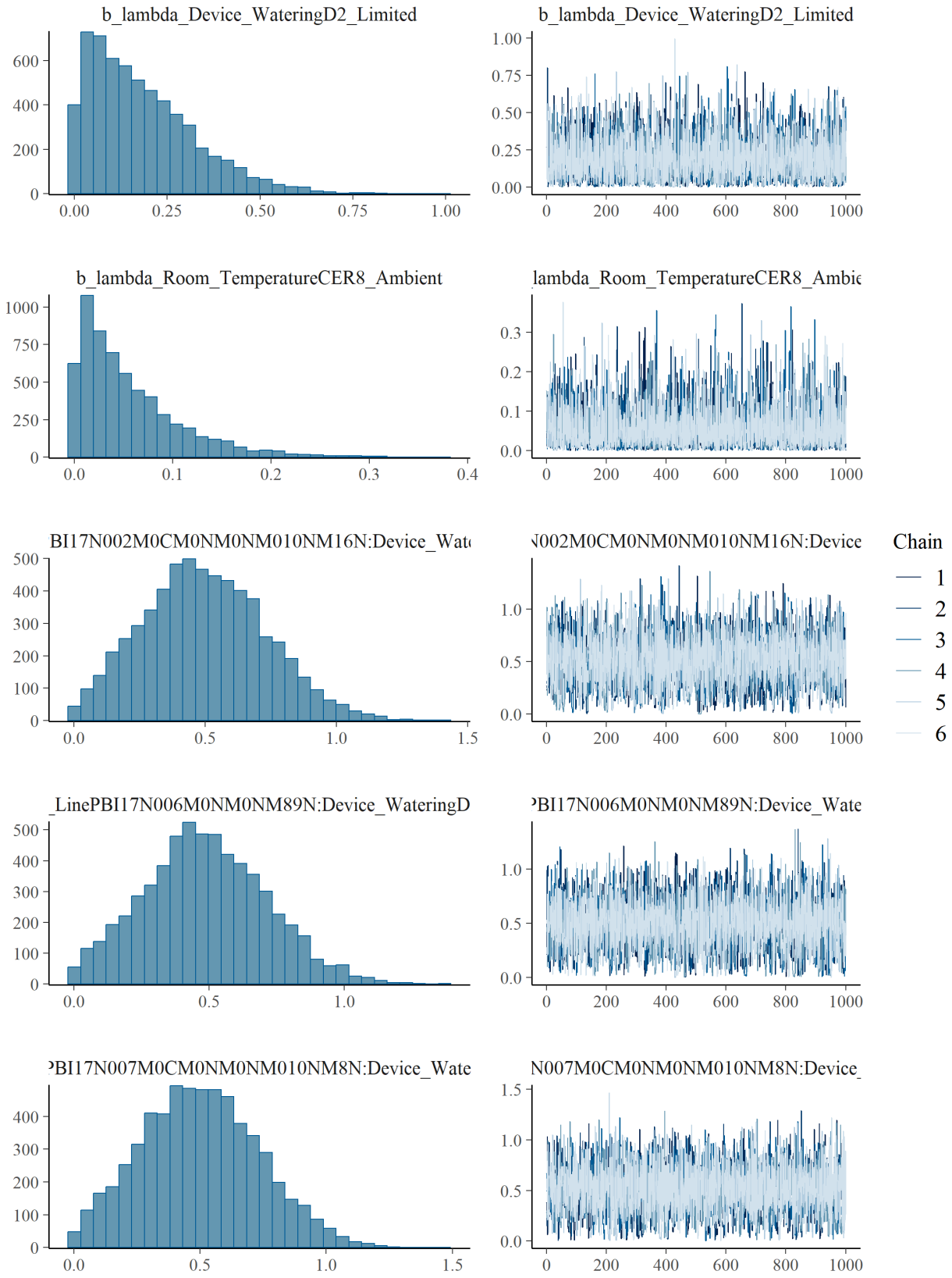


Figure 29: plot of chunk unnamed-chunk-4

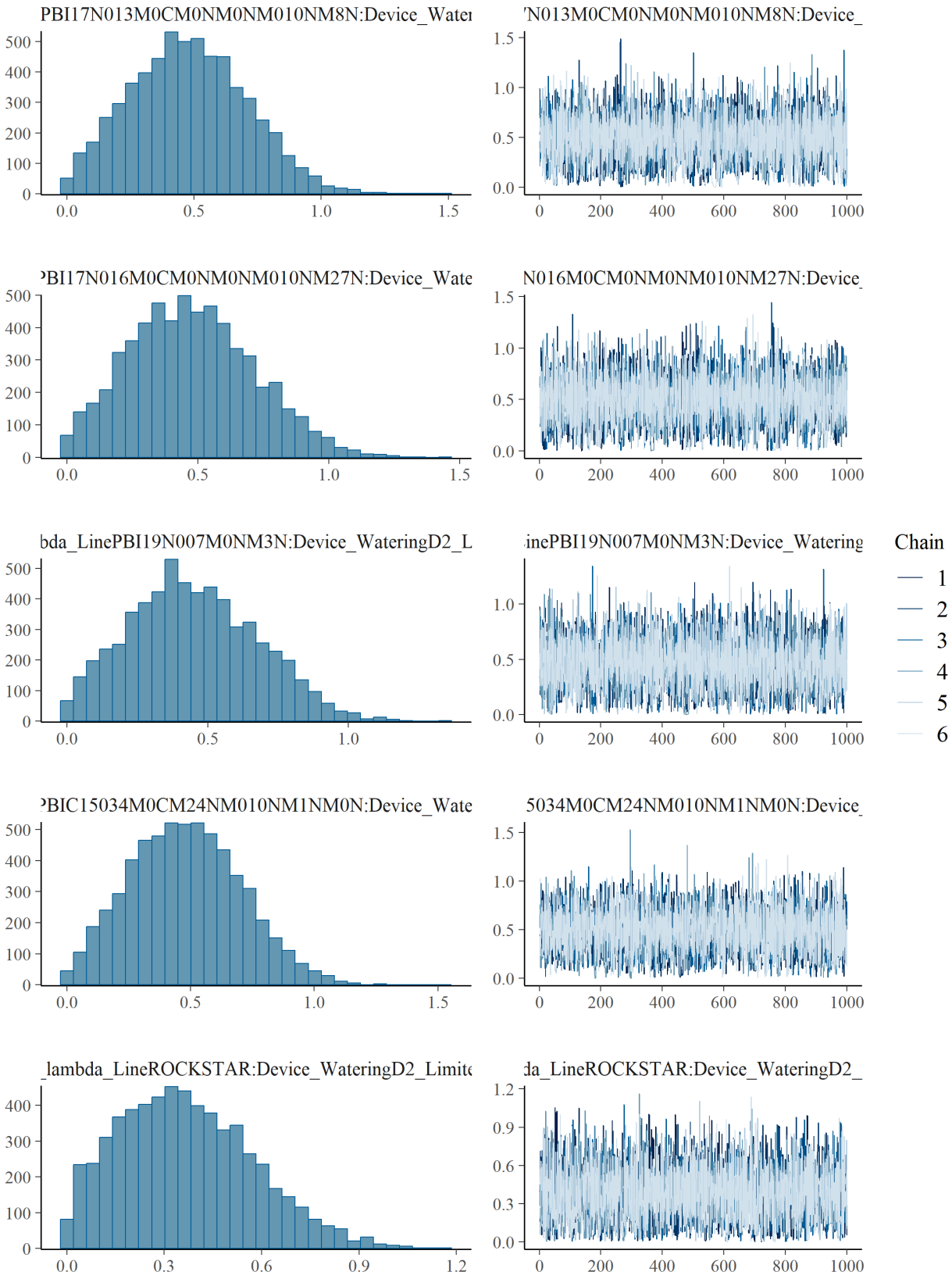


Figure 30: plot of chunk unnamed-chunk-4

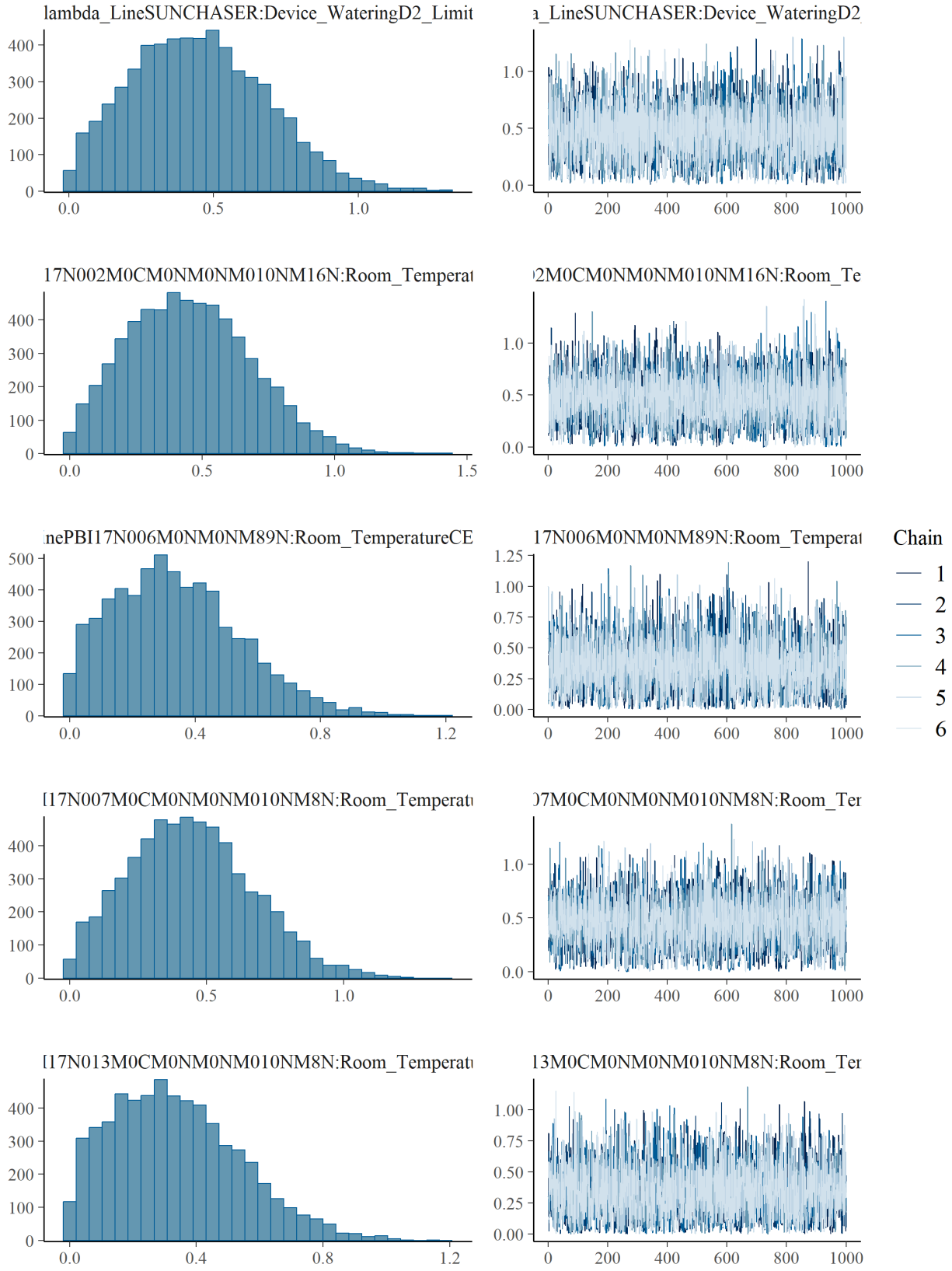


Figure 31: plot of chunk unnamed-chunk-4

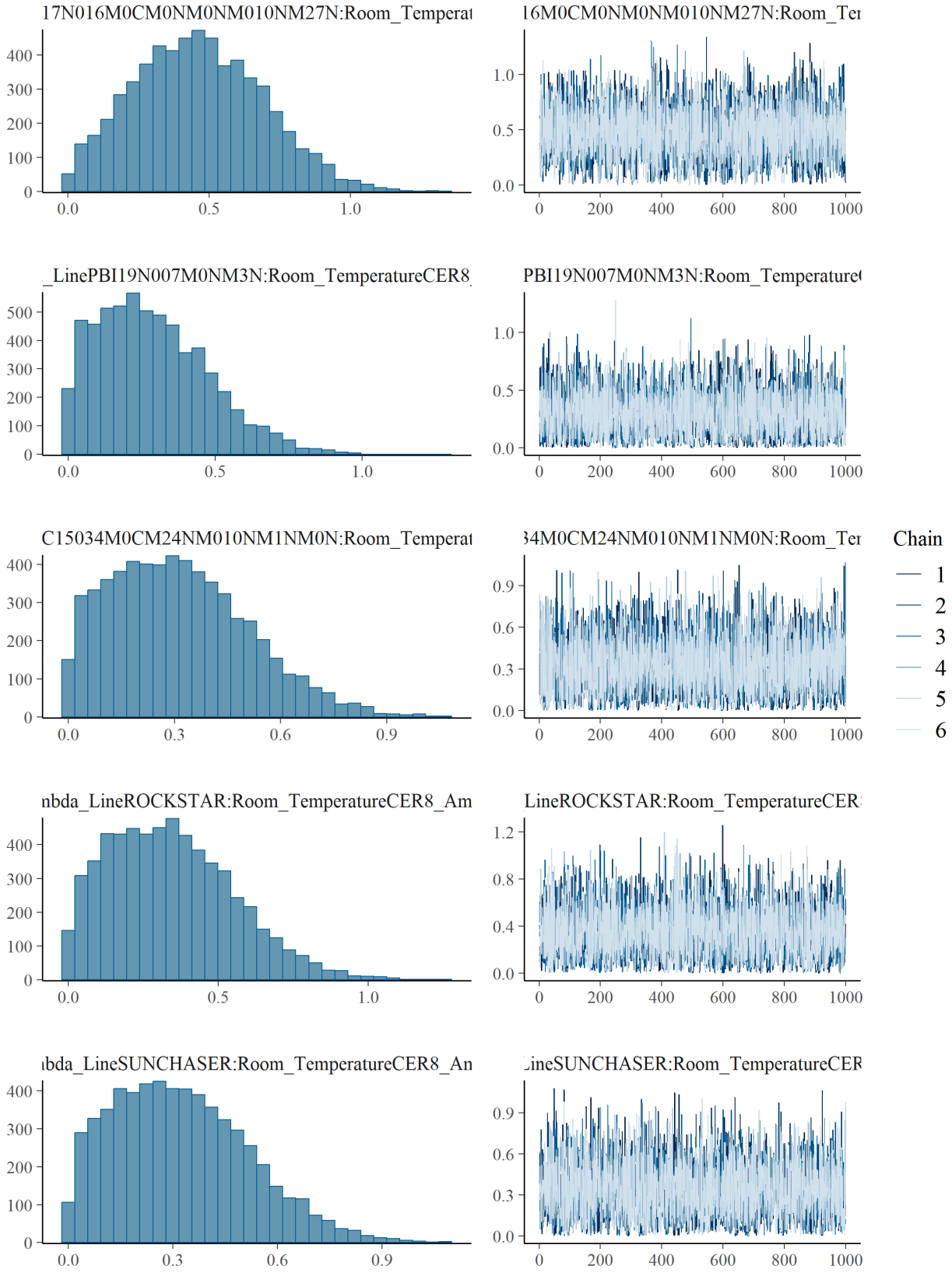


Figure 32: plot of chunk unnamed-chunk-4

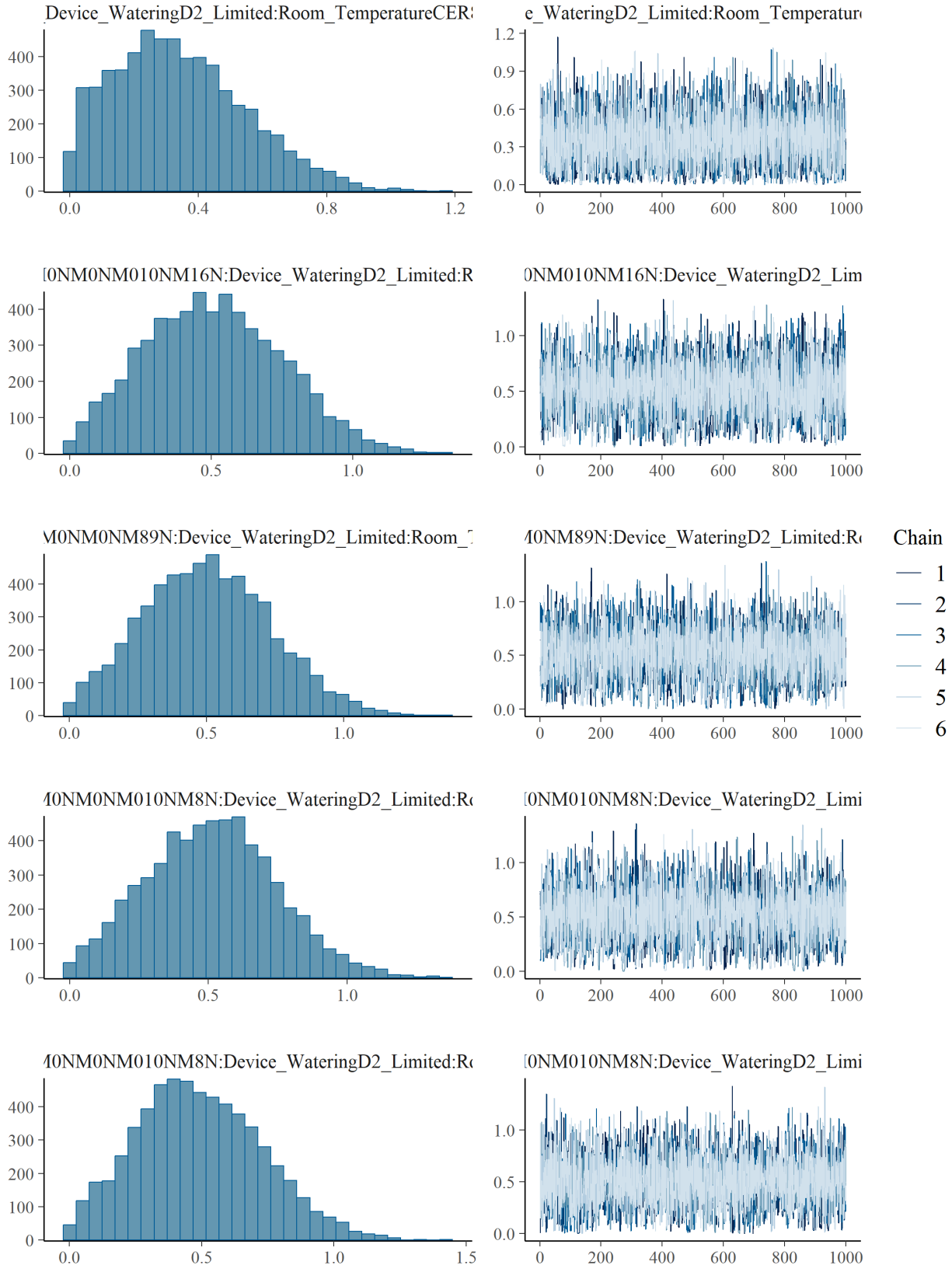


Figure 33: plot of chunk unnamed-chunk-4

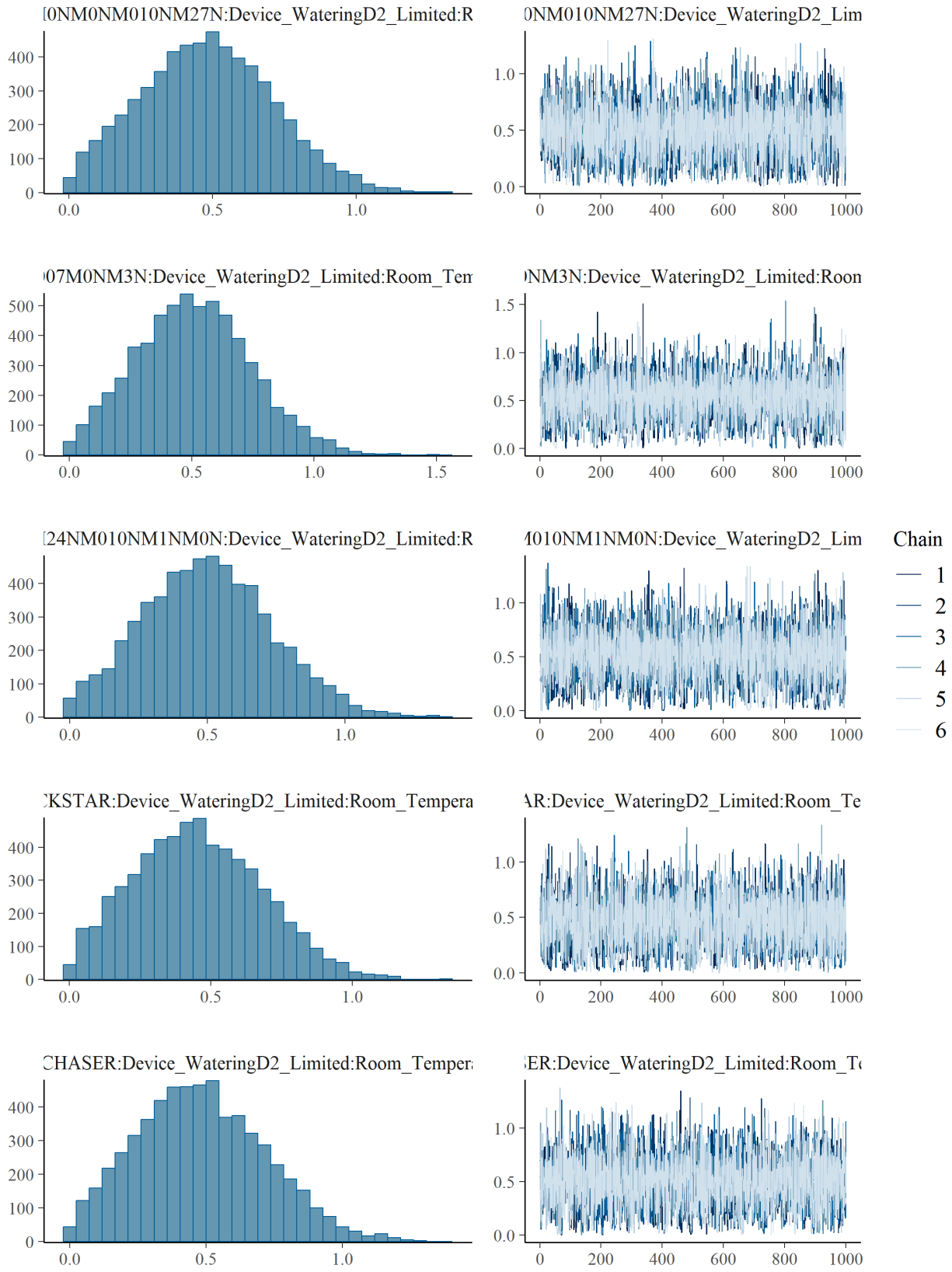


Figure 34: plot of chunk unnamed-chunk-4

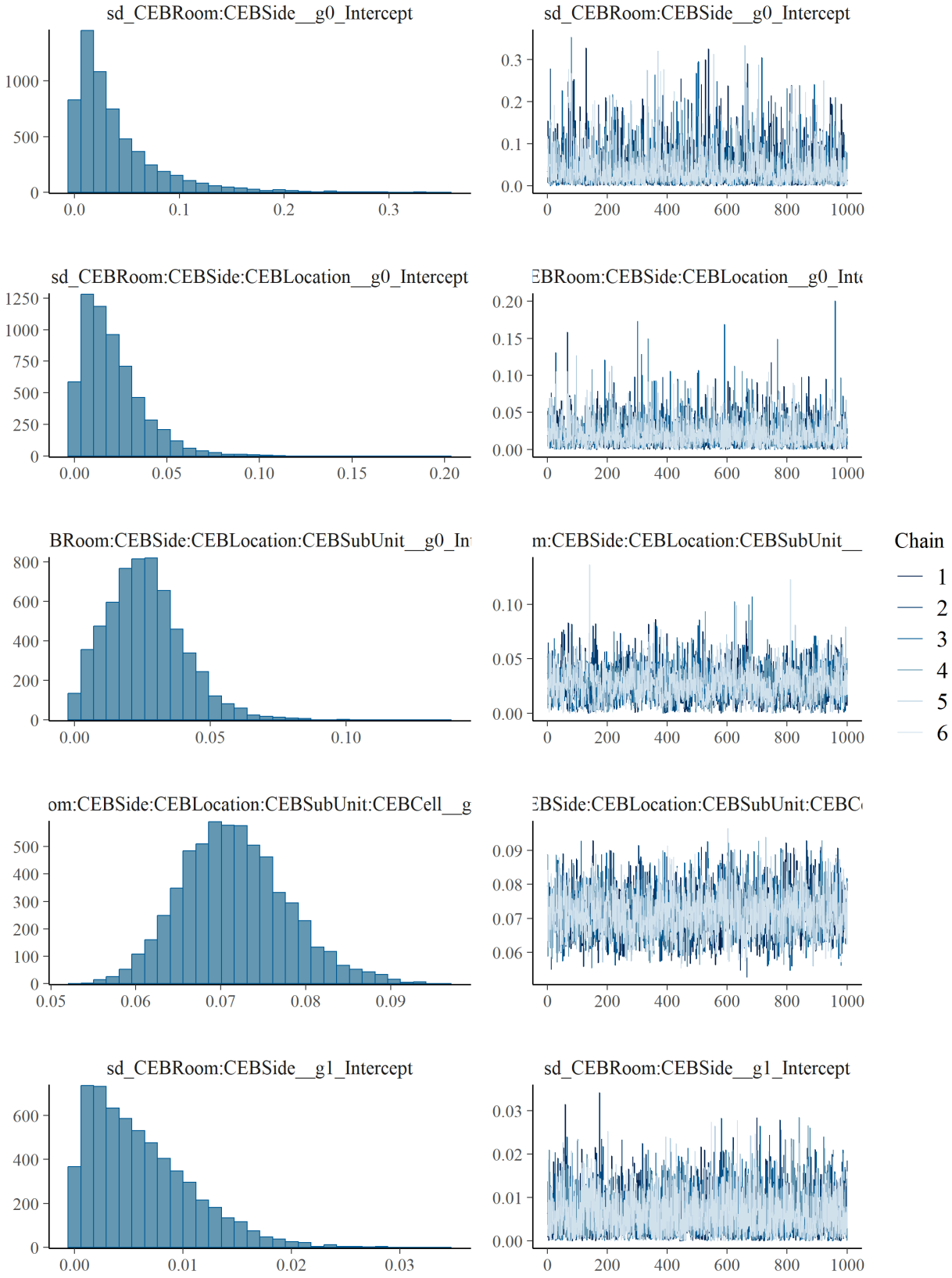


Figure 35: plot of chunk unnamed-chunk-4

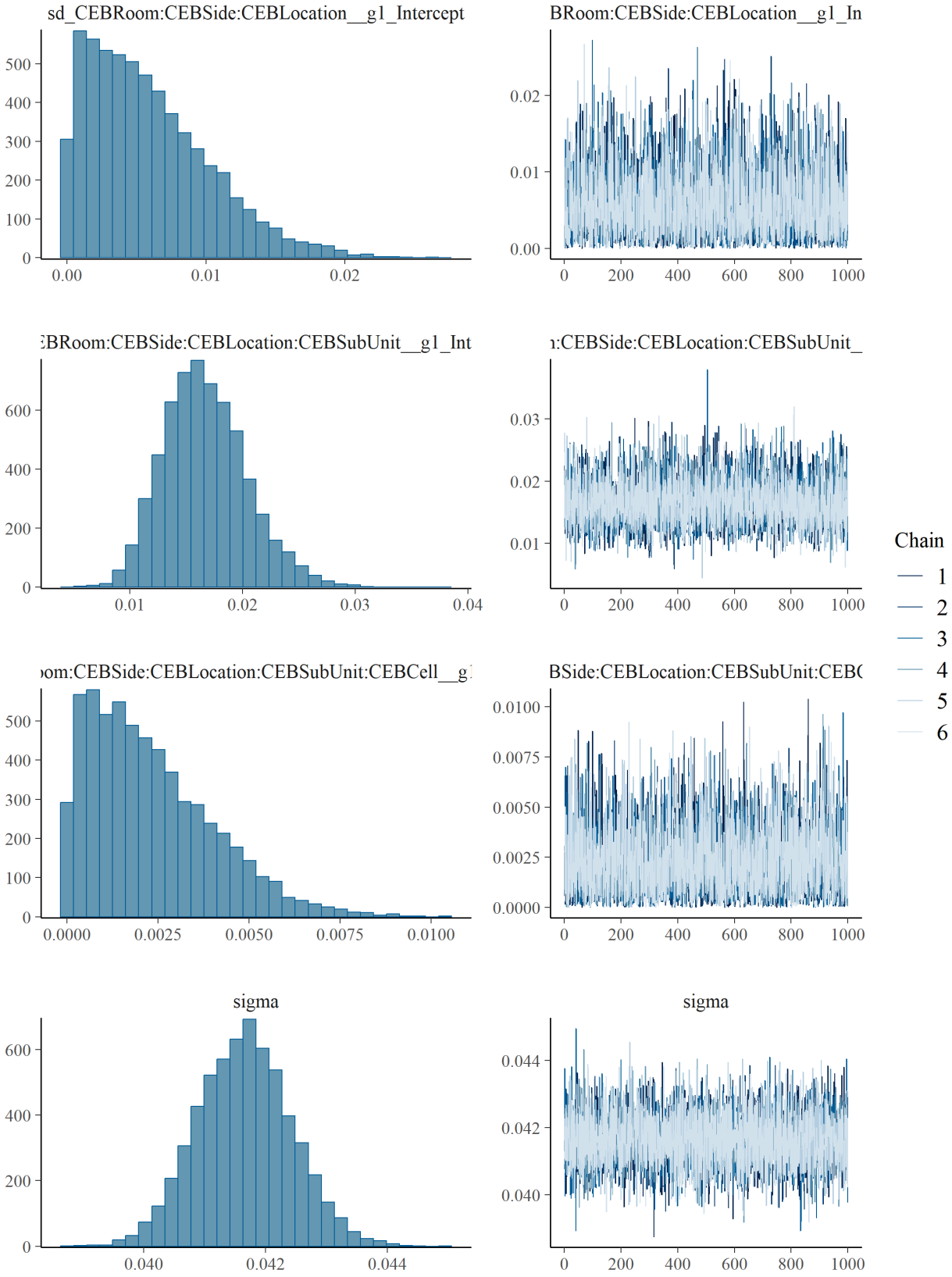


Figure 36: plot of chunk unnamed-chunk-4

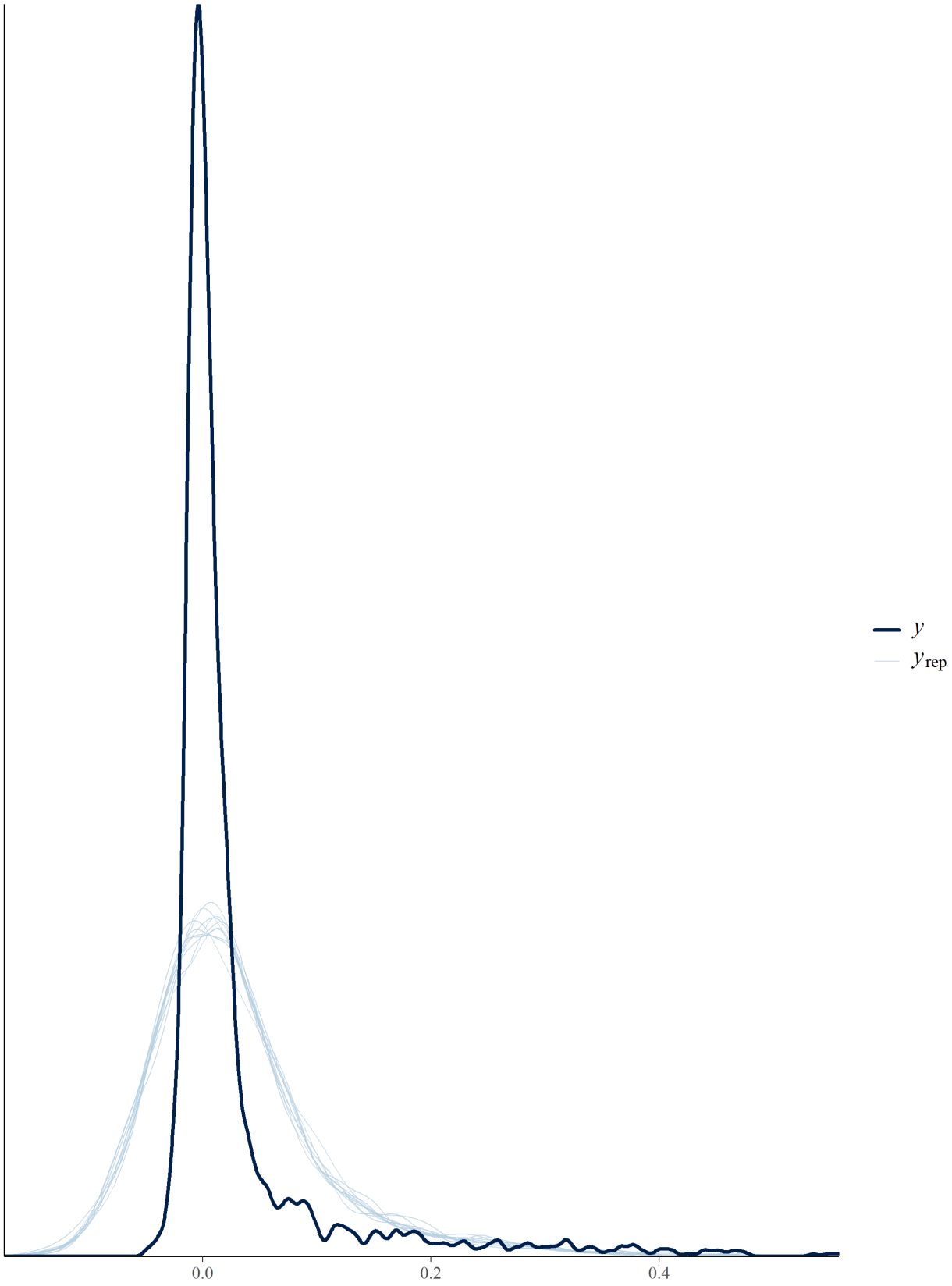


Figure 37: plot of chunk unnamed-chunk-4

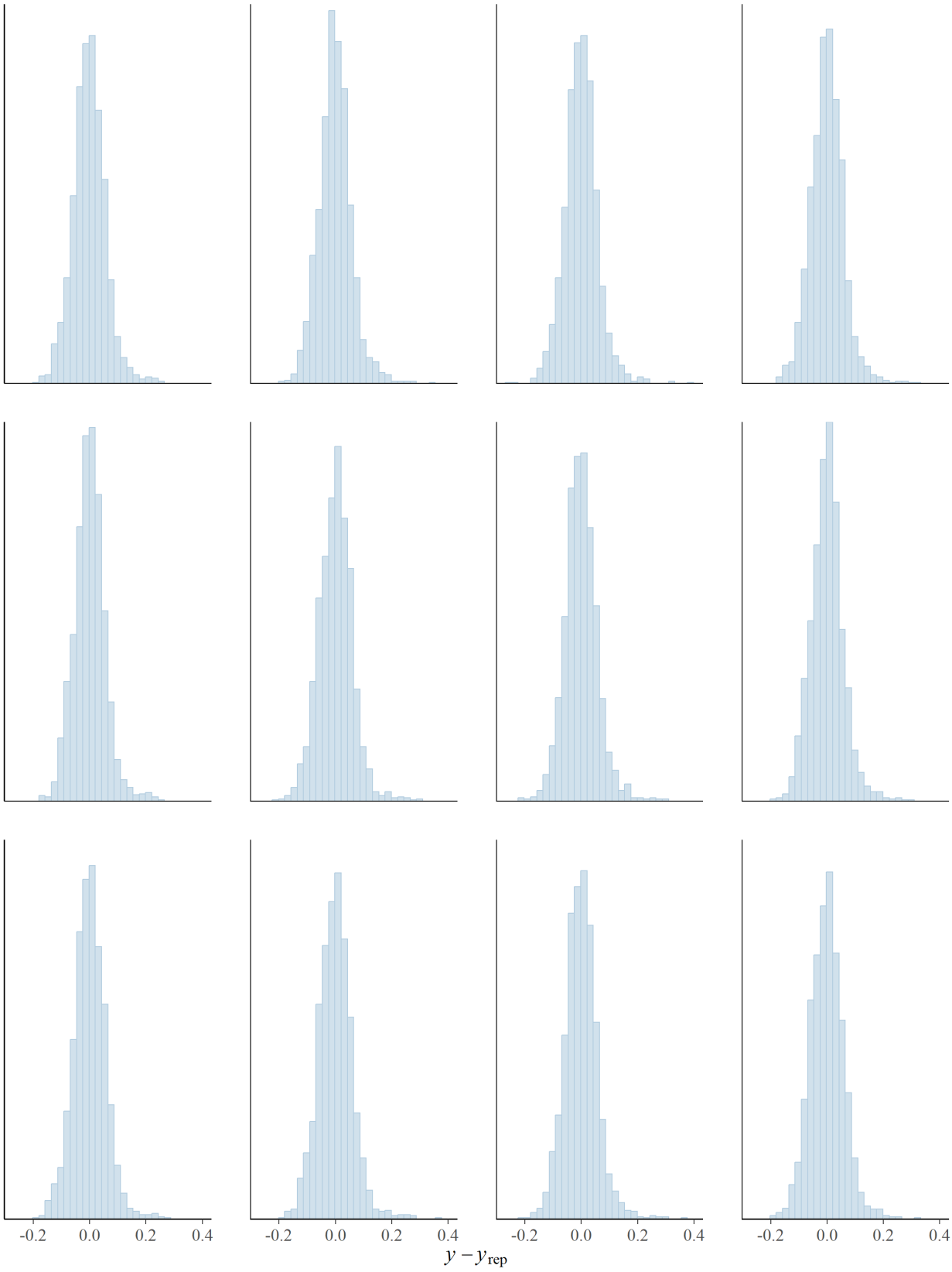


Figure 38: plot of chunk unnamed-chunk-4

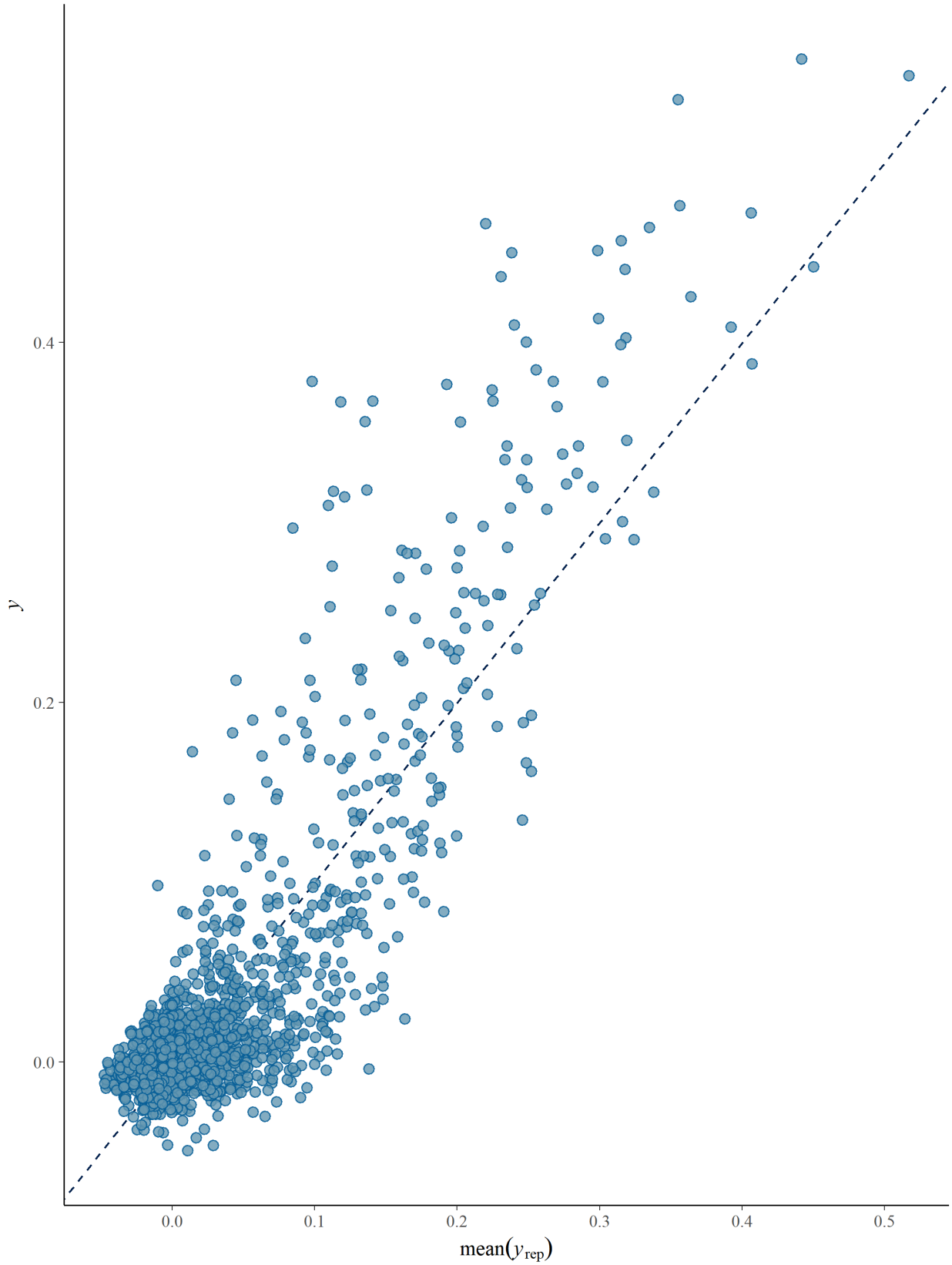


Figure 39: plot of chunk unnamed-chunk-4

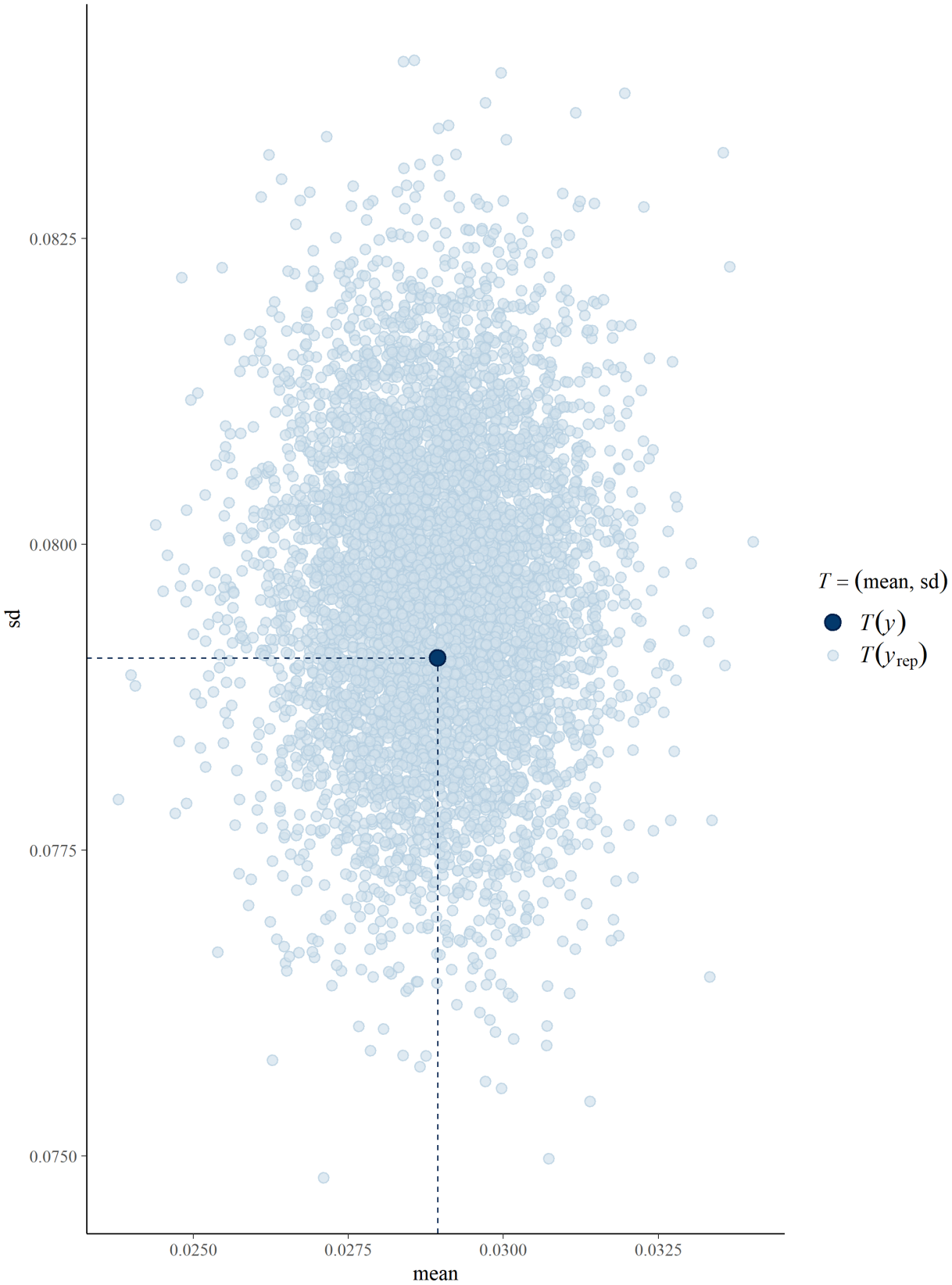


Figure 40: plot of chunk unnamed-chunk-4

```
pp_check(fit_off_max, type = "loo_ribbon")
```

```
Using all posterior draws for ppc type 'loo_ribbon' by default.
```

```
Warning: Found 61 observations with a pareto_k > 0.7 in model '.x1'. We recommend to set 'moment_match =
problematic observations.
```

```
plot(gsw~time.adj, li600.off.dat, pch=16)
points(fitted(fit_off_max)[,1]~time.adj, li600.off.dat, pch=16, col="red")
```

```
##make the above plot prettier
li600.off.plot <- li600.off.dat
li600.off.plot$fitted <- fitted(fit_off_max)[,1]
li600.off.plot$fitted.fixed <- fitted(fit_off_max, re_formula = NA)[,1]

ggplot(li600.off.plot, aes(x= time.adj, y = gsw, group = Replicate)) +
  geom_point() +
  facet_grid(Room_Temperature + Device_Watering ~ Line) +
  geom_point(aes(y=fitted), colour = "red") +
  geom_line(aes(y=fitted), colour = "red")
```

```
ggplot(li600.off.plot, aes(x= time.adj, y = gsw, group = Replicate)) +
  geom_point() +
  facet_grid(Room_Temperature + Device_Watering ~ Line) +
  geom_point(aes(y=fitted), colour = "red") +
  geom_line(aes(y=fitted), colour = "red") +
  geom_point(aes(y=fitted.fixed), colour = "blue") +
  geom_line(aes(y=fitted.fixed), colour = "blue")
```

```
##Testing contribution of effects in the model
# nlform.reduced <- bf(gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - time.adj)/k)+1)),
#                      g0 ~ 1 + Line*Device_Watering*Room_Temperature +
#                        (1 | CEBRoom:CEBSide:CEBLocation +
#                           CEBRoom:CEBSide:CEBLocation:CEBSubUnit +
#                           CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell),
#                      g1 ~ 1 + Line + Device_Watering + #Room_Temperature +
#                        Line:Device_Watering +
#                         #Line:Room_Temperature + # +
#                        #Device_Watering:Room_Temperature +
#                        (1 | CEBRoom:CEBSide:CEBLocation +
#                           CEBRoom:CEBSide:CEBLocation:CEBSubUnit +
#                           CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell),
#                      k ~ 1  + Room_Temperature,#+ Line +  Device_Watering,# +
#                        #Line:Device_Watering,# + Line:Room_Temperature,# +
#                        #Device_Watering:Room_Temperature,
#                      lambda ~ 1 + Device_Watering,#+ Line + Room_Temperature,# +
#                        #Line:Device_Watering,# + Line:Room_Temperature,
#                      nl = TRUE)
#
# # Fit the model
# fit_off_reduced <- brm(
#   formula = nlform.reduced,
#   data = li600.off.dat,
#   prior = priors,
#   chains = 6,
```

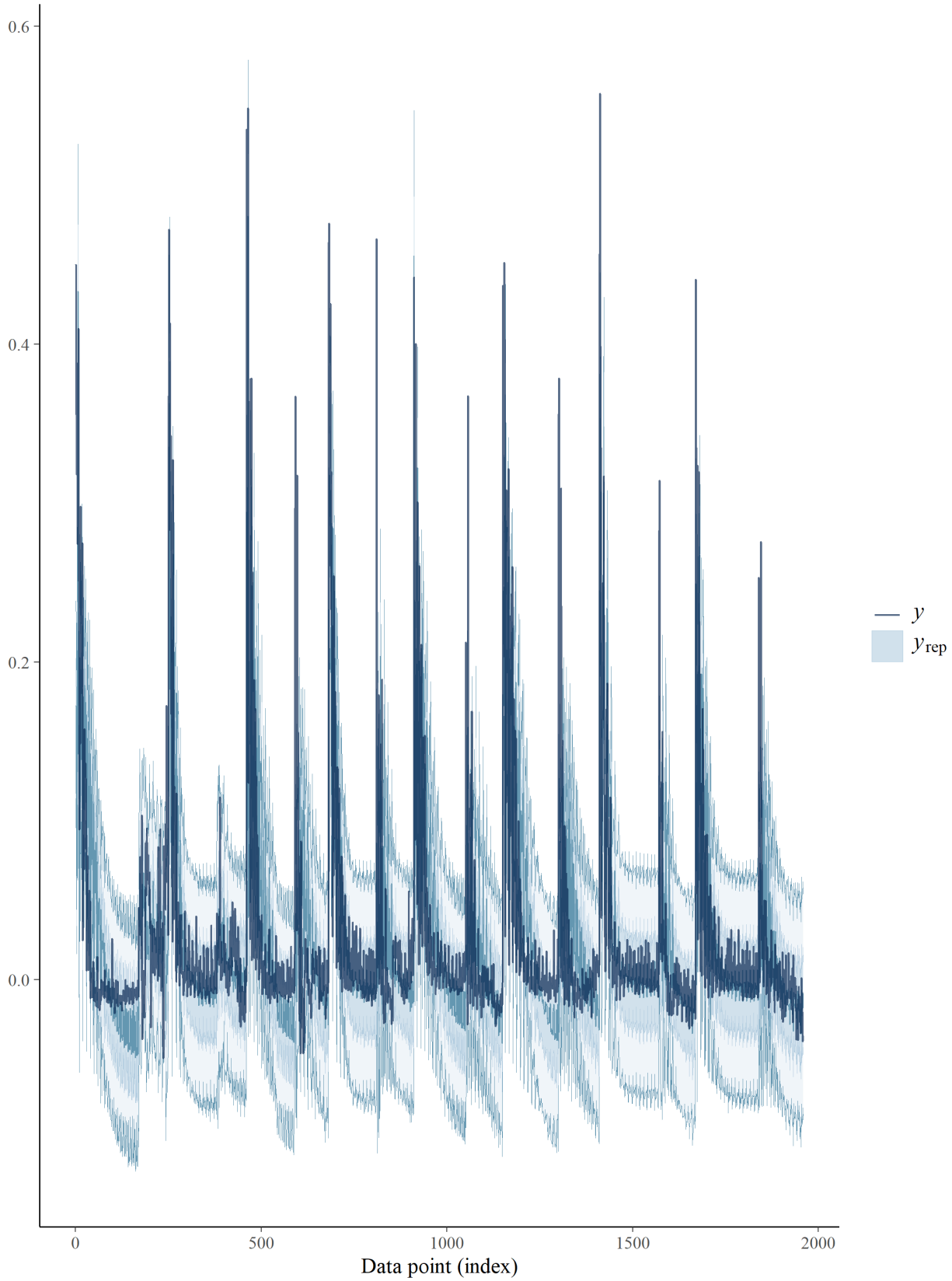


Figure 41: plot of chunk unnamed-chunk-4

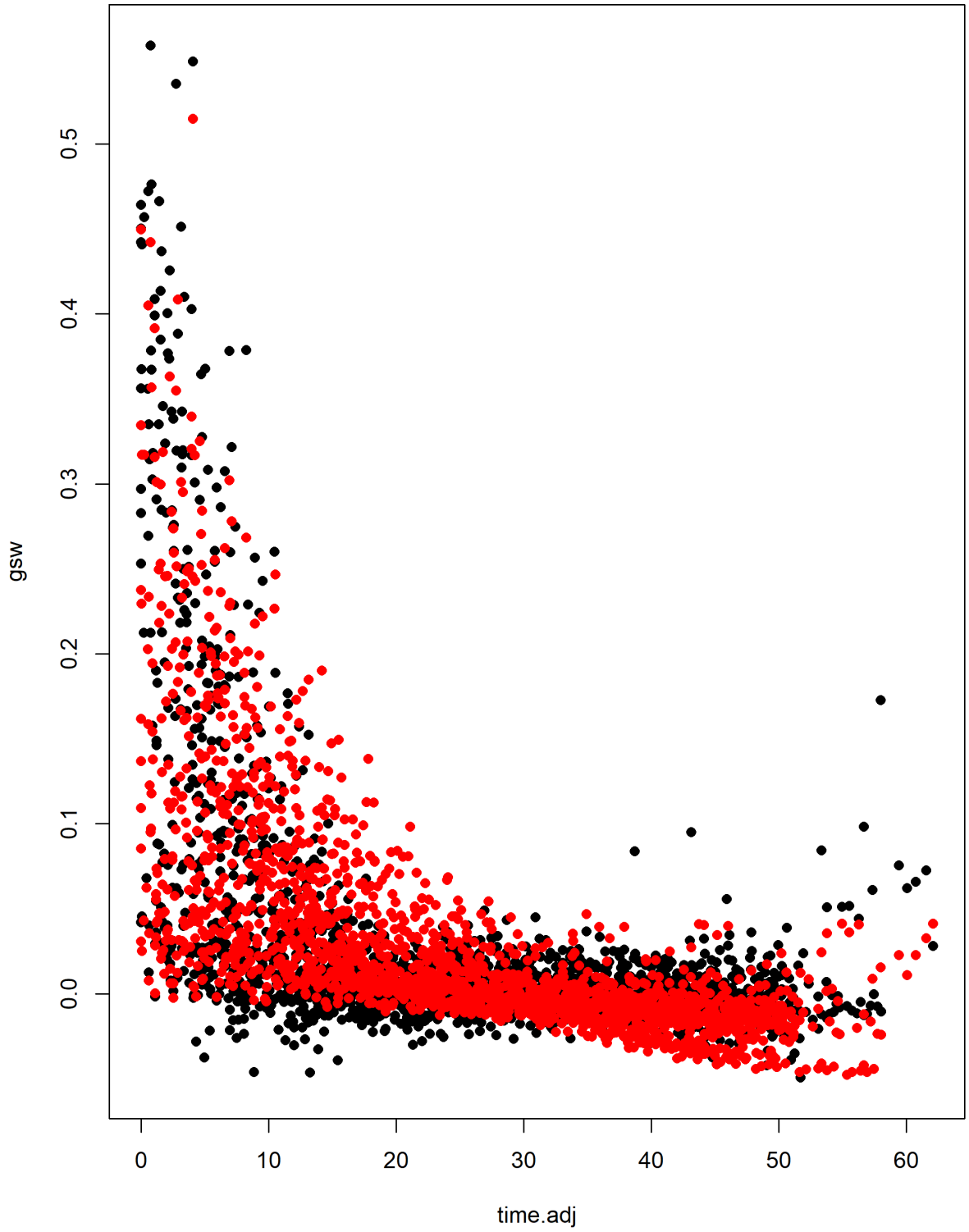


Figure 42: plot of chunk unnamed-chunk-4

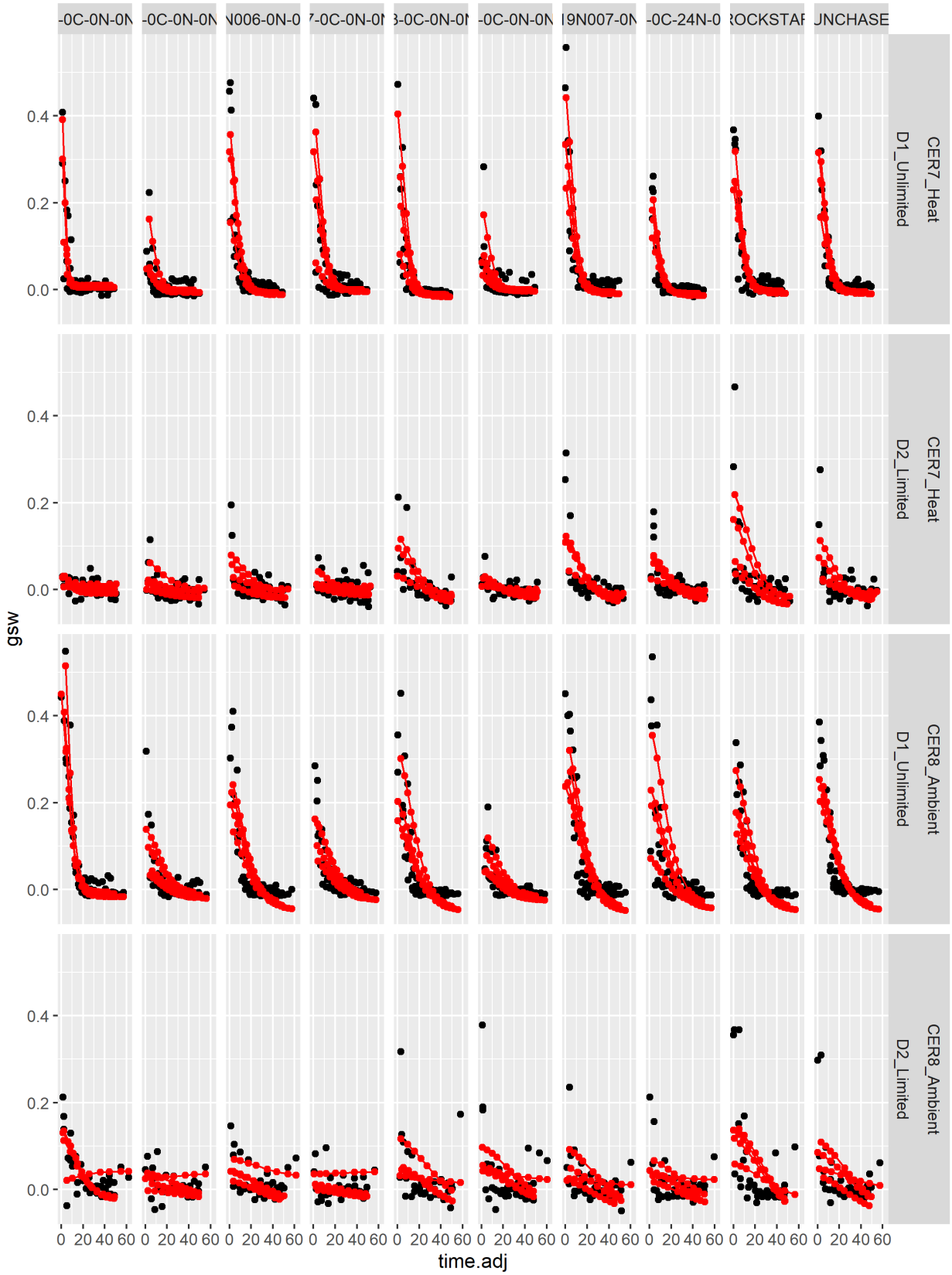


Figure 43: plot of chunk unnamed-chunk-4

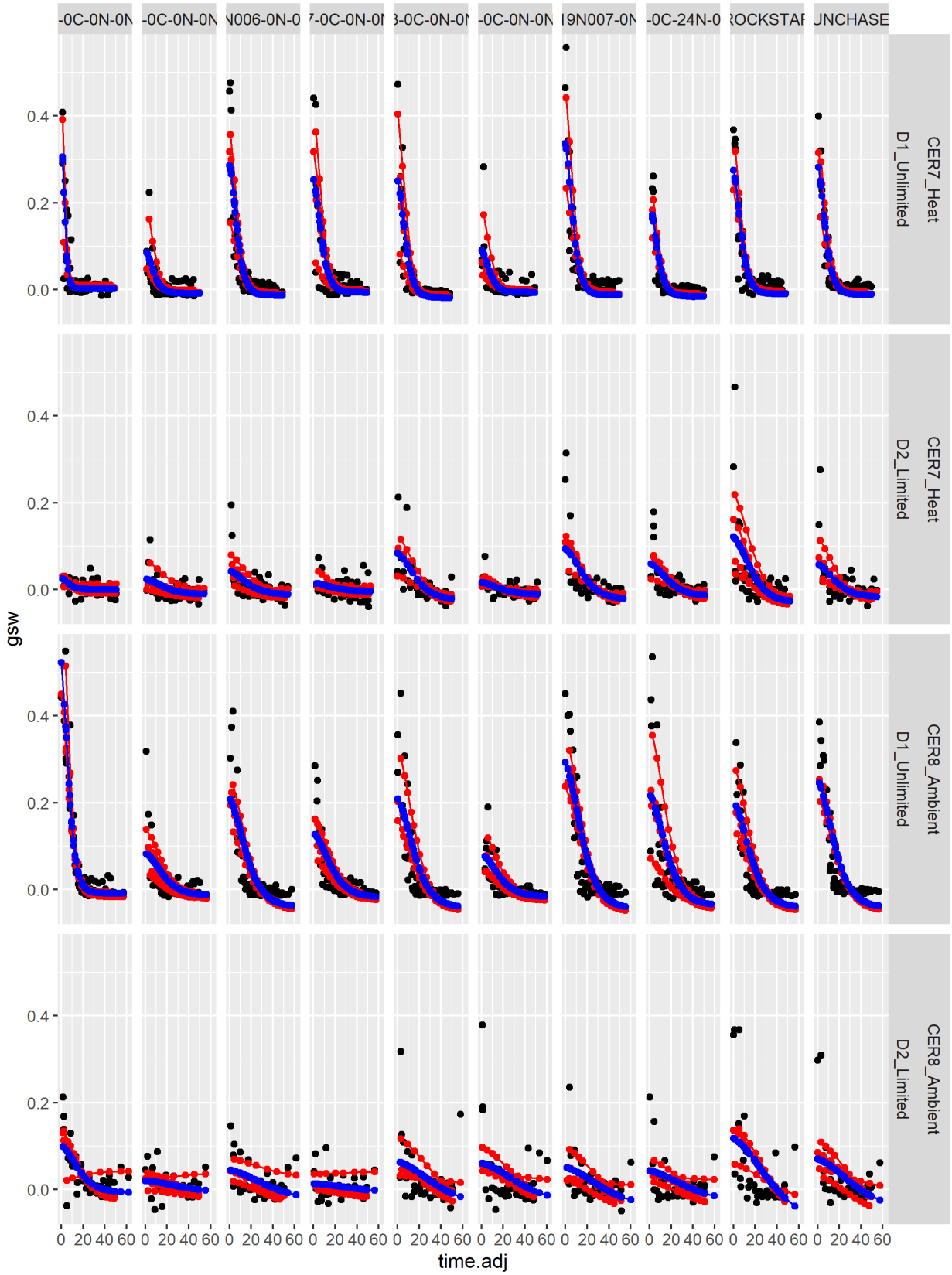


Figure 44: plot of chunk unnamed-chunk-4

```
#   cores = 6,
#   iter = 10000,
#   warmup = 6000,
#   control = list(
#     adapt_delta = 0.99,
#     max_treedepth = 12,
#     metric = "diag_e"
#   )
# )
#
# print(summary(fit_off_reduced))
# loo(fit_off_reduced)
# loo(fit_off_current)
# rstan::check_hmc_diagnostics(fit_off_reduced$fit)
# plot(gsw~time.adj, li600.on.dat, pch=16)
# points(fitted(fit_off_reduced)[,1]~time.adj, li600.on.dat, pch=16, col="red")
# length(fixef(fit_off_reduced))
#
# fit_off_current <- add_criterion(fit_off_current, "loo")
# fit_off_reduced <- add_criterion(fit_off_reduced, "loo")
#
# loo_compare(fit_off_current, fit_off_reduced)
##THREE WAY INTERACTION
##Lambda: Drop Room_Temperature:Device_Watering:Line interaction
##k: Drop Room_Temperature:Device_Watering:Line interaction
##g0: Retain Room_Temperature:Device_Watering:Line interaction
##g1: Drop Room_Temperature:Device_Watering:Line interaction

##TWO WAY INTERACTION
##lambda: Drop Room_Temperature:Device_Watering
##k: Drop Room_Temperature:Device_Watering
##g1: Drop Room_Temperature:Device_Watering
##lambda: Drop Line:Room_Temperature
##lambda: Drop Line:Device_Watering
##k: Drop Line:Room_Temperature
##k: Drop Line:Device_Watering
##g1: Retain Line:Device_Watering
##g1: Drop Line:Room_Temperature

##MAIN EFFECTS
##lambda: Drop Room_Temperature
##lambda: Retain Device_Watering
##lambda: Drop Line
##k: Retain Room_Temperature
##k: Drop Device_Watering
##k: Drop Line
##g1: Drop Room_Temperature
# fit_off_current <- fit_off_reduced
# print(summary(fit_off_current))
```

```
##Final model:
##g0: Room_Temperature:Device_Watering:Line interaction
```

```
##g1: Line:Device_Watering
##lambda: Device_Watering
##k: Room_Temperature

nlform_final <- bf(gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - time.adj)/k)+1)),
                   g0 ~ 1 + Line*Device_Watering*Room_Temperature +
                     (1 | CEBRoom:CEBSide:CEBLocation +
                        CEBRoom:CEBSide:CEBLocation:CEBSubUnit +
                        CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell),
                   g1 ~ 1 + Line + Device_Watering + Line:Device_Watering +
                     (1 | CEBRoom:CEBSide:CEBLocation +
                        CEBRoom:CEBSide:CEBLocation:CEBSubUnit +
                        CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell),
                   k ~ 1 + Room_Temperature,
                   lambda ~ 1+ Device_Watering,
                   nl = TRUE)

# Fit the model
fit_off_final <- brm(
  formula = nlform_final,
  data = li600.off.dat,
  prior = priors,
  chains = 6,
  cores = 6,
  iter = 10000,
  warmup = 6000,
  control = list(
    adapt_delta = 0.99,
    max_treedepth = 15,
    metric = "diag_e"
  )
)
```

Fit final model

```
Compiling Stan program...
```

```
Start sampling
```

```
print(summary(fit_off_final))
```

```
 Family: gaussian
  Links: mu = identity
Formula: gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - time.adj)/k) + 1))
         g0 ~ 1 + Line * Device_Watering * Room_Temperature + (1 | CEBRoom:CEBSide:CEBLocation + CEBRoom
         g1 ~ 1 + Line + Device_Watering + Line:Device_Watering + (1 | CEBRoom:CEBSide:CEBLocation + CEB
         k ~ 1 + Room_Temperature
         lambda ~ 1 + Device_Watering
   Data: li600.off.dat (Number of observations: 1959)
  Draws: 6 chains, each with iter = 10000; warmup = 6000; thin = 1;
         total post-warmup draws = 24000

Multilevel Hyperparameters:
~CEBRoom:CEBSide:CEBLocation (Number of levels: 8)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.04      0.03     0.00     0.11 1.00    11691    14175
```

```
sd(g1_Intercept)     0.00      0.00     0.00     0.01 1.00     9771    14474

~CEBRoom:CEBSide:CEBLocation:CEBSubUnit (Number of levels: 16)
                 Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.06      0.02     0.01     0.11 1.00     6481     6006
sd(g1_Intercept)     0.01      0.00     0.01     0.02 1.00    13702    17666

~CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell (Number of levels: 159)
                 Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.14      0.01     0.12     0.16 1.00     8078    14856
sd(g1_Intercept)     0.01      0.00     0.01     0.01 1.00    11004    15780

Regression Coefficients:
                                                                                                  Estimate Est.
g0_Intercept                                                                                          0.36
g0_LinePBI17N002M0CM0NM0NM010NM16N                                                                   -0.17
g0_LinePBI17N006M0NM0NM89N                                                                            0.06
g0_LinePBI17N007M0CM0NM0NM010NM8N                                                                     0.02
g0_LinePBI17N013M0CM0NM0NM010NM8N                                                                     0.02
g0_LinePBI17N016M0CM0NM0NM010NM27N                                                                   -0.20
g0_LinePBI19N007M0NM3N                                                                                0.13
g0_LinePBIC15034M0CM24NM010NM1NM0N                                                                    0.04
g0_LineROCKSTAR                                                                                       0.05
g0_LineSUNCHASER                                                                                      0.12
g0_Device_WateringD2_Limited                                                                         -0.31
g0_Room_TemperatureCER8_Ambient                                                                       0.17
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited                                          0.22
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited                                                  0.03
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited                                          -0.00
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited                                           0.08
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited                                          0.21
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited                                                      0.09
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited                                          0.15
g0_LineROCKSTAR:Device_WateringD2_Limited                                                             0.15
g0_LineSUNCHASER:Device_WateringD2_Limited                                                            0.03
g0_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient                                      -0.20
g0_LinePBI17N006M0NM0NM89N:Room_TemperatureCER8_Ambient                                              -0.18
g0_LinePBI17N007M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient                                       -0.29
g0_LinePBI17N013M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient                                       -0.16
g0_LinePBI17N016M0CM0NM0NM010NM27N:Room_TemperatureCER8_Ambient                                      -0.14
g0_LinePBI19N007M0NM3N:Room_TemperatureCER8_Ambient                                                  -0.12
g0_LinePBIC15034M0CM24NM010NM1NM0N:Room_TemperatureCER8_Ambient                                       -0.15
g0_LineROCKSTAR:Room_TemperatureCER8_Ambient                                                          -0.19
g0_LineSUNCHASER:Room_TemperatureCER8_Ambient                                                         -0.17
g0_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                                             -0.09
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient              0.07
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                      0.06
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient               0.16
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient               0.12
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient              0.16
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                         -0.10
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient             -0.05
g0_LineROCKSTAR:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                                 0.19
g0_LineSUNCHASER:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                                0.09
```

```
g1_Intercept                                                                  0.00
g1_LinePBI17N002M0CM0NM0NM010NM16N                                           -0.00
g1_LinePBI17N006M0NM0NM89N                                                   -0.00
g1_LinePBI17N007M0CM0NM0NM010NM8N                                             0.01
g1_LinePBI17N013M0CM0NM0NM010NM8N                                            -0.00
g1_LinePBI17N016M0CM0NM0NM010NM27N                                           -0.00
g1_LinePBI19N007M0NM3N                                                        0.00
g1_LinePBIC15034M0CM24NM010NM1NM0N                                           -0.00
g1_LineROCKSTAR                                                               0.00
g1_LineSUNCHASER                                                              0.00
g1_Device_WateringD2_Limited                                                  0.00
g1_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited                 -0.00
g1_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited                         -0.00
g1_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited                  -0.01
g1_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited                   0.00
g1_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited                 -0.01
g1_LinePBI19N007M0NM3N:Device_WateringD2_Limited                              -0.01
g1_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited                 -0.00
g1_LineROCKSTAR:Device_WateringD2_Limited                                     -0.01
g1_LineSUNCHASER:Device_WateringD2_Limited                                    -0.01
k_Intercept                                                                   2.62
k_Room_TemperatureCER8_Ambient                                                1.97
lambda_Intercept                                                              0.02
lambda_Device_WateringD2_Limited                                              0.03

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.02      0.00     0.02     0.02 1.00    28903    19031

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
pp_check(fit_off_final)
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

```
rstan::check_hmc_diagnostics(fit_off_final$fit)
```

```
Divergences:
0 of 24000 iterations ended with a divergence.

Tree depth:
0 of 24000 iterations saturated the maximum tree depth of 15.

Energy:
E-BFMI indicated no pathological behavior.
```

```
plot(fit_off_final, ask = FALSE)
```

```
##make the above plot prettier
li600.off.plot <- li600.off.dat
li600.off.plot$fitted <- fitted(fit_off_final)[,1]
```

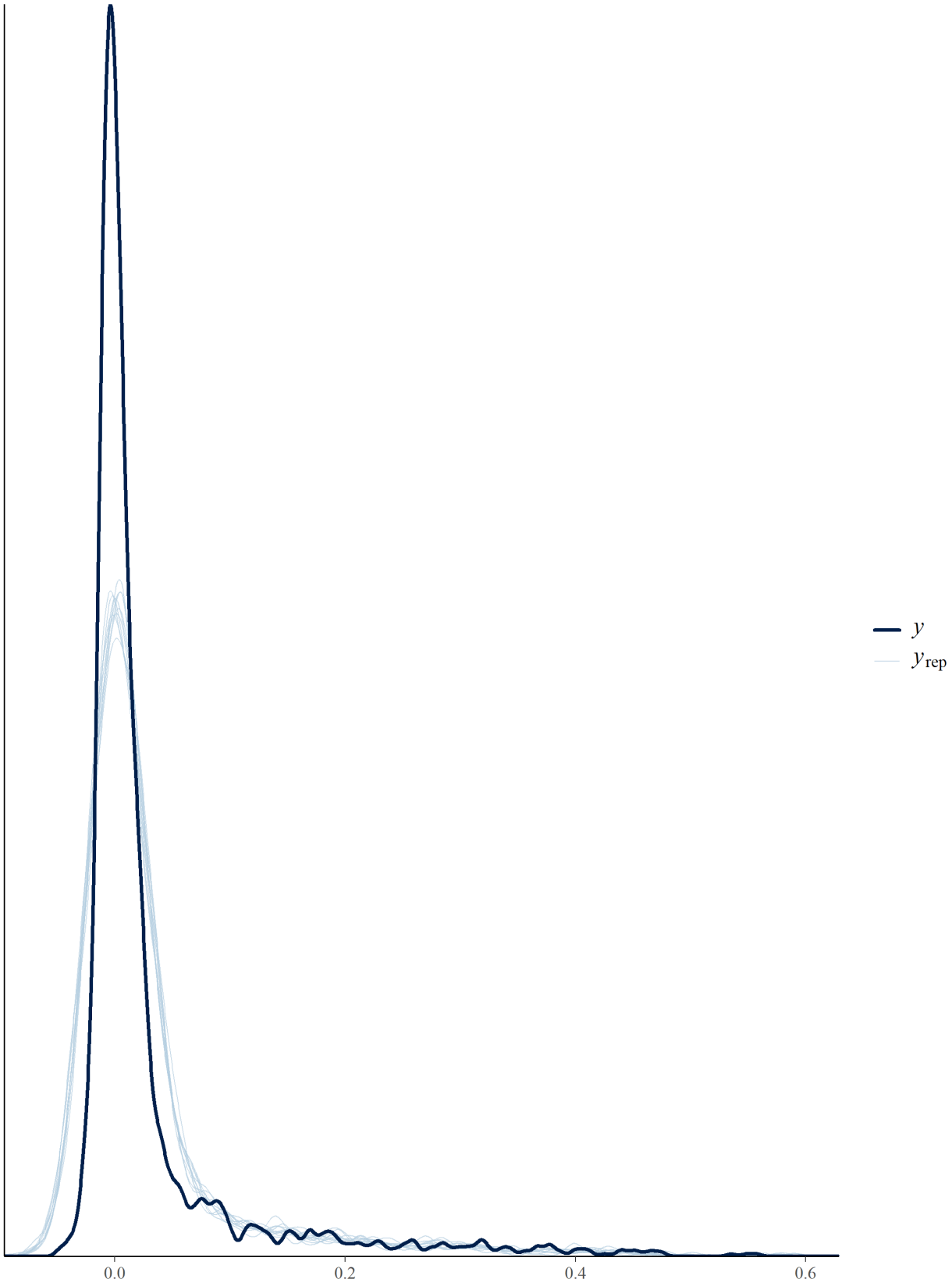


Figure 45: plot of chunk unnamed-chunk-5

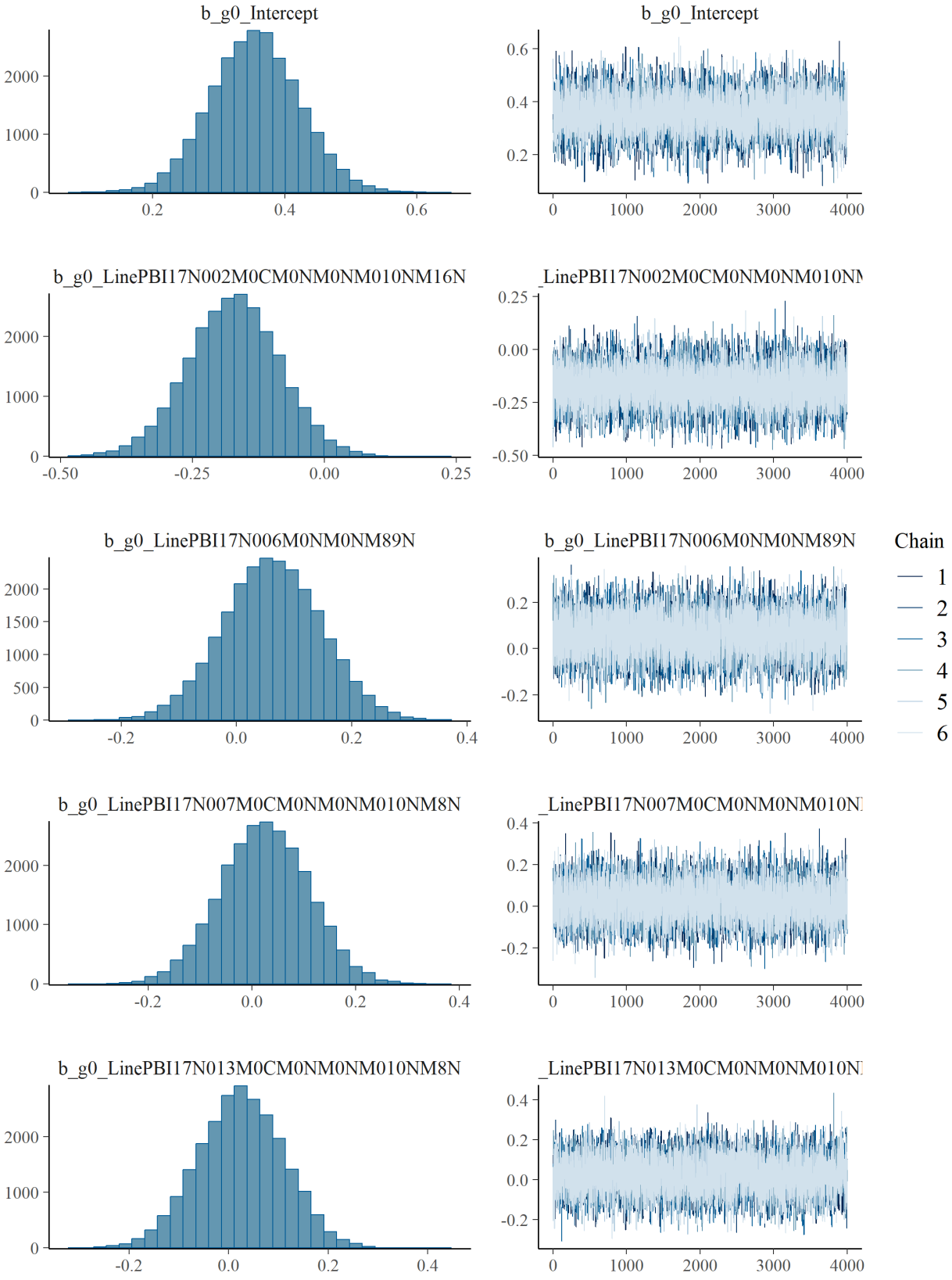


Figure 46: plot of chunk unnamed-chunk-5

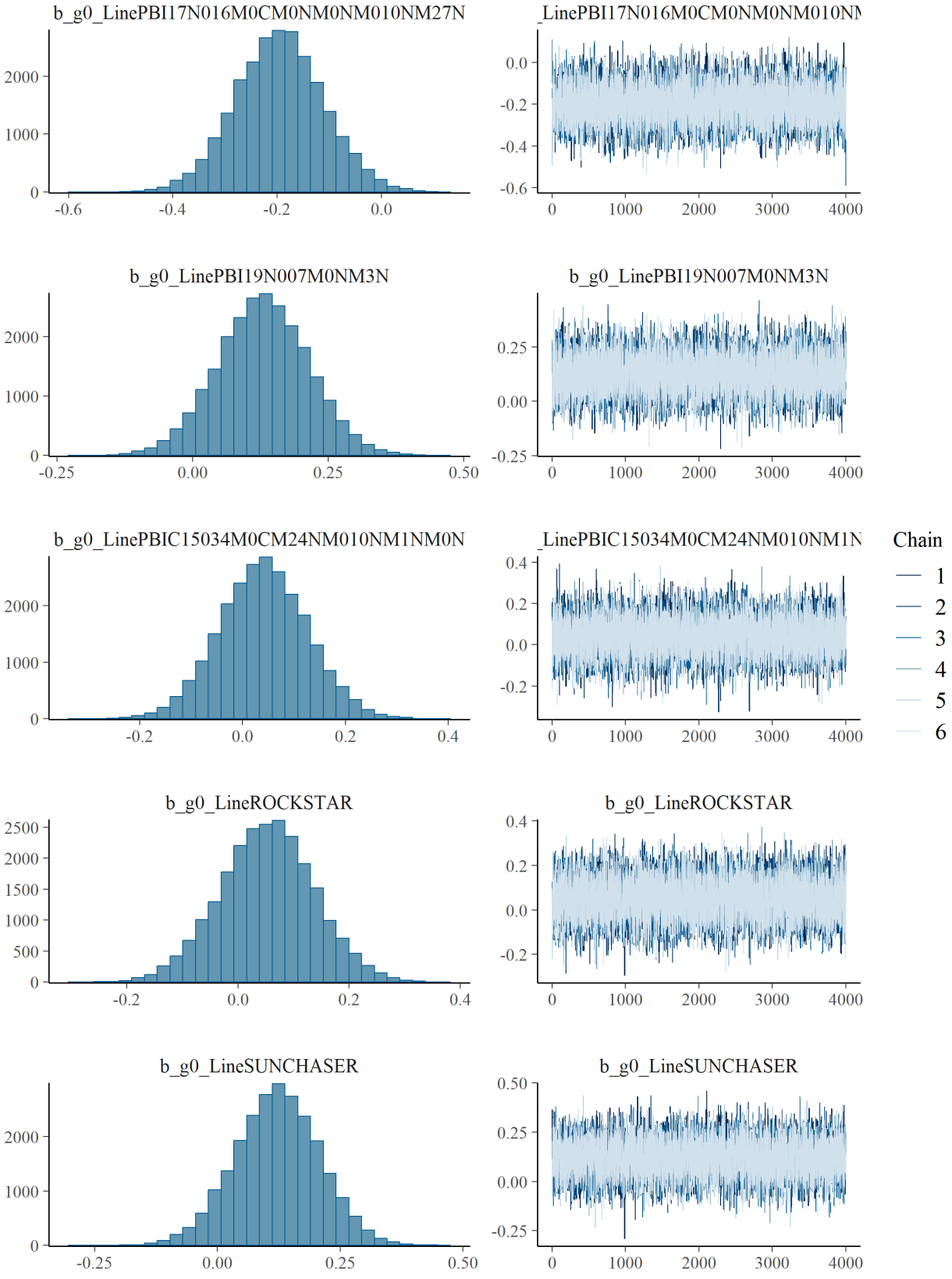


Figure 47: plot of chunk unnamed-chunk-5

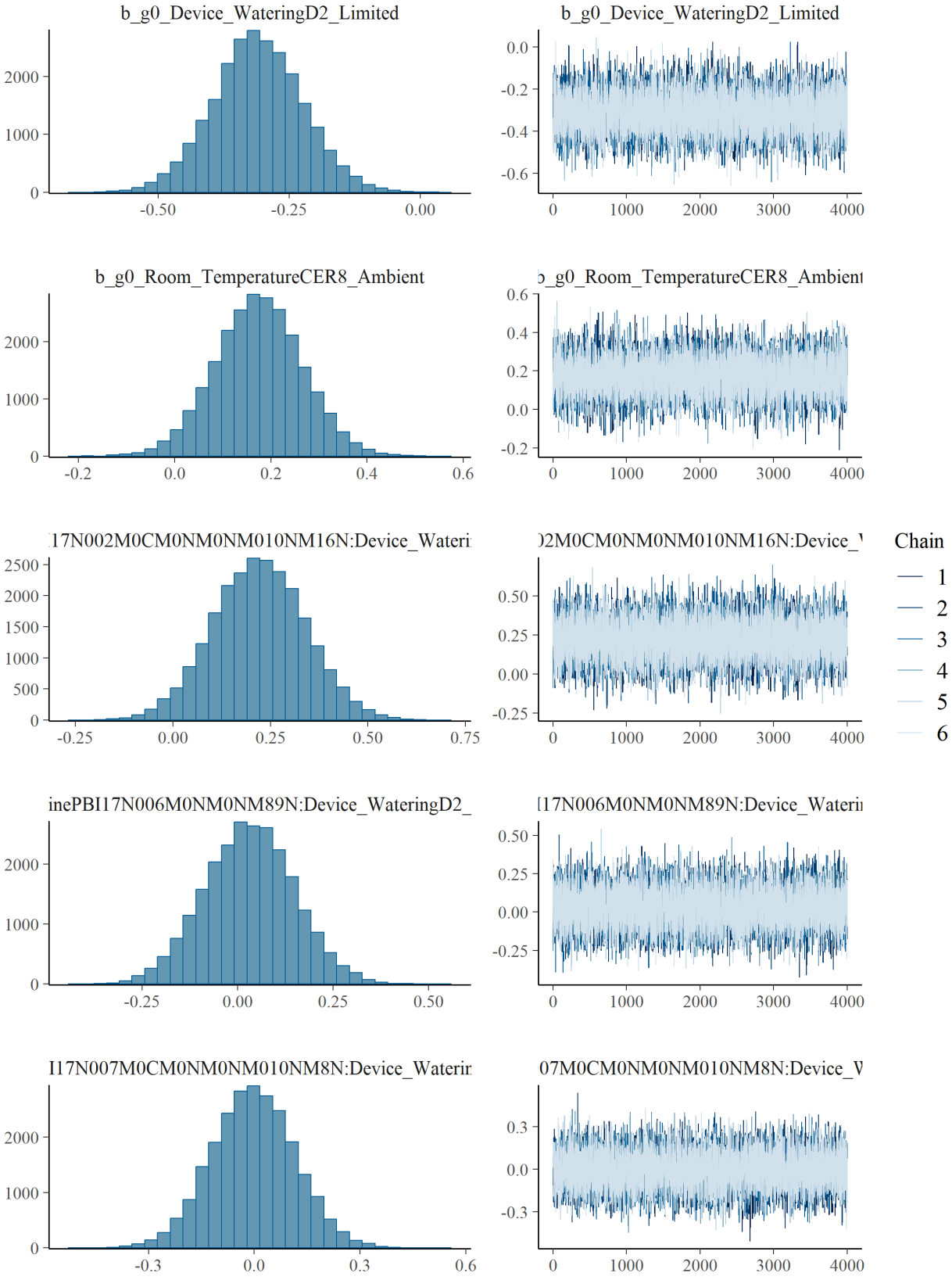


Figure 48: plot of chunk unnamed-chunk-5

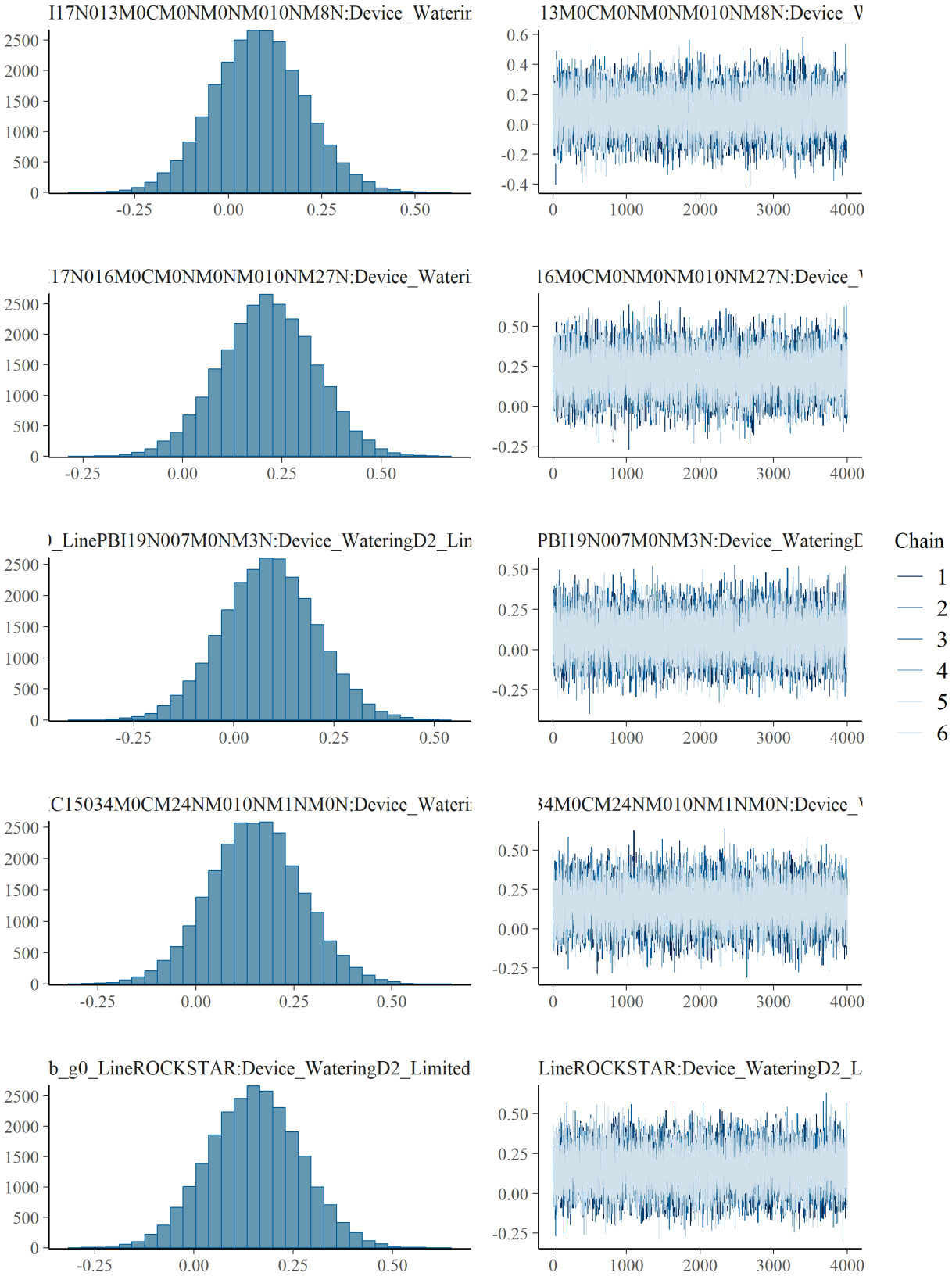


Figure 49: plot of chunk unnamed-chunk-5

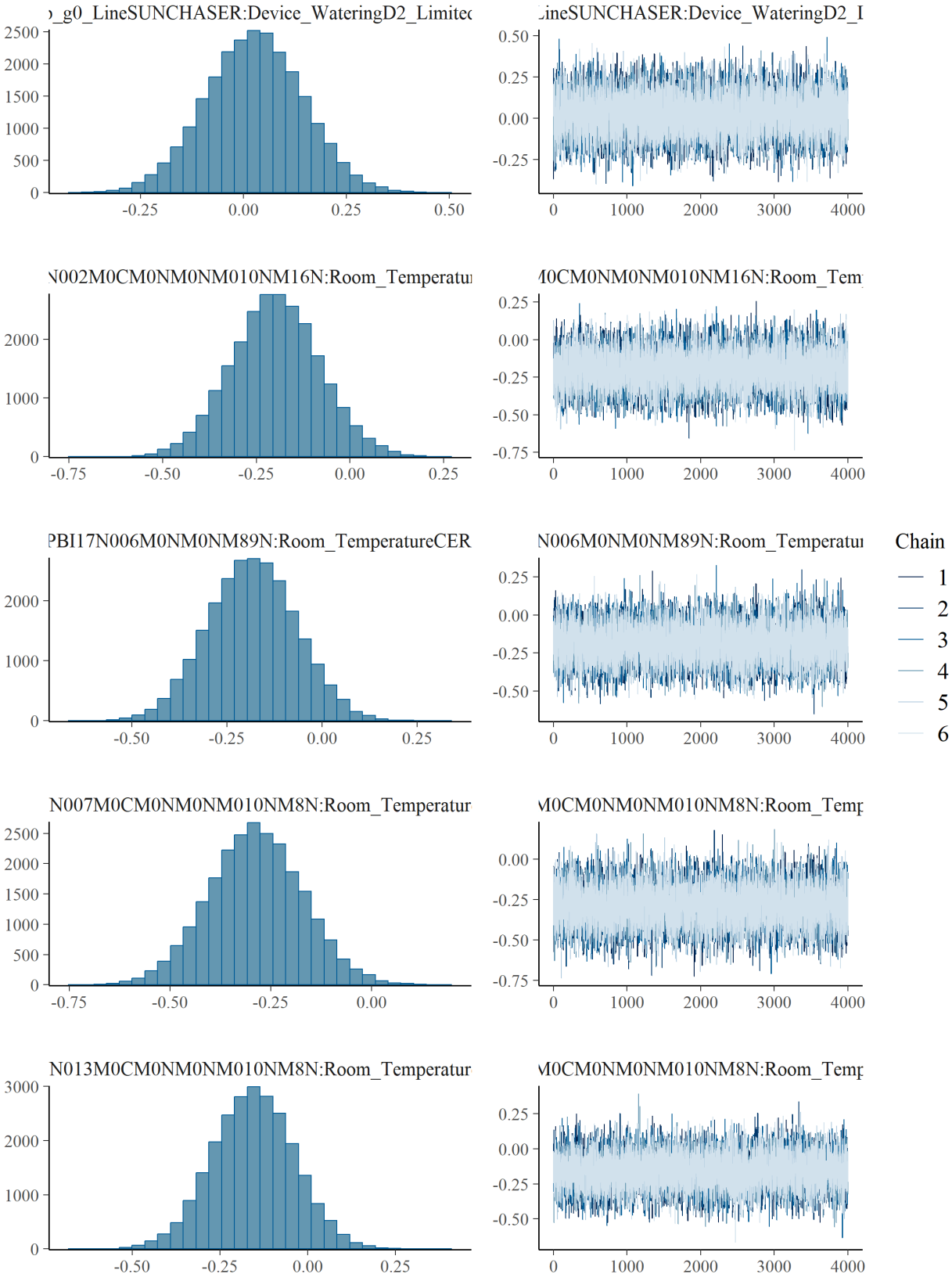


Figure 50: plot of chunk unnamed-chunk-5

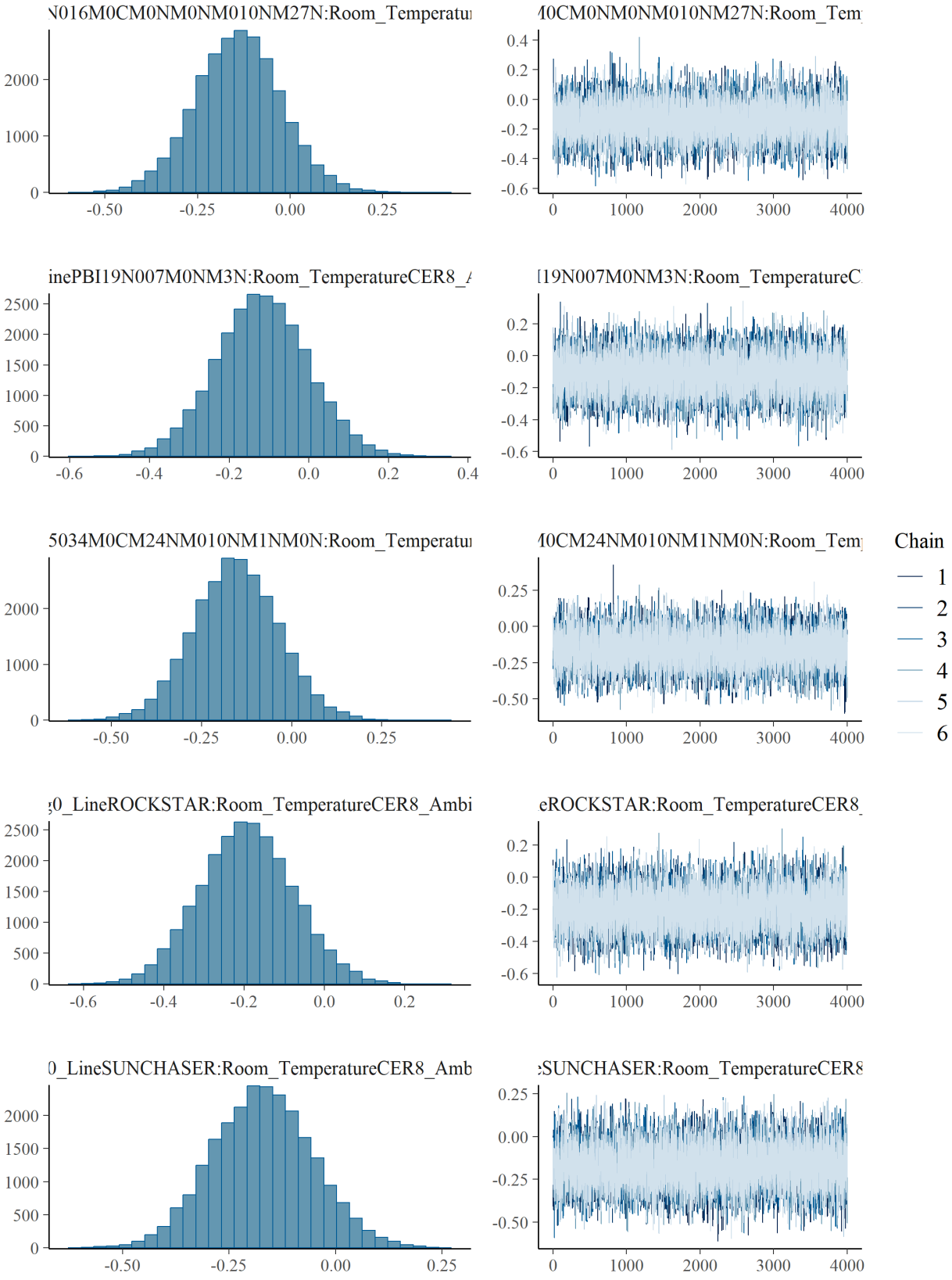


Figure 51: plot of chunk unnamed-chunk-5

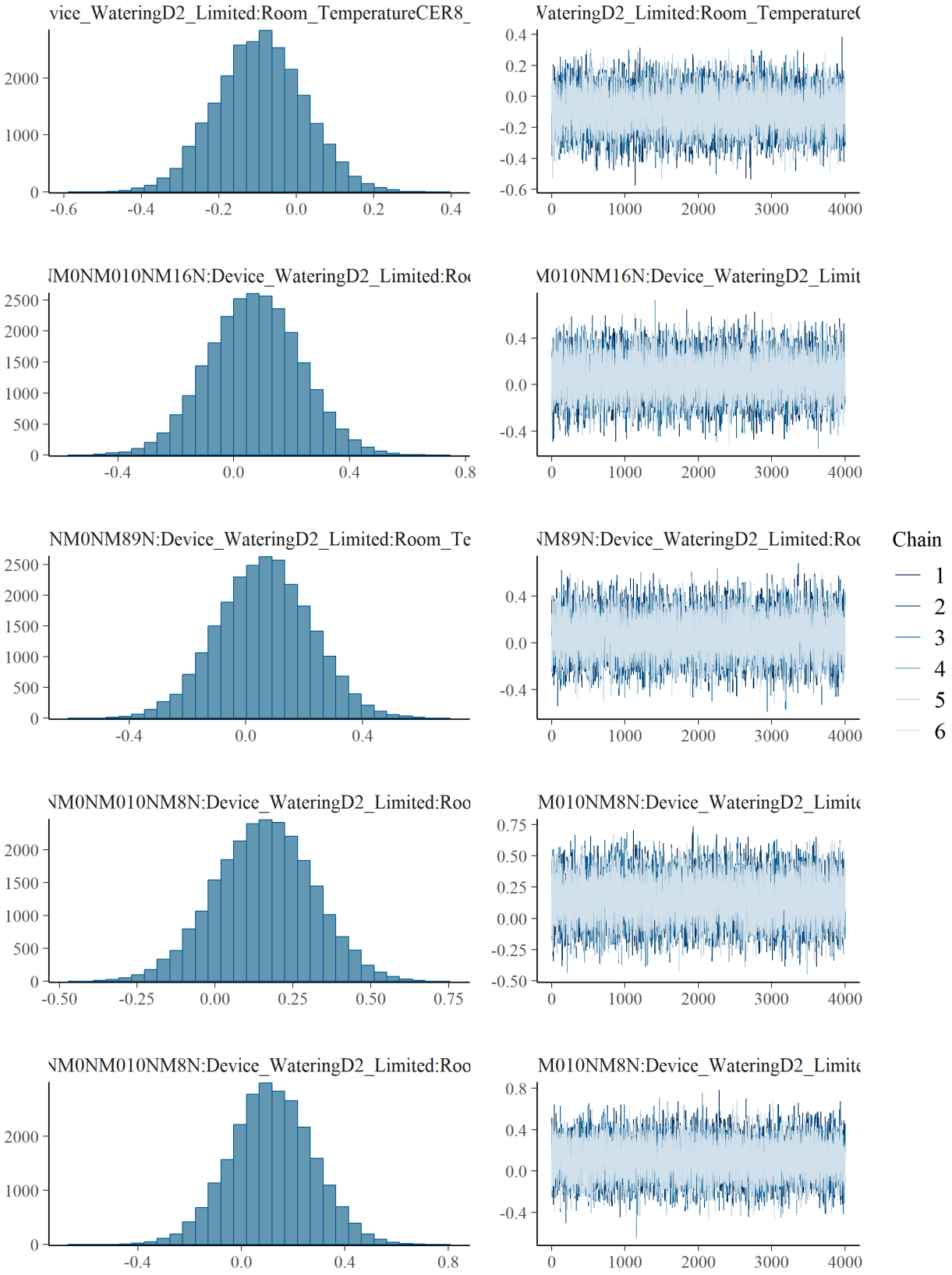


Figure 52: plot of chunk unnamed-chunk-5

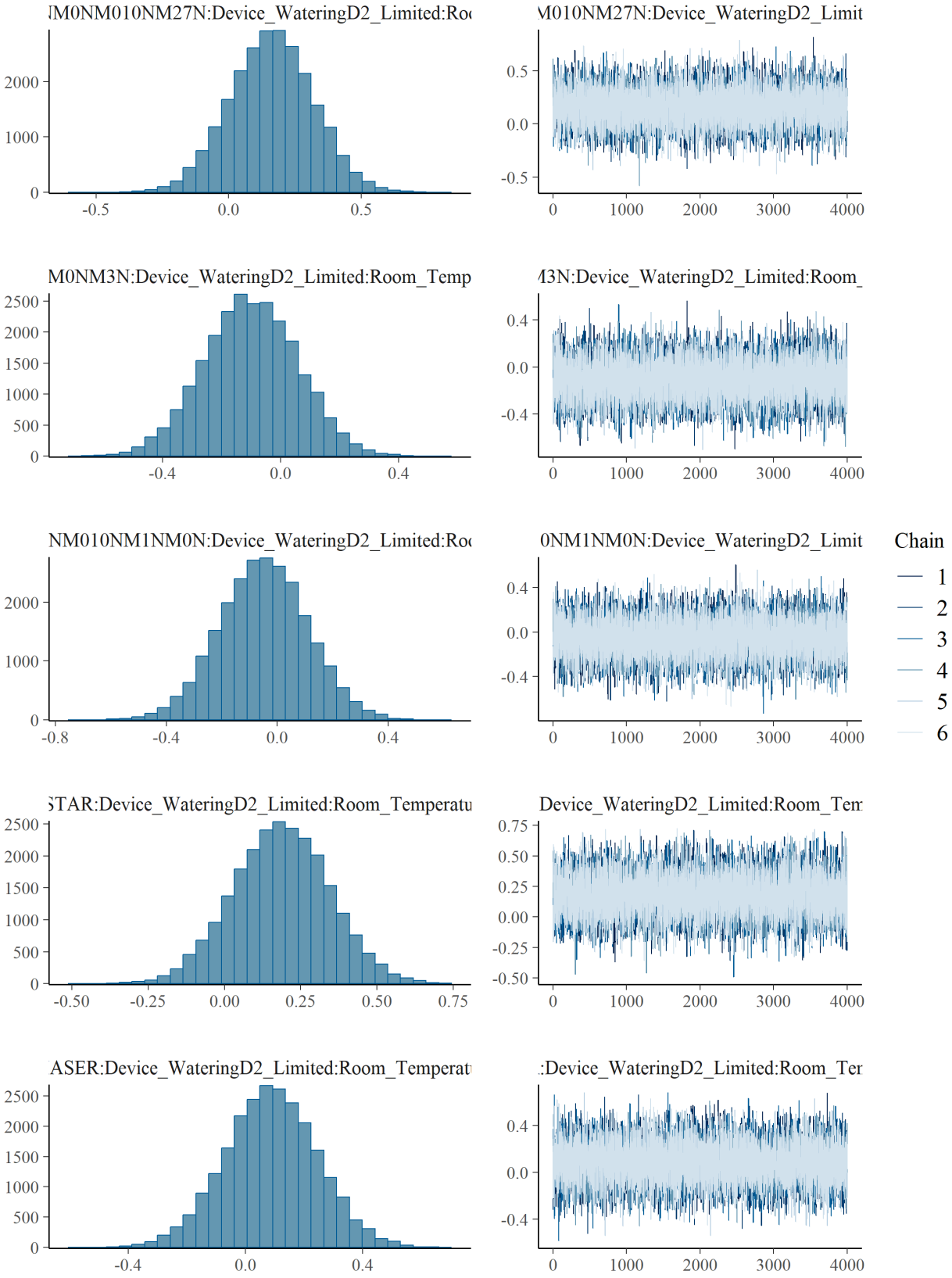


Figure 53: plot of chunk unnamed-chunk-5

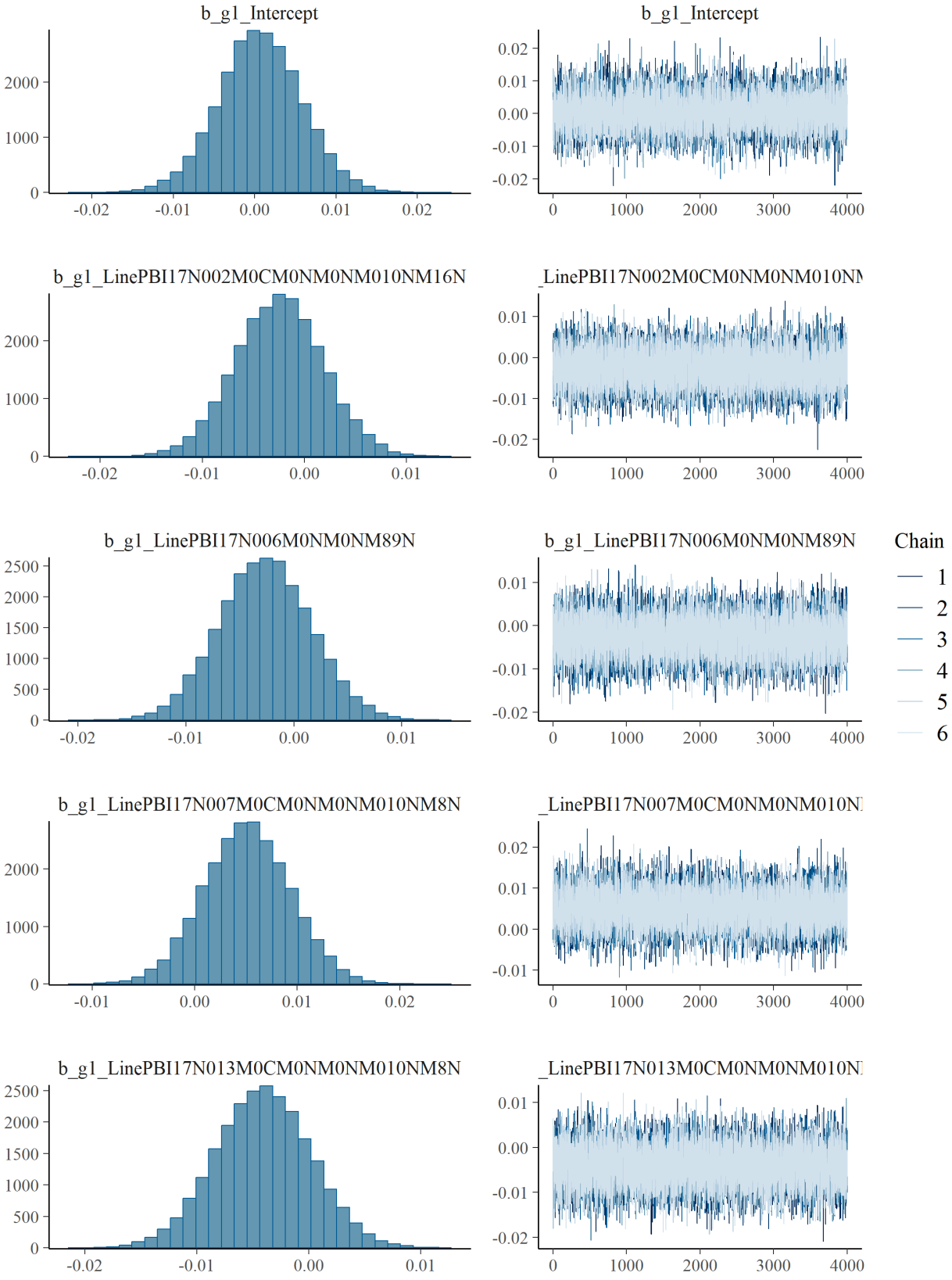


Figure 54: plot of chunk unnamed-chunk-5

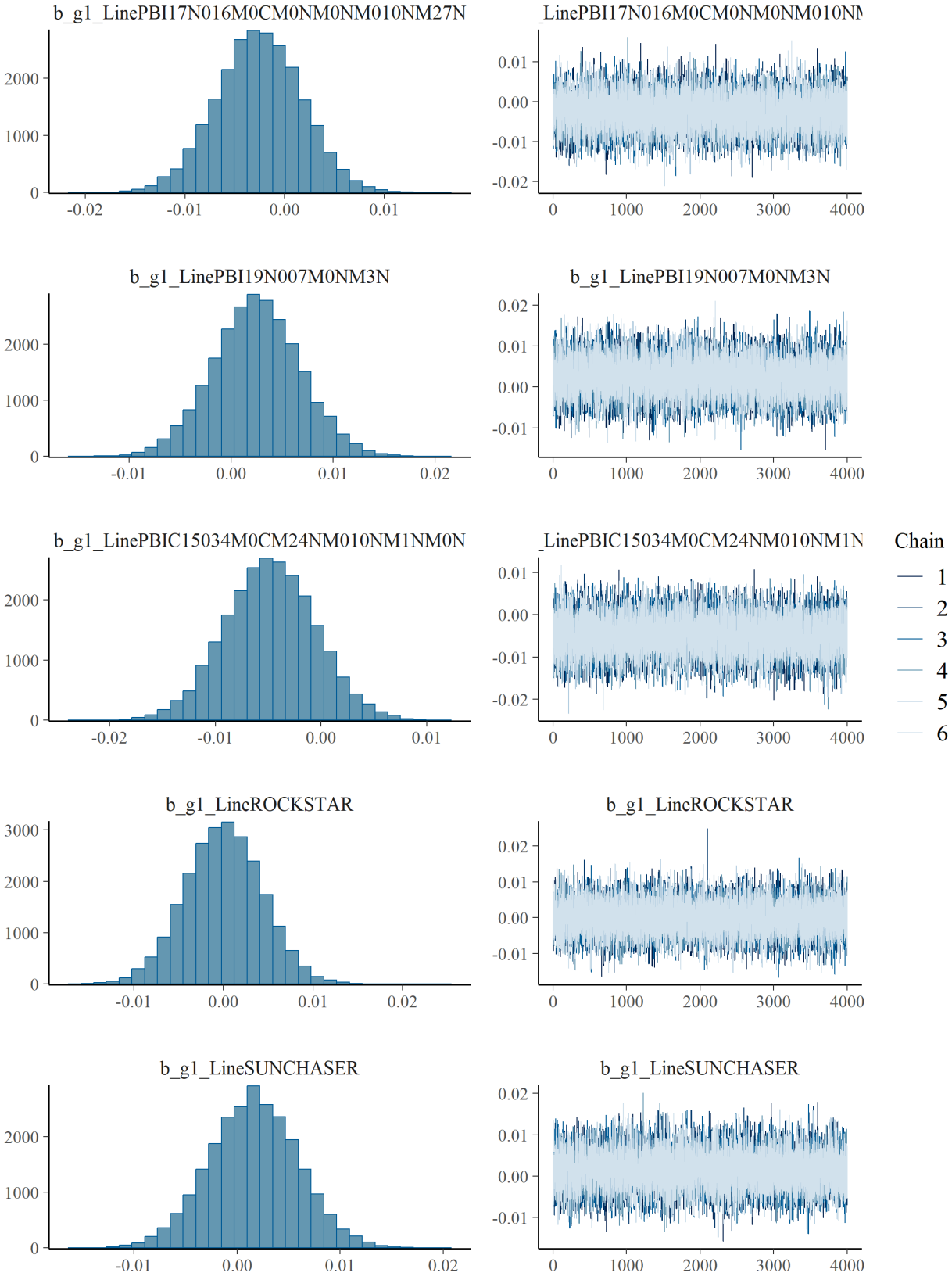


Figure 55: plot of chunk unnamed-chunk-5

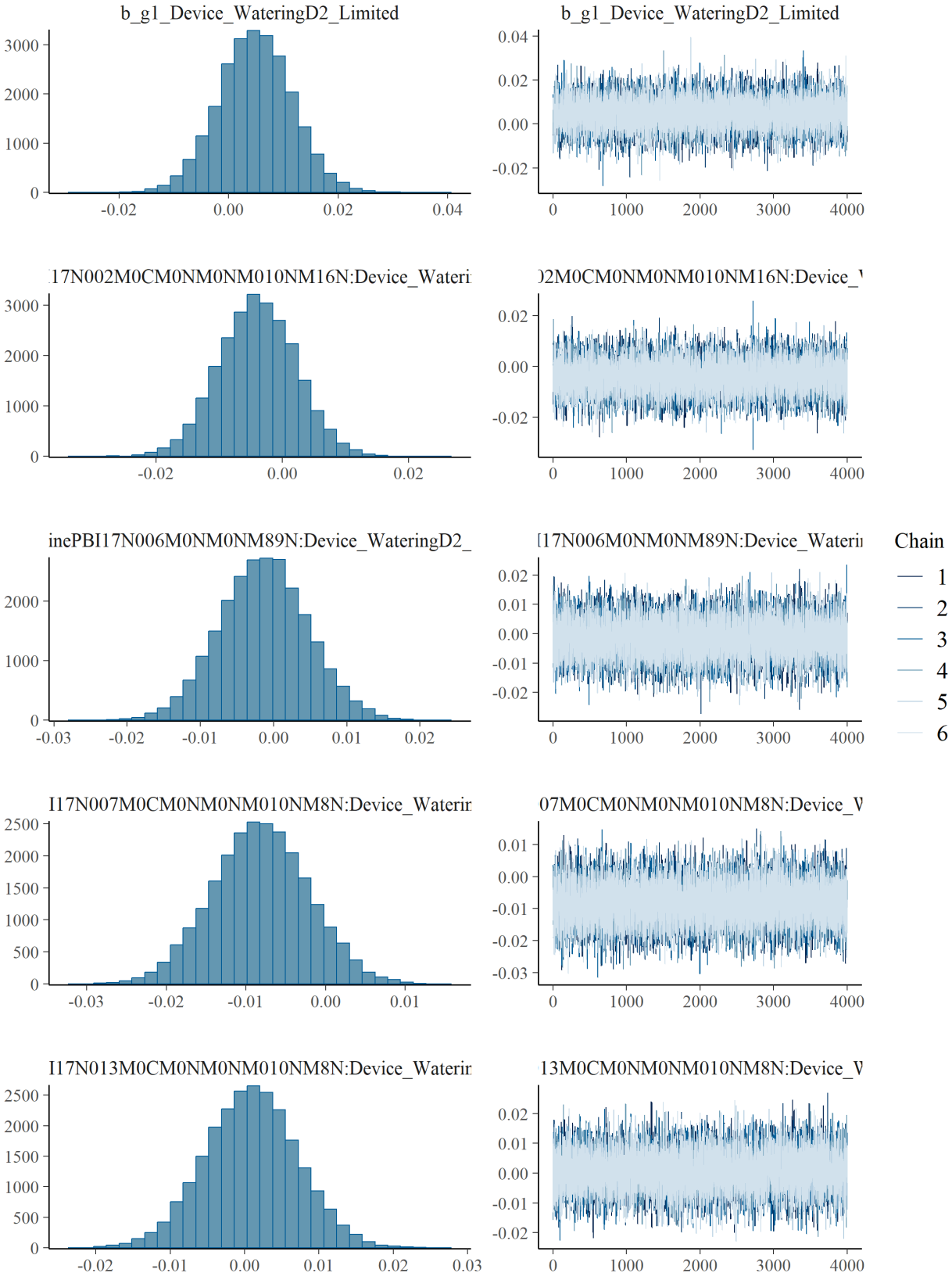


Figure 56: plot of chunk unnamed-chunk-5

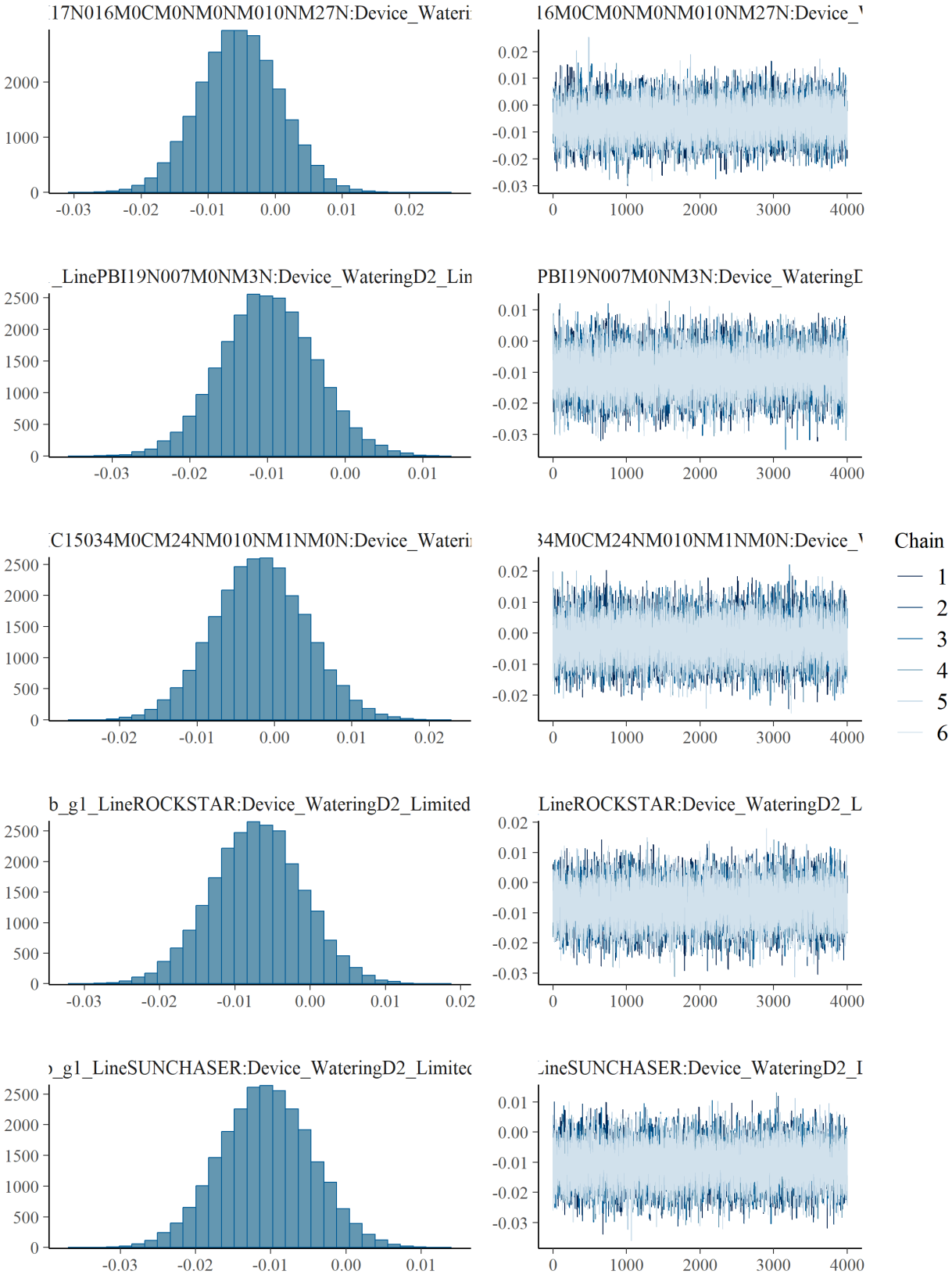


Figure 57: plot of chunk unnamed-chunk-5

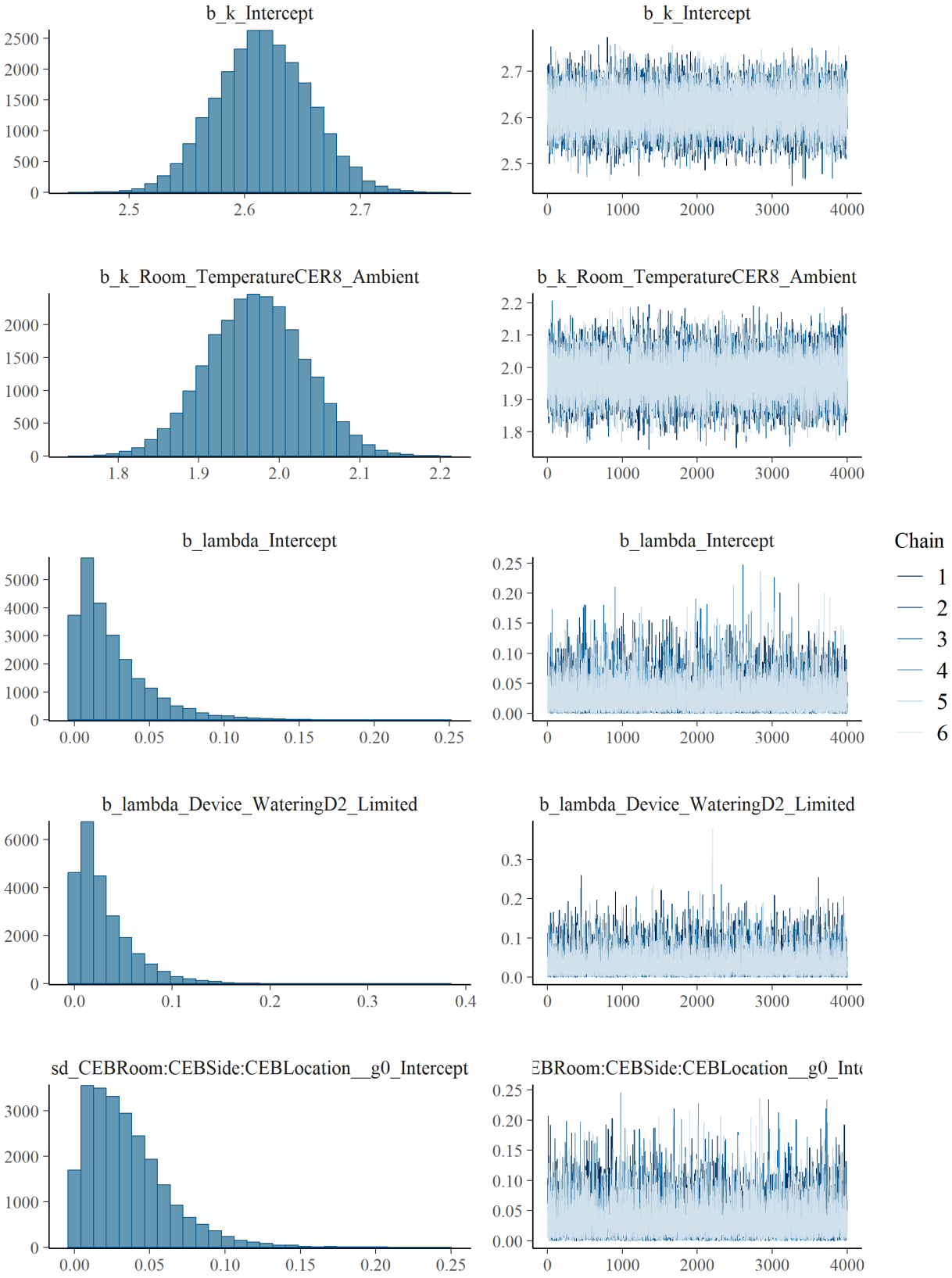


Figure 58: plot of chunk unnamed-chunk-5

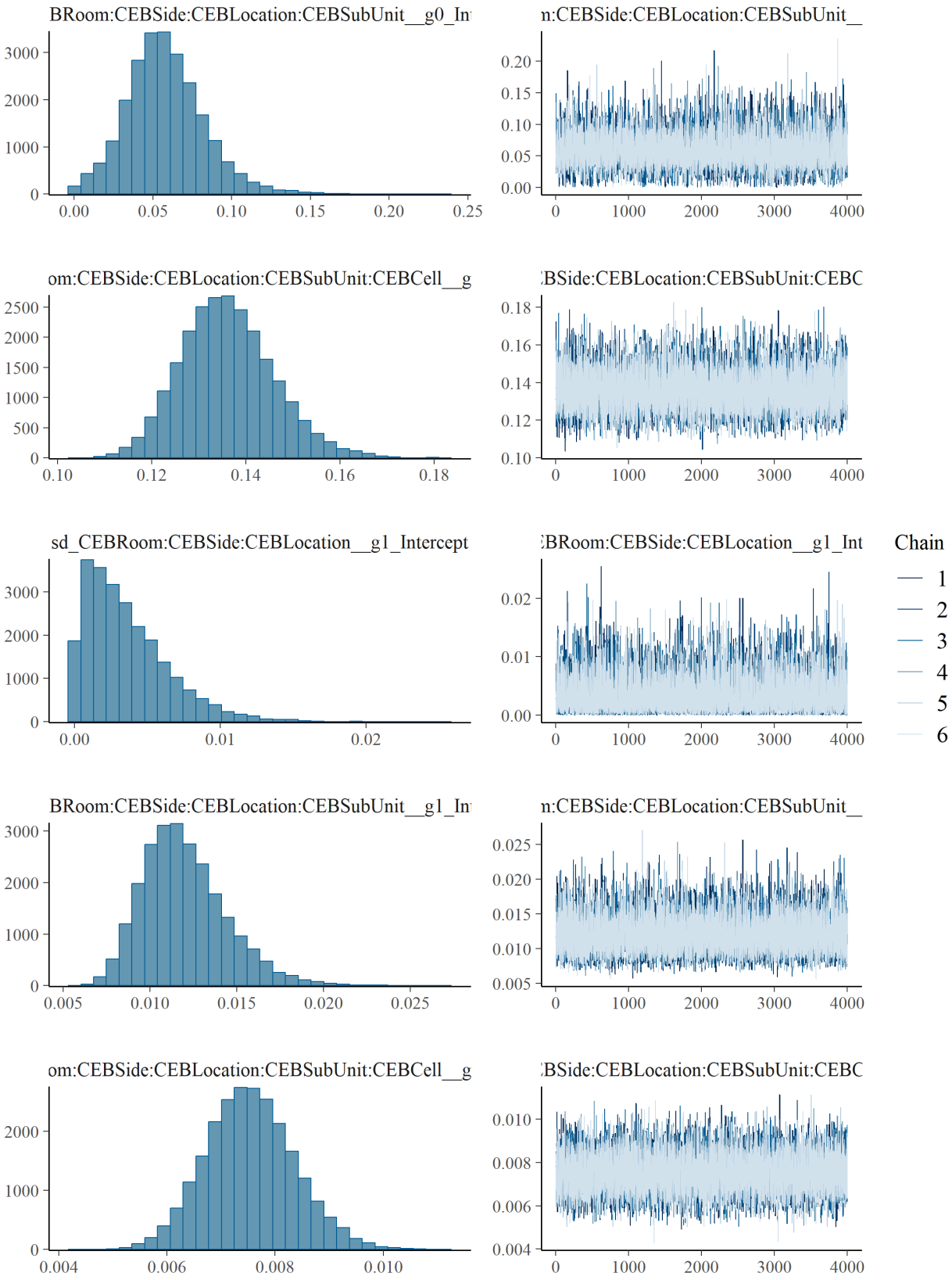


Figure 59: plot of chunk unnamed-chunk-5

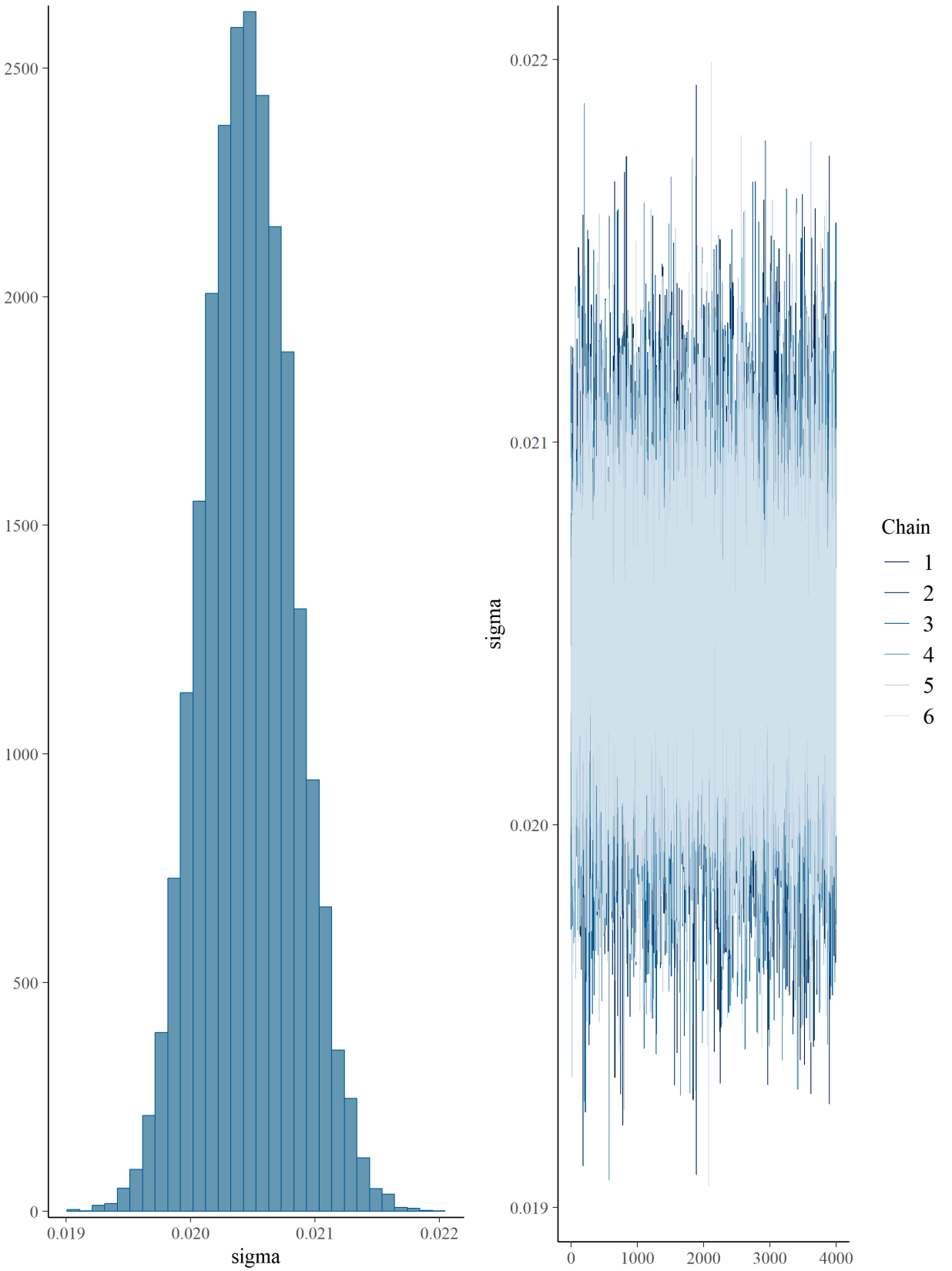


Figure 60: plot of chunk unnamed-chunk-5

```
li600.off.plot$fitted.fixed <- fitted(fit_off_final, re_formula = NA)[,1]

ggplot(li600.off.plot, aes(x= time.adj, y = gsw, group = Replicate)) +
  geom_point() +
  facet_grid(Room_Temperature + Device_Watering ~ Line) +
  geom_point(aes(y=fitted), colour = "red") +
  geom_line(aes(y=fitted), colour = "red")
```

```
ggplot(li600.off.plot, aes(x= time.adj, y = gsw, group = Replicate)) +
  geom_point() +
  facet_grid(Room_Temperature + Device_Watering ~ Line) +
  geom_point(aes(y=fitted), colour = "red") +
  geom_line(aes(y=fitted), colour = "red") +
  geom_point(aes(y=fitted.fixed), colour = "blue") +
  geom_line(aes(y=fitted.fixed), colour = "blue")
```

Save data

```
save.image(saveFile)
```

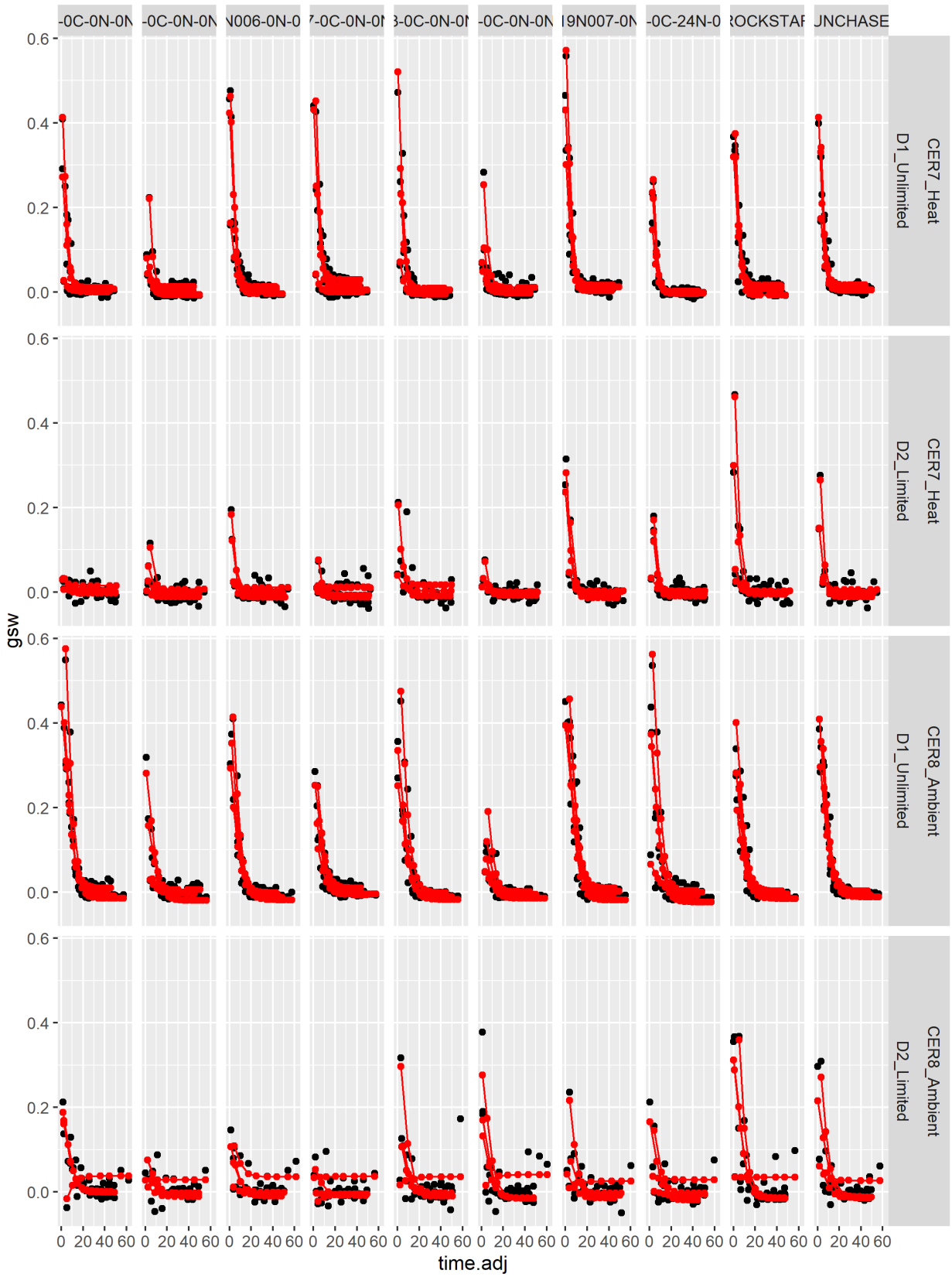


Figure 61: plot of chunk unnamed-chunk-5

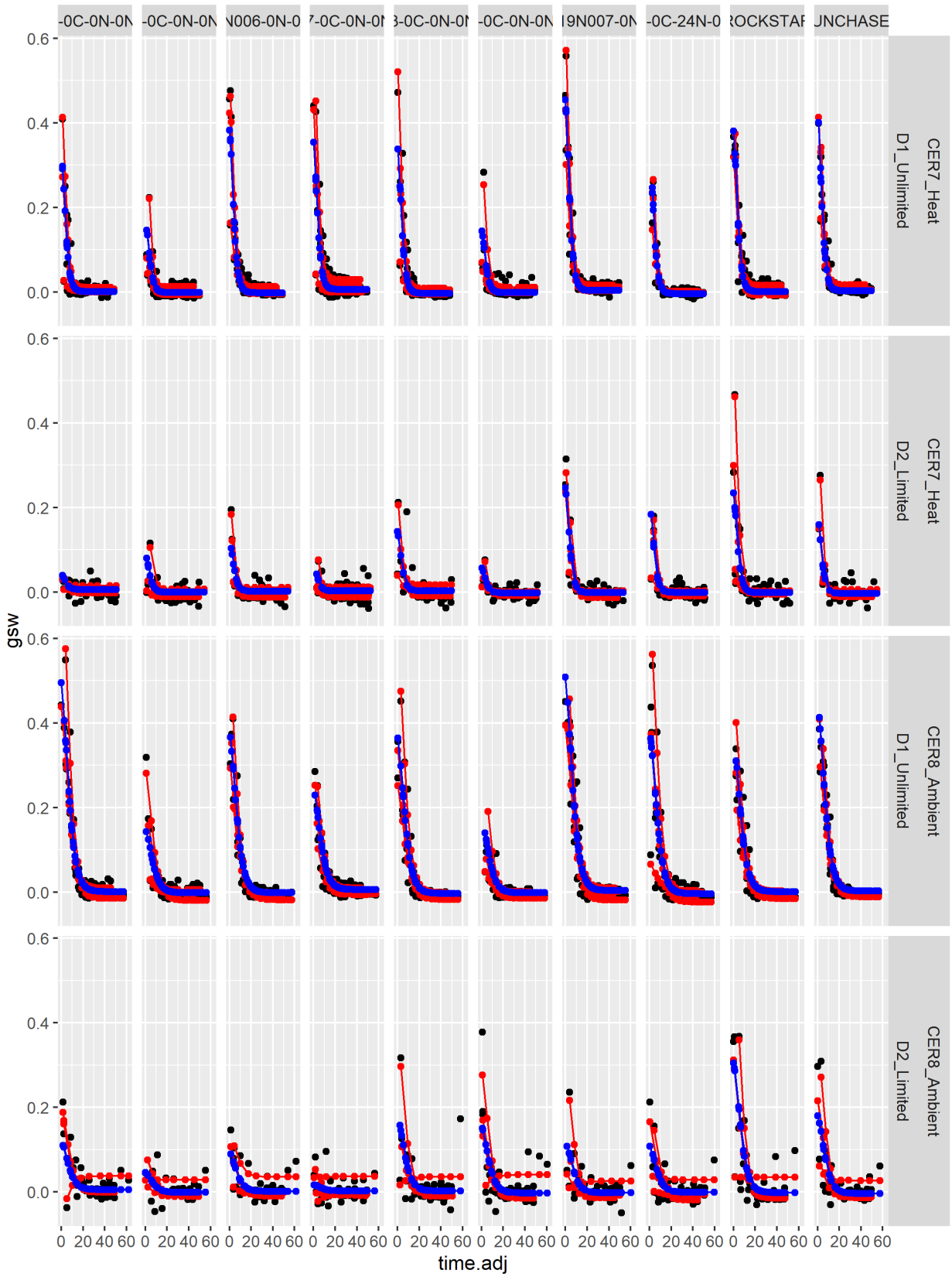


Figure 62: plot of chunk unnamed-chunk-5